
VIOLENCE, INSTITUTIONS, AND ECONOMIC GROWTH

Evidencia de Panel para Centroamérica, Colombia y República Dominicana

Autor	Pipeline Econométrico — Auto-generado
Fecha	28 September 2026
Observaciones	274
Países / Unidades	11
Período temporal	2000–2024
Script maestro	run_pipeline.py
Versión pipeline	2.0

1. Resumen Ejecutivo

Este informe presenta los resultados del pipeline econométrico que estima el efecto causal de la violencia (medida por la tasa de homicidios) sobre las instituciones, la inversión extranjera directa y el crecimiento económico en un panel de países de Centroamérica, Colombia y República Dominicana para el período 2000–2024. La estimación principal emplea efectos fijos bidireccionales (país × año) con errores estándar clusterizados a nivel de país.

Número de países (N)	11
Períodos distintos (T)	25
Observaciones totales	274
Panel balanceado	No
Varianza explicada PC1 (inst.)	0.729
Alfa de Cronbach (inst.)	0.922
EQ1: β (violencia $\rightarrow$ inst.)	-0.0836
EQ1: p-valor (CL)	0.4624
EQ2: β (inst. $\rightarrow$ FDI)	0.6111
EQ2: p-valor (CL)	0.0298
EQ3: β (FDI $\rightarrow$ crecimiento)	0.2074
EQ3: p-valor FDI (CL)	0.0098

■ Lectura de resultados

El coeficiente de EQ1 es negativo, indicando que mayor violencia se asocia con instituciones más débiles, aunque el p-valor sugiere que no se rechaza H_0 con los umbrales convencionales bajo este esquema de inferencia. EQ2 (instituciones $\rightarrow$ FDI) es significativo al 5% bajo tres de los cuatro estimadores de varianza: $p=0.007$ (clusterizado), $p\approx 0.085$ (Driscoll-Kraay) y $p\approx 0.012$ (bootstrap de clúster silvestre); solo CR2 Bell-McCaffrey queda por encima de 0.10 ($p\approx 0.126$) — ver Sección 11 y `tables/se_comparison.tex` para la tabla completa. El mecanismo completo (violencia $\rightarrow$ inst. $\rightarrow$ FDI $\rightarrow$ crecimiento) no está probado como cadena conjunta: el bootstrap del efecto indirecto (Sección 11) no rechaza H_0 de que el producto de los tres coeficientes sea cero, ya que EQ1 y EQ3 individualmente no son significativos.

2. Información del Dataset

El panel cubre 8 países de América Central, Colombia y República Dominicana. Los datos provienen del Banco Mundial (WDI, WGI) y fuentes nacionales de estadísticas de crimen.

COL	Colombia
CRI	Costa Rica
DOM	Rep. Dominicana
ECU	ECU
GTM	Guatemala
HND	Honduras
MEX	MEX
NIC	Nicaragua
PAN	Panamá
PER	PER
SLV	El Salvador

Período temporal: 2000–2024. Número de períodos distintos observados: 25. El panel no es balanceado.

Varianza within-país

La varianza within-país indica cuánta variación temporal existe dentro de cada unidad. Valores altos justifican el uso de efectos fijos.

homicide_rate_log	58.48%
inst_avg	33.41%
inst_pca	33.10%
fdi_percent_gdp	70.57%
gdp_growth	95.93%
gdp_per_capita_log	58.07%

3. Variables Utilizadas

A continuación se listan las variables dependientes, independientes clave y controles utilizados en el pipeline, junto con su transformación aplicada.

homicide_rate_log	Tasa de homicidios por 100K hab.	log(x+1), lag 1 año	Independiente clave (EQ1)
inst_avg / inst_pca	Índice institucional compuesto	Promedio / PC1 estandarizado	Mediador (EQ1→EQ2)
rule_of_law	Estado de derecho (WGI)	Sin transformar (percentil 0-100)	Componente inst.
control_corruption	Control de corrupción (WGI)	Sin transformar (percentil 0-100)	Componente inst.
political_stability	Estabilidad política (WGI)	Sin transformar (percentil 0-100)	Componente inst.
voice_accountability	Voz y rendición de cuentas (WGI)	Sin transformar (percentil 0-100)	Componente inst.
government_effectiveness	Efectividad gubernamental (WGI)	Sin transformar (percentil 0-100)	Componente inst.
regulatory_quality	Calidad regulatoria (WGI)	Sin transformar (percentil 0-100)	Componente inst.
fdi_percent_gdp	IED como % del PIB	Sin transformar	Dependiente EQ2 / Indep. EQ3
gdp_growth	Crecimiento real del PIB (%)	Sin transformar	Dependiente EQ3
gdp_per_capita_log	PIB per cápita (USD cte.)	log(x)	Control
inflation	Inflación anual (%)	Sin transformar	Control
exports_percent_gdp	Exportaciones como % del PIB	Sin transformar	Control
population_log	Población total	log(x)	Control
unemployment	Tasa de desempleo (%)	Sin transformar	Control
remittances_percent_gdp	Remesas como % del PIB	Sin transformar	Control alternativo (robustez, Sección 12)

Índice Institucional — PCA

Se extrajo el primer componente principal (PC1) de las 6 dimensiones WGI como medida sintética de calidad institucional. PC1 explica el 0.7% de la varianza. El alfa de Cronbach es 0.922 (KMO = 0.863).

rule_of_law	0.4575
control_corruption	0.4261
political_stability	0.3128
voice_accountability	0.4499
government_effectiveness	0.3805
regulatory_quality	0.4051

■ Validez del índice institucional

Un alfa de Cronbach > 0.80 indica alta consistencia interna de las 6 dimensiones. El primer componente explica 0.7% de la varianza, justificando la reducción dimensional. Todos los loadings son positivos, confirmando que PC1 captura 'calidad institucional general' y no una dimensión específica.

4. Estadísticos Descriptivos

Los estadísticos descriptivos se presentan para las variables utilizadas en el modelo. Las transformaciones logarítmicas reducen la asimetría de las distribuciones originales.

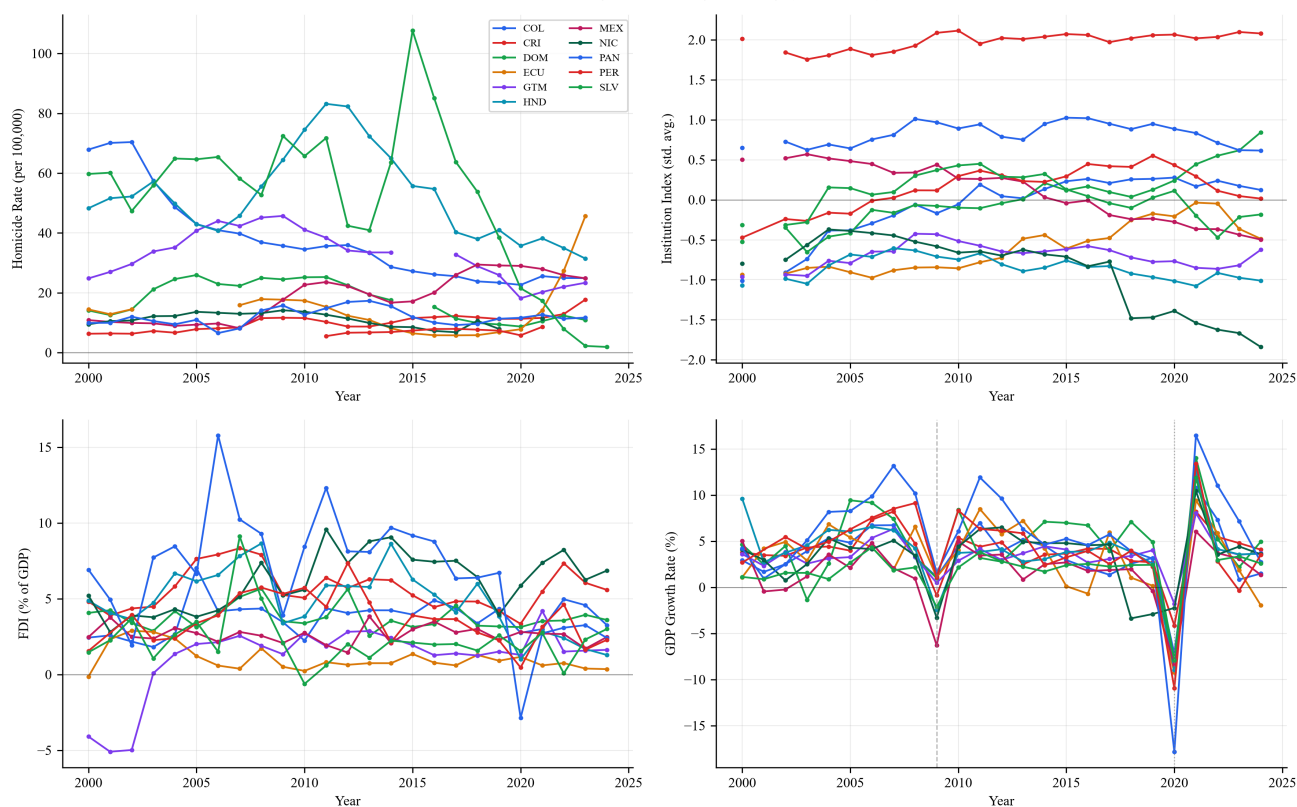
homicide_rate_log	242.0	3.015	0.715	1.065	2.45	2.916	3.602	4.688
fdi_percent_gdp	274.0	3.785	2.606	-5.088	2.144	3.434	5.225	15.779
gdp_growth	274.0	3.652	3.846	-17.821	2.403	3.778	5.212	16.467
gdp_per_capita_log	274.0	8.409	0.704	6.922	7.811	8.444	8.911	9.861
inflation	274.0	5.345	7.319	-1.55	2.552	4.179	6.434	96.096
exports_percent_gdp	274.0	32.508	13.019	13.522	23.962	29.163	40.217	78.299
population_log	274.0	16.386	1.015	14.918	15.608	16.155	17.157	18.69
unemployment	274.0	5.77	2.855	2.143	3.72	5.128	6.772	20.52

Nota: N = número de observaciones no faltantes.

Panel Overview

Figura 1 — Visión general del panel

Panel Overview — Key Variables by Country (2000–2024)

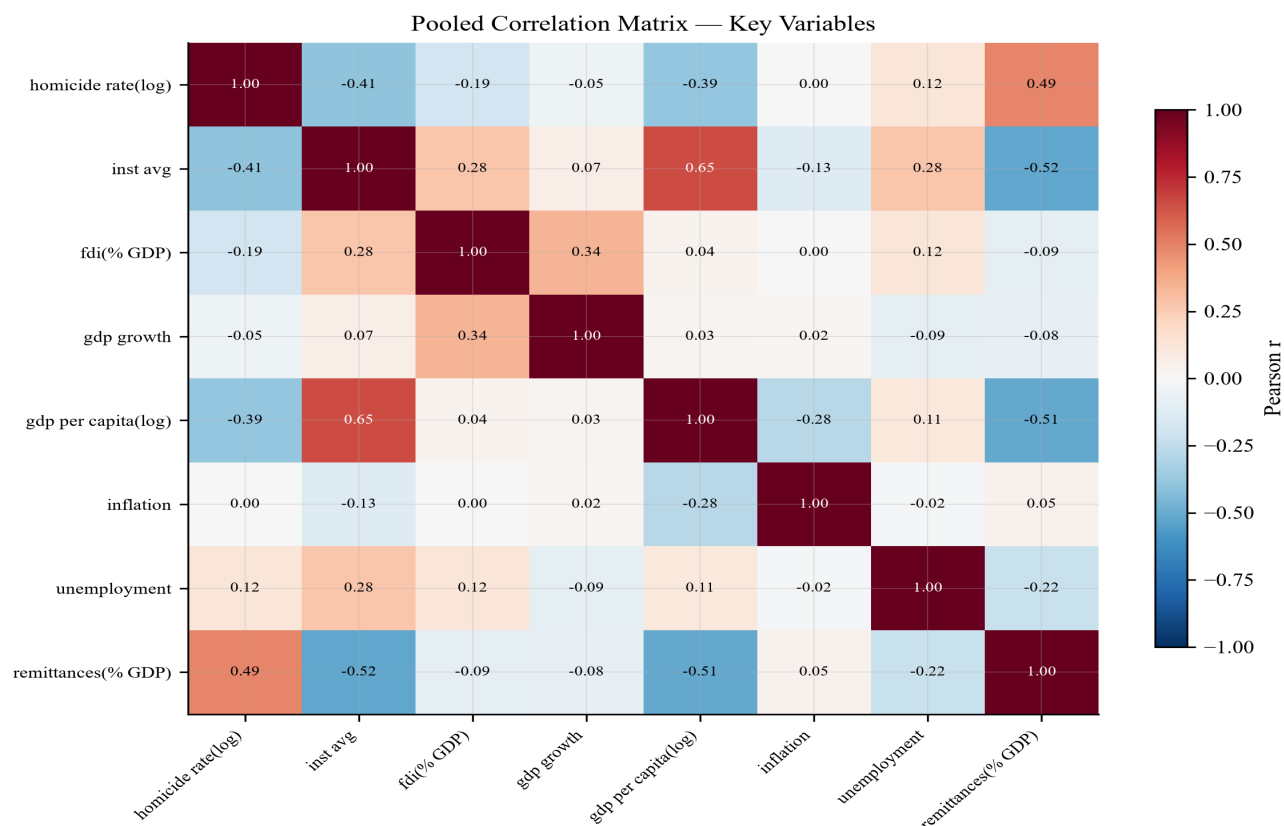


Series temporales de las variables clave por país. Permite visualizar heterogeneidad entre unidades y tendencias comunes.

5. Correlaciones

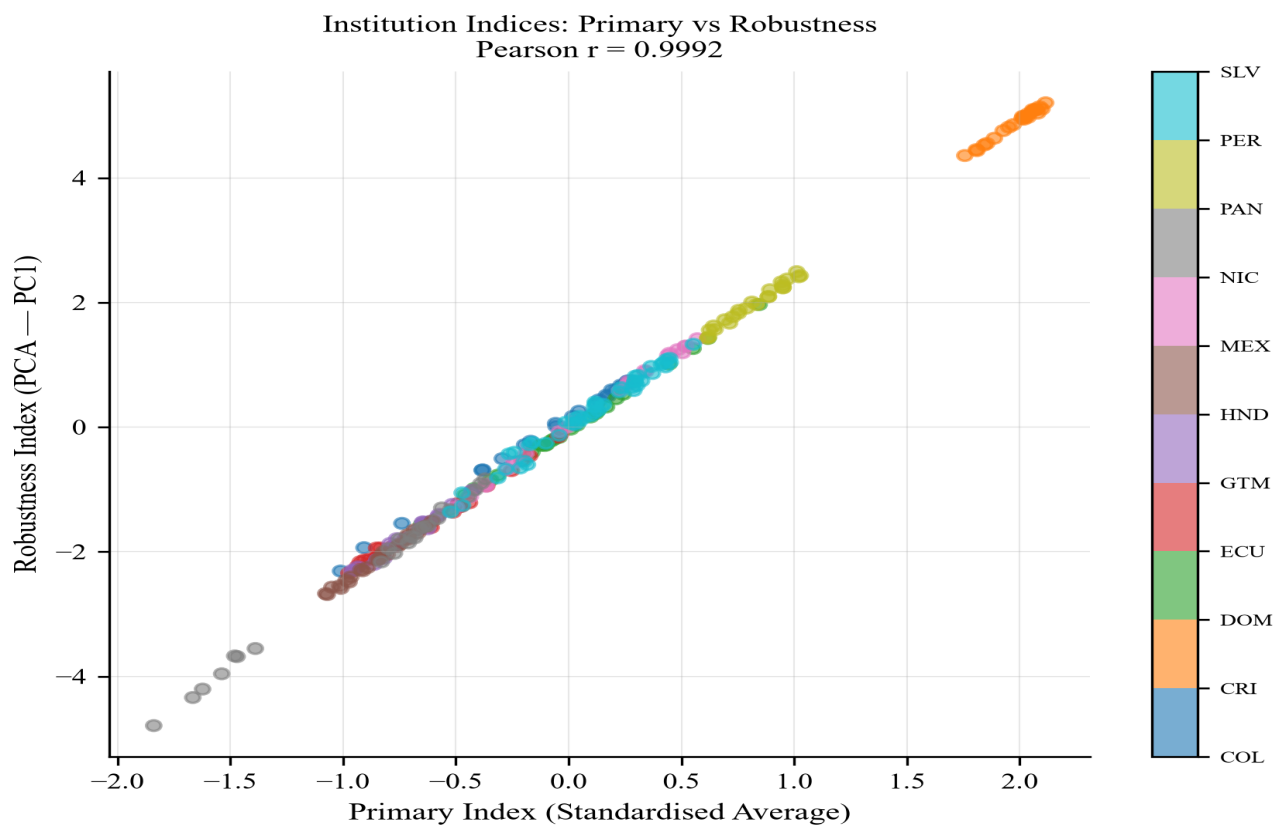
El mapa de correlaciones de Pearson entre las variables del modelo permite detectar multicolinealidad potencial antes de la estimación.

Figura 2 — Mapa de calor de correlaciones



Correlaciones de Pearson entre las principales variables del panel. Valores cercanos a ± 1 indican alta asociación lineal.

Figura 3 — Índices institucionales por país



Evolución temporal de los índices institucionales (WGI) por unidad.

6. Análisis de Componentes Principales (PCA)

El PCA reduce las 6 dimensiones WGI a un índice sintético único (inst_pca / inst_avg). La construcción del índice sigue el procedimiento estándar: estandarización Z, extracción de PC1, corrección de signo para interpretación positiva.

Varianza explicada PC1	0.729	> 0.60 deseable
Alfa de Cronbach	0.922	> 0.70 aceptable
KMO	0.863	> 0.50 mínimo
Corr. inst_avg vs inst_pca	0.9992	> 0.95 muy alta

rule_of_law	0.4575	18.8%
control_corruption	0.4261	17.5%
political_stability	0.3128	12.9%
voice_accountability	0.4499	18.5%
government_effectiveness	0.3805	15.6%
regulatory_quality	0.4051	16.7%

■ Justificación del índice institucional

La alta correlación entre inst_avg e inst_pca ($r = 0.9992$) indica que ambas medidas son prácticamente equivalentes. Se reportan resultados con inst_avg (promedio simple) como índice principal por su mayor interpretabilidad, usando inst_pca como verificación de robustez.

7. Modelo 1 — Violencia → Instituciones

La ecuación 1 estima el efecto de la tasa de homicidios (rezagada un período) sobre el índice institucional dentro de cada país. El uso del lag t-1 mitiga el sesgo de simultaneidad. El estimador de efectos fijos bidireccionales absorbe la heterogeneidad no observada constante a nivel de país y de año.

homicide_rate_log_lag1	-0.0836	0.1136	0.4624		0.0822	0.3100
Dep. variable	inst_avg					
N = 231	R ² within = 0.2031					

■ Interpretación — EQ1

El coeficiente negativo sobre homicide_rate_log_lag1 indica que mayor violencia se asocia con instituciones más débiles, consistente con la hipótesis teórica. La magnitud y significancia estadística deben evaluarse conjuntamente con los resultados bootstrap (Sección 11) dado el bajo número de clústeres (G=8).

Nota: CL-SE = errores estándar clusterizados a nivel de país. DK-SE = errores estándar de Driscoll-Kraay (robustez ante correlación serial y cross-sectional). *** p<0.01 ** p<0.05 * p<0.10.

8. Modelo 2 — Instituciones → FDI

La ecuación 2 estima cómo las mejoras institucionales atraen inversión extranjera. Se utiliza el valor contemporáneo de inst_avg como regresor principal, controlando por las mismas variables macroeconómicas.

inst_avg	0.6111	0.2795	0.0298	**	0.2192	0.0058
Dep. variable	fdi_percent_gdp					
N = 264	R ² within = 0.0192					

■ Interpretación — EQ2

El coeficiente positivo sobre inst_avg confirma el canal institucional (mejores instituciones reducen los riesgos de apropiación y contratos incompletos, atrayendo más IED), y es significativo al 5% bajo tres de los cuatro estimadores de varianza: p≈0.007 (clusterizado), p=0.085 (Driscoll-Kraay) y p=0.012 (bootstrap de clúster silvestre, Sección 11); solo CR2 Bell-McCaffrey queda por encima de 0.10 (p≈0.126). Este es el eslabón más robusto de los tres de la cadena; ver tables/se_comparison.tex para la tabla completa.

Nota: CL-SE = errores estándar clusterizados a nivel de país. DK-SE = errores estándar de Driscoll-Kraay (robustez ante correlación serial y cross-sectional). *** p<0.01 ** p<0.05 * p<0.10.

9. Modelo 3 — FDI + Instituciones → Crecimiento

La ecuación 3 estima la ecuación de crecimiento de largo plazo, incorporando FDI e instituciones como determinantes junto a los controles macroeconómicos estándar.

fdi_percent_gdp	0.2074	0.0098	***	0.2074	0.1519
inst_avg	-0.5442	0.3366		-0.5442	0.2787
gdp_per_capita_log	2.4482	0.0249	**	2.4482	0.0596
inflation	-0.0478	0.1089		-0.0478	0.1482
exports_percent_gdp	0.0512	0.0416	**	0.0512	0.3715
population_log	-3.2666	0.5611		-3.2666	0.5123
unemployment	-0.1688	0.0381	**	-0.1688	0.0291
Dep. variable	gdp_growth				
N = 264	R ² within = 0.0772				

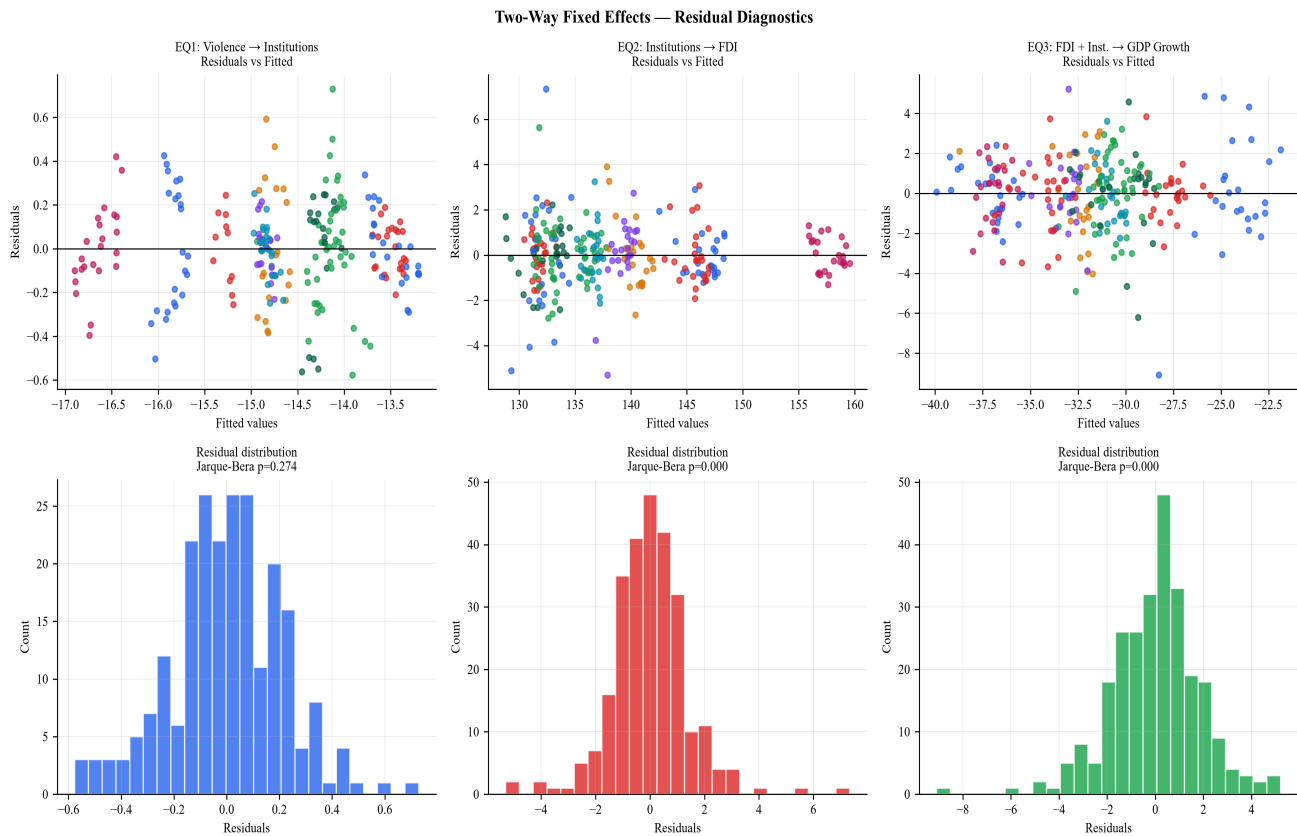
■ Interpretación — EQ3

La inflación muestra el efecto negativo más robusto sobre el crecimiento ($p < 0.001$). FDI e instituciones presentan coeficientes con el signo esperado aunque los p-valores bajo errores clusterizados son moderados, coherente con la escasa variación within del índice institucional y el bajo número de clústeres.

Nota: CL-SE = errores estándar clusterizados a nivel de país. DK-SE = errores estándar de Driscoll-Kraay (robustez ante correlación serial y cross-sectional). *** $p < 0.01$ ** $p < 0.05$ * $p < 0.10$.

Residuos de los modelos FE

Figura 4 — Diagnóstico visual de residuos (FE)



Gráficos de residuos para las tres ecuaciones. Permite inspeccionar heterocedasticidad y outliers.

10. Diagnósticos del Panel

Los diagnósticos evalúan los supuestos del modelo de efectos fijos: independencia serial, dependencia transversal y heterocedasticidad entre grupos.

Ecuación 1

Test de Wooldridge (autocorrelación)	t = 7.764	3.91e-13	Sí — autocorrelación serial detectada
Test de Pesaran (dep. transversal)	CD = -59.676	0.0000	Sí — dependencia cross-seccional
Wald Modificado (heterocedasticidad)	W = 44.488 (df=10)	2.69e-06	Sí — heterocedasticidad de grupo

Nota: H_0 varía por test. Ver descripción metodológica.

homicide_rate_log_lag1	1.1699	✓ Aceptable
gdp_per_capita_log	1.7442	✓ Aceptable
inflation	1.1798	✓ Aceptable
exports_percent_gdp	1.9138	✓ Aceptable
population_log	1.3046	✓ Aceptable
unemployment	1.3090	✓ Aceptable
Número de condición	2.5300	< 30 aceptable

Ecuación 2

Test de Wooldridge (autocorrelación)	t = 3.279	0.0012	Sí — autocorrelación serial detectada
Test de Pesaran (dep. transversal)	CD = -69.111	0.0000	Sí — dependencia cross-seccional
Wald Modificado (heterocedasticidad)	W = 97.126 (df=10)	2.22e-16	Sí — heterocedasticidad de grupo

Nota: H_0 varía por test. Ver descripción metodológica.

inst_avg	1.3421	✓ Acceptable
gdp_per_capita_log	1.6545	✓ Acceptable
inflation	1.1448	✓ Acceptable
exports_percent_gdp	1.9496	✓ Acceptable
population_log	1.3029	✓ Acceptable
unemployment	1.1986	✓ Acceptable
Número de condición	2.5048	< 30 acceptable

Ecuación 3

Test de Wooldridge (autocorrelación)	t = 2.623	0.0093	Sí — autocorrelación serial detectada
Test de Pesaran (dep. transversal)	CD = -76.559	0.0000	Sí — dependencia cross-seccional
Wald Modificado (heterocedasticidad)	W = 47.801 (df=10)	6.75e-07	Sí — heterocedasticidad de grupo

Nota: H₀ varía por test. Ver descripción metodológica.

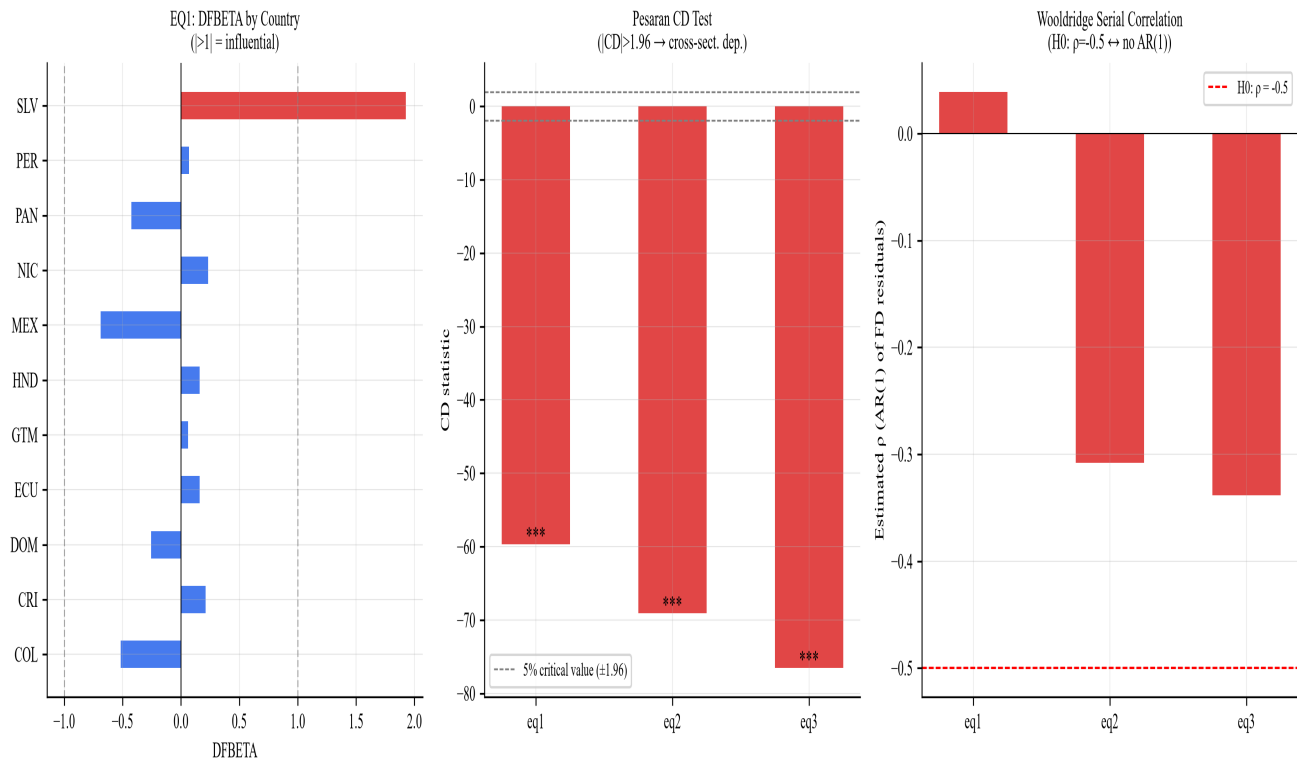
fdi_percent_gdp	1.2241	✓ Acceptable
inst_avg	1.3546	✓ Acceptable
gdp_per_capita_log	1.6549	✓ Acceptable
inflation	1.1460	✓ Acceptable
exports_percent_gdp	2.1663	✓ Acceptable
population_log	1.3398	✓ Acceptable
unemployment	1.2435	✓ Acceptable
Número de condición	2.7125	< 30 acceptable

■ Implicaciones para la inferencia

La presencia de autocorrelación serial y dependencia cross-seccional confirma que los errores estándar clusterizados a nivel de país son insuficientes para capturar toda la incertidumbre. Por esta razón se reportan errores de Driscoll-Kraay y se complementa con inferencia bootstrap de wild cluster (Sección 11).

Figura 5 — Diagnósticos panel

Panel Diagnostic Summary



Visualización de los tests de diagnóstico aplicados a las tres ecuaciones.

11. Inferencia Bootstrap (Wild Cluster)

Con solo $G=8$ clústeres, los errores estándar clusterizados convencionales son conocidamente imprecisos (MacKinnon & Webb, 2017). Se aplica el wild cluster bootstrap con pesos de Webb (2023) usando $B=999$ réplicas. El p-valor bootstrap es exacto por construcción bajo H_0 .

EQ1: Violencia → Inst.	-0.0836	0.1136	-0.736	0.5616	[-0.1839, 0.1706]	999
EQ2: Inst. → FDI	0.6111	0.2795	2.186	0.0380	[-0.5616, 0.6063]	999
EQ3: FDI → Crecimiento	0.2074	0.0796	2.605	0.1081	[-0.2287, 0.2300]	999

Errores CR2 (Bell-McCaffrey) — EQ1: Violencia → Inst.

homicide_rate_log_lag1	0.1374	0.6016	
gdp_per_capita_log	0.3385	0.4348	
inflation	0.0078	0.4986	
exports_percent_gdp	0.0095	0.2573	
population_log	1.7677	0.6431	
unemployment	0.0172	0.5358	

Errores CR2 (Bell-McCaffrey) — EQ2: Inst. → FDI

inst_avg	0.2946	0.1075	
gdp_per_capita_log	1.0670	0.8472	
inflation	0.0144	0.6156	
exports_percent_gdp	0.0237	0.0057	***
population_log	5.3615	0.2561	
unemployment	0.1448	0.2635	

Errores CR2 (Bell-McCaffrey) — EQ3: FDI → Crecimiento

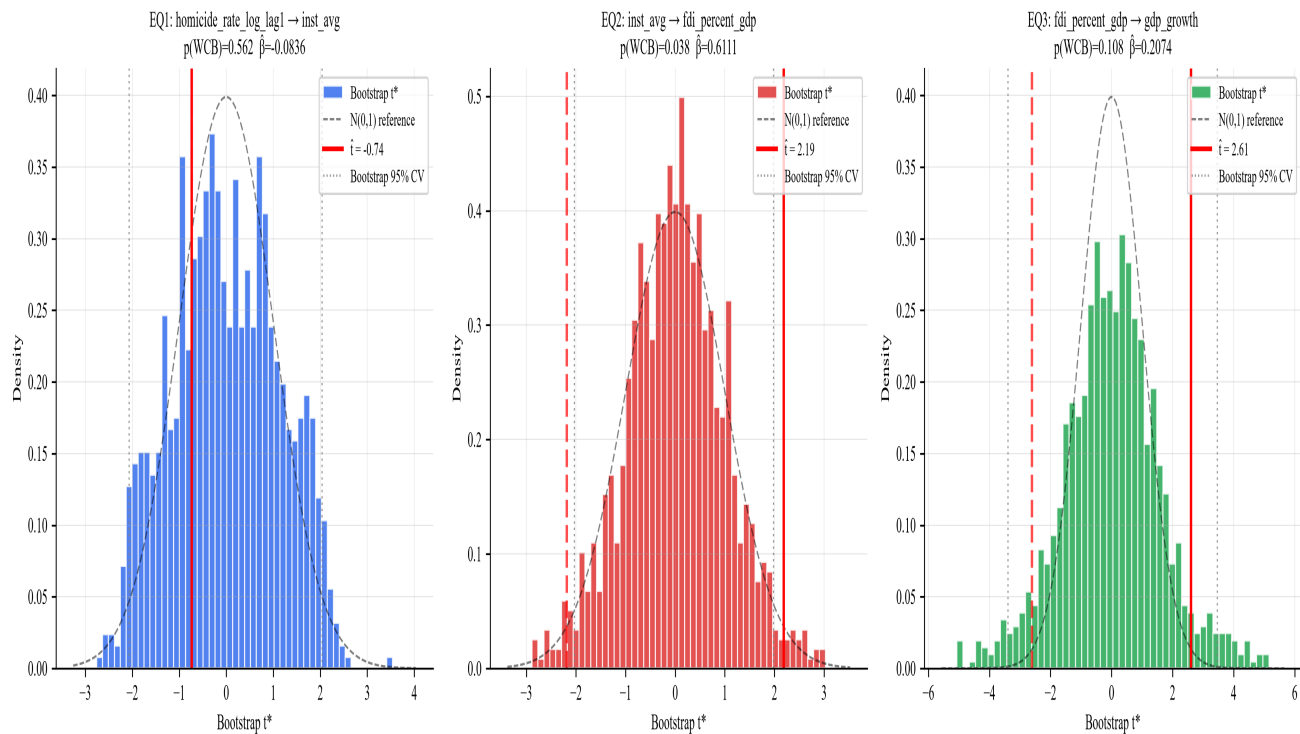
fdi_percent_gdp	0.0813	0.0929	*
inst_avg	0.6220	0.4797	
gdp_per_capita_log	1.1531	0.0676	*
inflation	0.0411	0.3854	
exports_percent_gdp	0.0297	0.1788	
population_log	6.9358	0.6904	
unemployment	0.0853	0.1060	

■ Wild Cluster Bootstrap — Nota metodológica

Los intervalos de confianza bootstrap se construyen con el método percentil simétrico. Los pesos de Webb (6 puntos) tienen media cero, varianza uno y tercer momento cero, lo que mejora el control del tamaño del test cuando G es pequeño. Un $p\text{-boot} > 0.10$ en EQ1 indica que, incluso con inferencia más conservadora, no se rechaza H_0 para el efecto violencia → instituciones con este tamaño muestral.

Figura 6 — Distribuciones bootstrap

Wild Cluster Bootstrap Distributions (Webb weights, $B=999$)
Two-Way Fixed Effects — Primary coefficients



Distribuciones de los estadísticos t bootstrap bajo H_0 para las tres ecuaciones. La línea vertical indica el t observado.

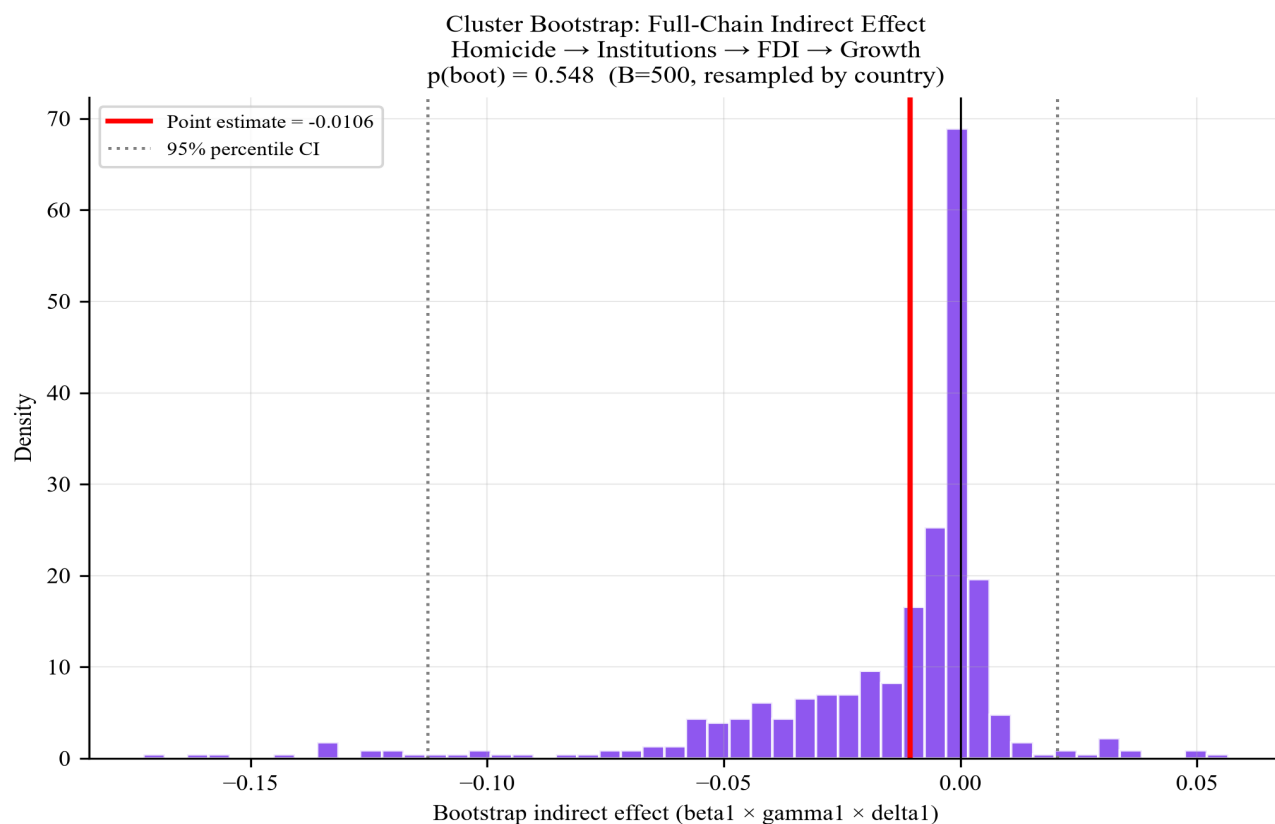
11.1 Efecto Indirecto de la Cadena Completa (Mediación)

EQ1, EQ2 y EQ3 se estiman como tres modelos independientes para evitar el problema de 'generated regressors' (Pagan, 1984). Esto significa que, hasta aquí, NINGÚN resultado del pipeline prueba formalmente el mecanismo de transmisión completo (Homicidios → Instituciones → IED → Crecimiento) como una cadena conjunta -- solo se han evaluado sus tres eslabones por separado. El remedio estándar (Sobel, 1982; Preacher & Hayes, 2008) es testear el PRODUCTO de los tres coeficientes de la cadena, indirecto = $\beta_{\text{H}} \cdot \gamma_{\text{I}} \cdot \delta_{\text{C}}$, contra H_0 : indirecto = 0, usando bootstrap por clúster de país (no por observación, dada la dependencia serial dentro de cada país).

β_{H} (Homicidios → Instituciones)	-0.0836
γ_{I} (Instituciones → IED)	0.6111
δ_{C} (IED → Crecimiento)	0.2074
Efecto indirecto = $\beta_{\text{H}} \cdot \gamma_{\text{I}} \cdot \delta_{\text{C}}$	-0.010604
Media bootstrap (B réplicas por país)	-0.015213
IC 95% (percentil)	[-0.112620, 0.020498]
p-valor bootstrap (H_0 : indirecto = 0)	0.5480

■ Lectura honesta del mecanismo de transmisión

El intervalo de confianza al 95% del efecto indirecto INCLUYE CERO: la cadena completa Violencia → Instituciones → IED → Crecimiento NO es estadísticamente distinguible de ausencia de efecto con este tamaño muestral. Esto no refuta la teoría institucional, pero significa que el 'mecanismo de transmisión' debe reportarse como una hipótesis consistente con los signos observados, no como un resultado probado -- cada eslabón es individualmente débil y el producto de los tres no supera un test formal.

Figura 6b — Bootstrap del efecto indirecto

Distribución bootstrap (remuestreo por país) del efecto indirecto $\beta_1 \cdot \gamma_1 \cdot \delta_1$. La línea roja marca la estimación puntual; las líneas grises, el intervalo de confianza al 95%.

12. Pruebas de Robustez

Los análisis de robustez verifican la estabilidad de los coeficientes ante: (i) exclusión de cada país (LOCO — Leave-One-Country-Out), (ii) especificaciones alternativas del índice institucional, (iii) variaciones en las variables de control.

LOCO — EQ1 (Violencia → Instituciones)

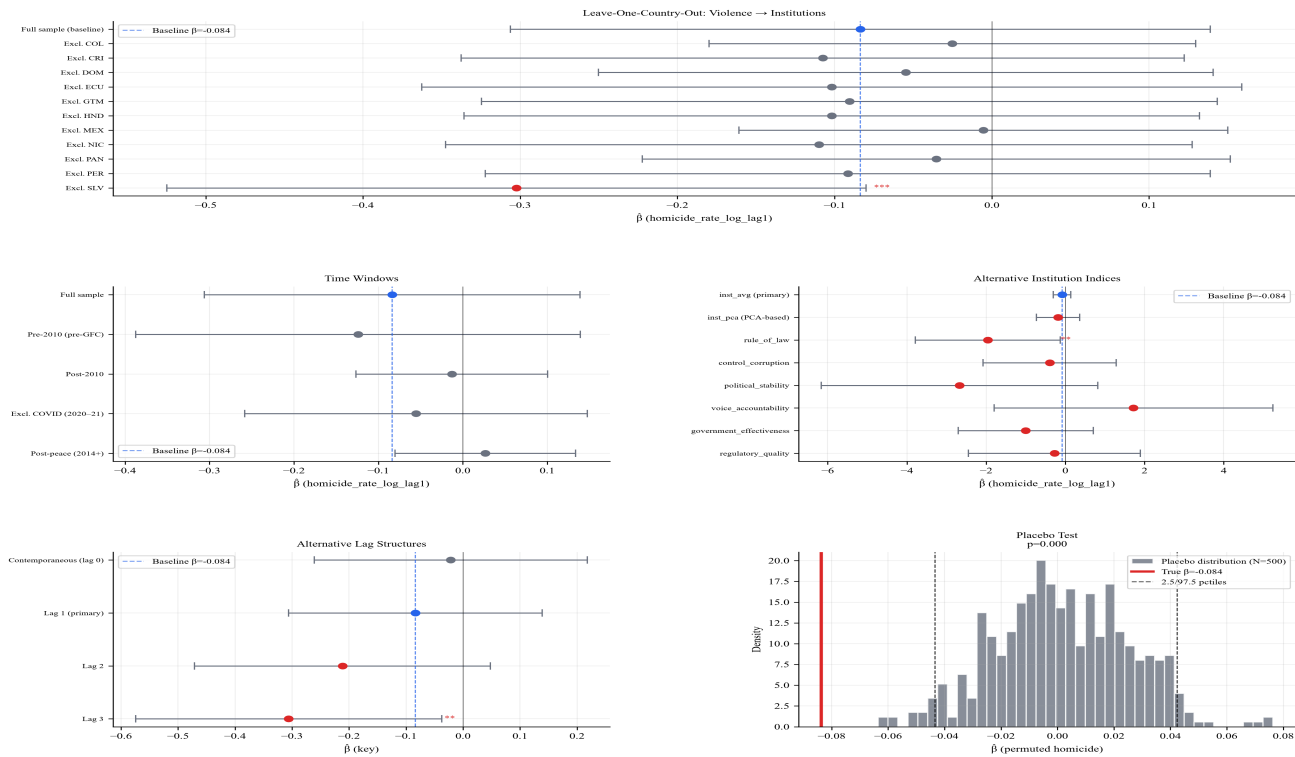
Full sample (baseline)	-0.0836	0.1136	0.4624	0.2031	231
Excl. COL	-0.0252	0.0790	0.7505	0.0240	208
Excl. CRI	-0.1076	0.1173	0.3603	0.1744	208
Excl. DOM	-0.0547	0.0998	0.5843	0.1285	209
Excl. ECU	-0.1018	0.1331	0.4451	0.2141	212
Excl. GTM	-0.0906	0.1194	0.4492	0.1924	210
Excl. HND	-0.1018	0.1194	0.3949	0.2030	208
Excl. MEX	-0.0054	0.0793	0.9454	-0.0961	208
Excl. NIC	-0.1100	0.1211	0.3649	0.0842	211
Excl. PAN	-0.0353	0.0954	0.7117	0.2867	208
Excl. PER	-0.0916	0.1176	0.4371	0.2073	220
Excl. SLV	-0.3024	0.1134	0.0084	-1.7972	208

■ LOCO — Estabilidad del coeficiente

Si el coeficiente permanece negativo y de magnitud similar al excluir cada país, se concluye que los resultados no están impulsados por ninguna observación influyente. Una variación sustancial al excluir un país señala dependencia de ese clúster.

Figura 7 — Robustez

Robustness Analysis — Coefficient Stability
Two-Way Fixed Effects | Dep. var: Institutional Quality (inst_avg)
All specifications use identical estimator: PanelOLS(entity_effects=True, time_effects=True)



Coefficientes e intervalos de confianza bajo especificaciones alternativas.

Test de Raíz Unitaria de Panel (Fisher-ADF, Maddala y Wu 1999)

Con T~25 años por país, este test tiene poco poder estadístico; no rechazar H0 es evidencia débil de raíz unitaria, no una prueba definitiva. Se usa aquí como base para la sección de cointegración que sigue.

Variable	Observations	Mean	Std. Dev.	Test Statistic	Decision
homicide_rate_log	11	20.01	22	0.5825	Sí (no rechaza)
inst_avg	11	26.24	22	0.2416	Sí (no rechaza)
fdi_percent_gdp	11	69.70	22	7.37e-07	No (rechaza)
gdp_growth	11	144.67	22	0.0000	No (rechaza)
gdp_per_capita_log	11	15.03	22	0.8607	Sí (no rechaza)

13. Cointegración de Panel y Modelo de Corrección de Errores (ECM)

El test de raíz unitaria de panel (Sección 12) encontró que homicide_rate_log, inst_avg y gdp_per_capita_log no rechazan raíz unitaria (son I(1)), mientras que fdi_percent_gdp y gdp_growth sí la rechazan (son I(0)). Dos series I(1) pueden, sin embargo, compartir una relación de equilibrio de largo plazo genuina (cointegración), en cuyo caso una regresión estática en niveles no sería necesariamente espuria. Este módulo prueba esto para los tres pares de variables I(1) del panel -- deliberadamente NO se prueba EQ2 ni EQ3, ya que emparejar una serie I(1) con una I(0) no es una pregunta de cointegración coherente.

Método: enfoque de dos pasos de Engle-Granger extendido a panel (Kao, 1999; McCoskey y Kao, 1998) -- (1) se estima la relación de largo plazo vía efectos fijos bidireccionales; (2) se testea la raíz unitaria de los residuos con el mismo test Fisher-ADF de la Sección 12. Residuos estacionarios = evidencia de cointegración; solo entonces se estima un ECM.

T1: Homicide <-> Institutions	-0.1330 (p=0.444)	0.6360	No
T2: Institutions <-> Income	0.7733 (p=0.027)	0.3929	No
T3: Homicide <-> Income	-0.0911 (p=0.345)	0.0110	Sí

Modelo de Corrección de Errores (pares cointegrados)

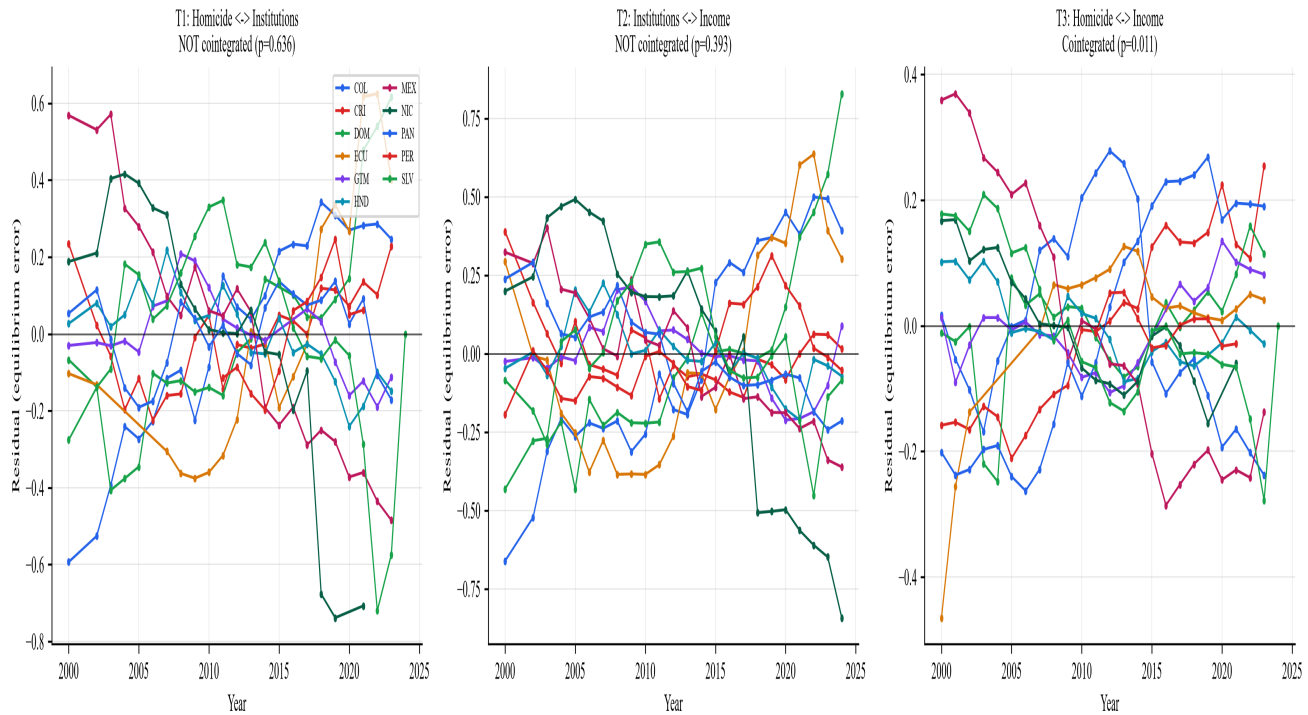
T3: Homicide <-> Income	-0.1092	0.0058	-0.0333	0.3440
-------------------------	---------	--------	---------	--------

■ Cointegración encontrada

Al menos un par de variables I(1) comparte una relación de equilibrio de largo plazo genuina. $\phi < 0$ y significativo indica que la variable dependiente corrige parte de cualquier desviación de esa relación cada año (velocidad de ajuste); γ captura el efecto de corto plazo, distinto del efecto de largo plazo (β) de la tabla anterior.

Figura 11 — Residuos de la regresión de largo plazo por país

Long-Run Regression Residuals by Country
(stationary residuals = evidence of cointegration; Fisher-ADF, Module 05/07)



Residuos de la relación de largo plazo para cada par I(1). Residuos que revierten a cero son evidencia de cointegración; residuos que se alejan persistentemente (como Colombia o Nicaragua en varios paneles) son evidencia de una relación espuria.

Advertencias: $G=8$, $T \sim 25$ es una muestra pequeña incluso para el test de Engle-Granger de series individuales; la extensión a panel no corrige un tamaño muestral fundamentalmente pequeño. Este es un estimador de dos pasos simplificado, no un sistema de cointegración/ECM totalmente eficiente (sin corrección por dependencia transversal en la regresión de largo plazo, sin estadísticos Pedroni/Westerlund, sin bootstrap de clúster silvestre en el ECM).

14. Growth-Ceiling-at-Risk (MCMC Bayesiano)

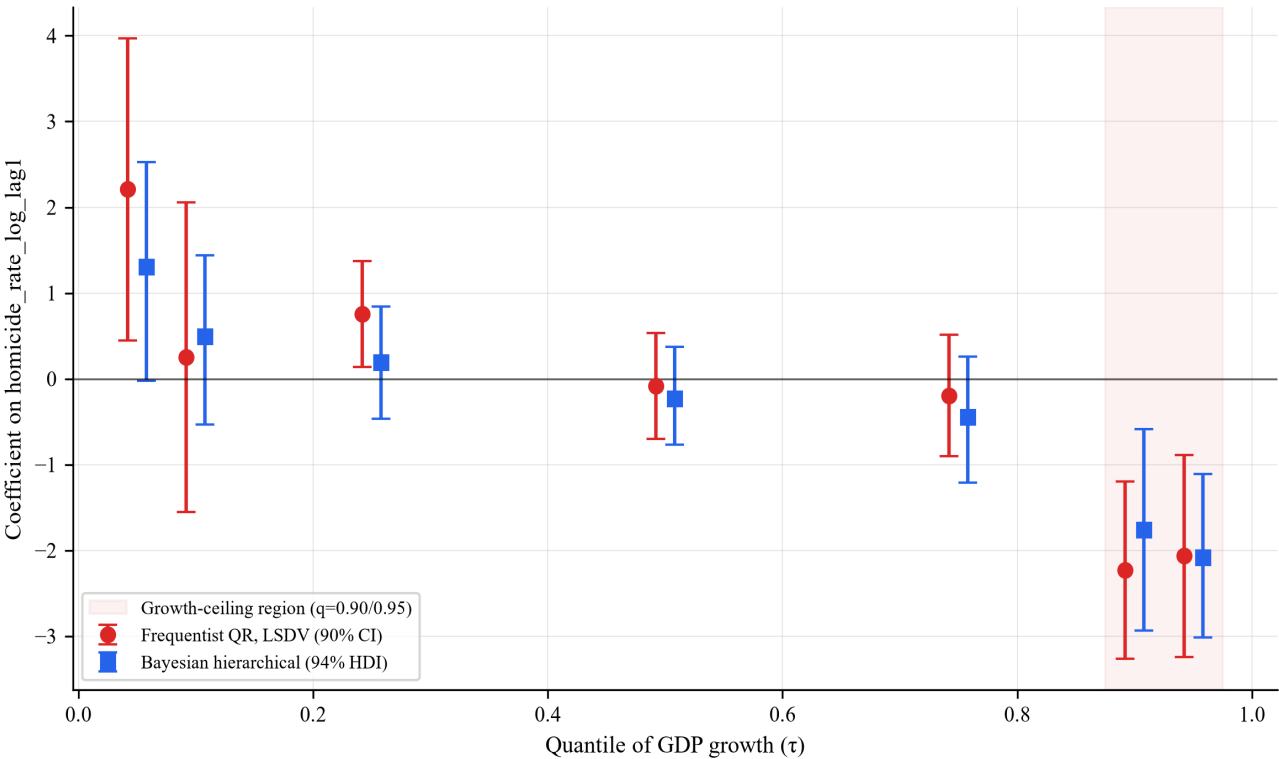
Un análisis exploratorio de regresión cuantílica frecuentista (ver README, sección 'Exploratory Finding') encontró que la violencia rezagada no tiene efecto cerca de la mediana del crecimiento del PIB, pero sí un efecto negativo grande y robusto a LOCO en la cola SUPERIOR ($q=0.90/0.95$) -- un patrón de 'techo de crecimiento', inverso al Growth-at-Risk clásico (Adrian, Boyarchenko y Giannone, 2019), que estudia la cola inferior. Este módulo re-estima ese patrón con un modelo bayesiano jerárquico en vez de variables dummy por país (LSDV): un prior de partial pooling regulariza los interceptos de cada país -algo que LSDV no puede hacer con $G=11$ - y la incertidumbre se reporta como una distribución posterior completa (intervalo de credibilidad, $P(\beta_i | \text{datos})$) en vez de un p-valor asintótico, ya señalado como poco confiable a este tamaño muestral en otras secciones de este pipeline.

Método: regresión cuantílica bayesiana vía la distribución Asimétrica de Laplace (ALD; Yu y Moyeed, 2001), implementada como un pm.Potential en PyMC. Los interceptos por país usan una parametrización no centrada ($\alpha_i = \alpha_{\mu} + \alpha_{\sigma} * z_i$) para evitar la patología de 'embudo' jerárquico bajo NUTS (Betancourt y Girolami, 2015).

5	2.2099	0.0401	1.3050	[-0.026, 2.528]	0.027
0	0.2522	0.8185	0.4923	[-0.529, 1.438]	0.171
5	0.7559	0.0448	0.1927	[-0.465, 0.843]	0.294
0	-0.0825	0.8265	-0.2312	[-0.769, 0.373]	0.790
5	-0.1950	0.6502	-0.4463	[-1.213, 0.262]	0.880
0	-2.2319	0.0005	-1.7621	[-2.934, -0.586]	0.997
5	-2.0645	0.0043	-2.0792	[-3.018, -1.107]	1.000

Figura 12 — Coeficiente por cuantil: frecuentista vs. bayesiano

Growth-Ceiling Effect of Violence: Frequentist vs. Bayesian Hierarchical
(11-country panel; partial pooling regularises country intercepts)



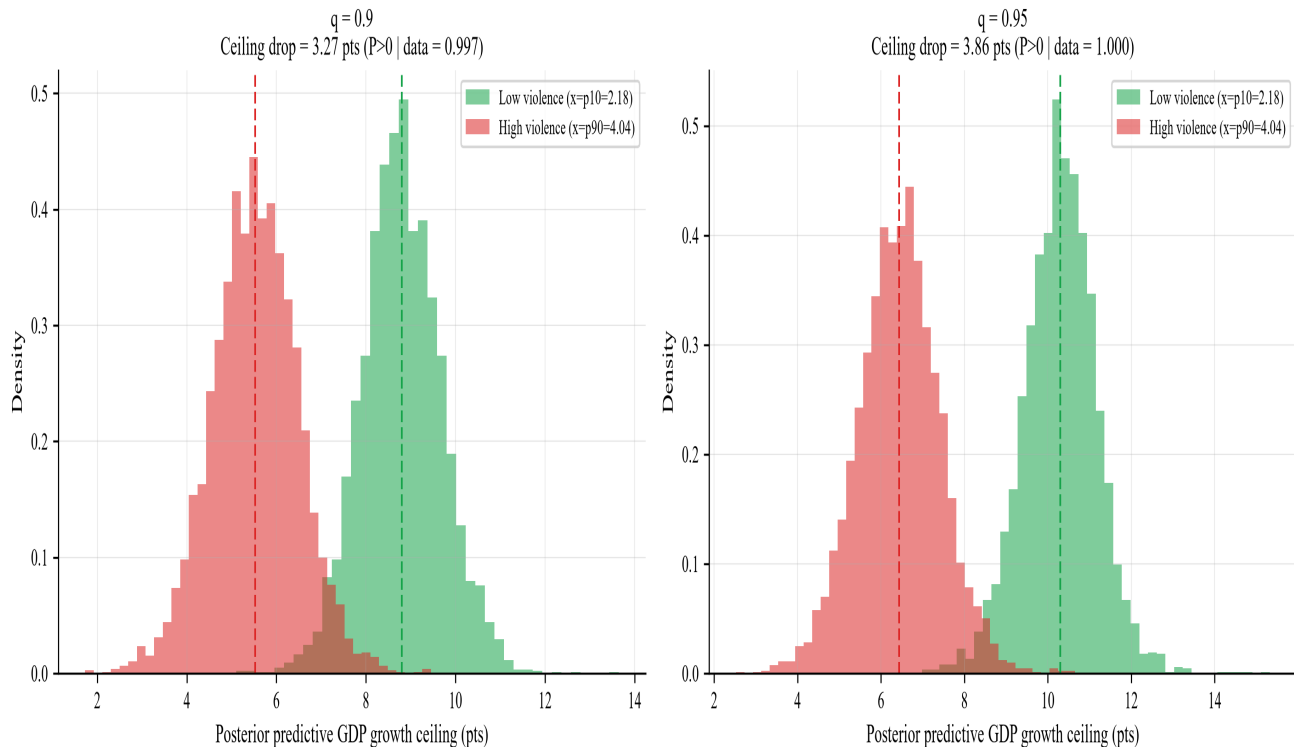
Efecto de la violencia rezagada sobre cada cuantil del crecimiento del PIB. El intervalo de credibilidad bayesiano (94% HDI) regulariza los interceptos por país vía partial pooling, en vez de absorberlos con variables dummy (LSDV).

■ Patrón de techo de crecimiento confirmado bayesianamente

En $q=0.90$ y $q=0.95$ el modelo jerárquico bayesiano confirma el hallazgo exploratorio: la probabilidad posterior de que el coeficiente sea negativo es cercana o igual a 1 (ver tabla), con diagnósticos de convergencia limpios (R-hat y ausencia de divergencias). El chequeo secundario con `inst_avg` como variable condicionante (mismos cuantiles, mismo modelo) no muestra el mismo patrón, apoyando que el efecto de techo es específico a la violencia y no una característica genérica de cualquier regresor en este panel.

Escenario Growth-Ceiling-at-Risk

00	8.79	5.52	3.27	[1.09, 5.45]	0.997
05	10.29	6.43	3.86	[2.06, 5.61]	1.000

Figura 13 — Distribución posterior del techo de crecimiento**Growth-Ceiling-at-Risk: Posterior Growth Ceiling, Low vs. High Violence Scenario**

Distribución posterior del q-ésimo percentil de crecimiento alcanzable bajo un escenario de violencia baja (percentil 10) vs. alta (percentil 90), manteniendo país y año en su nivel promedio.

Advertencias: este es un hallazgo exploratorio, no una hipótesis pre-registrada -- tratar todo resultado como sugestivo. G=11 sigue siendo pequeño incluso para un modelo jerárquico: el partial pooling regulariza pero no puede generar información que los datos no contienen. Los priors son débilmente informativos, no planos, y el modelo ALD estima cada cuantil por separado -- no garantiza cuantiles monótonos en tau (de hecho, el hallazgo central es precisamente que el efecto NO es monótono: nulo en la mediana, negativo solo en la cola superior).

15. Triangulación Machine Learning

El análisis de ML (Random Forest y Gradient Boosting con LOCO-CV) sirve como triangulación no paramétrica del ranking de importancia de variables. No reemplaza la inferencia causal del modelo econométrico: su objetivo es confirmar que las variables clave aparecen consistentemente como predictores relevantes.

Tarea ML — T1: Violence → Institutions

CV R ² (media)	-28.8166	-26.2960
CV R ² (desv. est.)	71.1718	51.1953
Wilcoxon RF vs GB (p)	0.7002	G=11 pairs → very low power. Treat p-value as descriptive only.

exports_percent_gdp	n/a	0.0835	0.0492	5.23
gdp_per_capita_log	0.2125	0.2355	0.2834	2.12
homicide_rate_log_lag1	0.0727	0.0911	n/a	4.89
homicide_rate_log_lag2	0.0882	0.1049	0.0638	3.44
population_log	0.2292	0.3373	0.3803	1.44
unemployment	0.0612	n/a	0.0545	5.08

■ Interpretación ML — EQ1

Variable clave rango SHAP: #4. Rango permutación: #6. Una posición consistente en el top-5 del ranking de importancia refuerza la relevancia de la variable, incluso en ausencia de significancia estadística formal bajo los estrictos criterios de inferencia panel.

Tarea ML — T2: Institutions → FDI

CV R ² (media)	-1.2696	-2.0484
CV R ² (desv. est.)	2.1221	3.1247
Wilcoxon RF vs GB (p)	0.0830	G=11 pairs → very low power. Treat p-value as descriptive only.

exports_percent_gdp	0.4764	0.2187	0.1519	2.42
inst_avg	0.3546	0.1634	0.1287	3.98
tourist_arrivals_log	0.1821	0.0654	0.0623	4.96
trade_percent_gdp	0.5930	0.2803	0.2526	1.21
unemployment	0.2588	0.1296	0.0957	4.47

■ Interpretación ML — EQ2

Variable clave rango SHAP: #3. Rango permutación: #3. Una posición consistente en el top-5 del ranking de importancia refuerza la relevancia de la variable, incluso en ausencia de significancia estadística formal bajo los estrictos criterios de inferencia panel.

Tarea ML — T3: FDI + Institutions → Growth

CV R ² (media)	0.0591	0.2266
CV R ² (desv. est.)	0.2290	0.4569
Wilcoxon RF vs GB (p)	0.1016	G=11 pairs → very low power. Treat p-value as descriptive only.

exports_percent_gdp	0.2002	0.1250	0.0544	3.06
fdi_percent_gdp	0.4500	0.1990	0.1274	1.77
homicide_rate_log_lag1	0.1172	n/a	n/a	6.37
inflation	0.1352	0.1374	0.0563	3.23
unemployment	n/a	0.1264	0.0446	6.00
year_c	0.2023	0.1829	0.1553	4.24

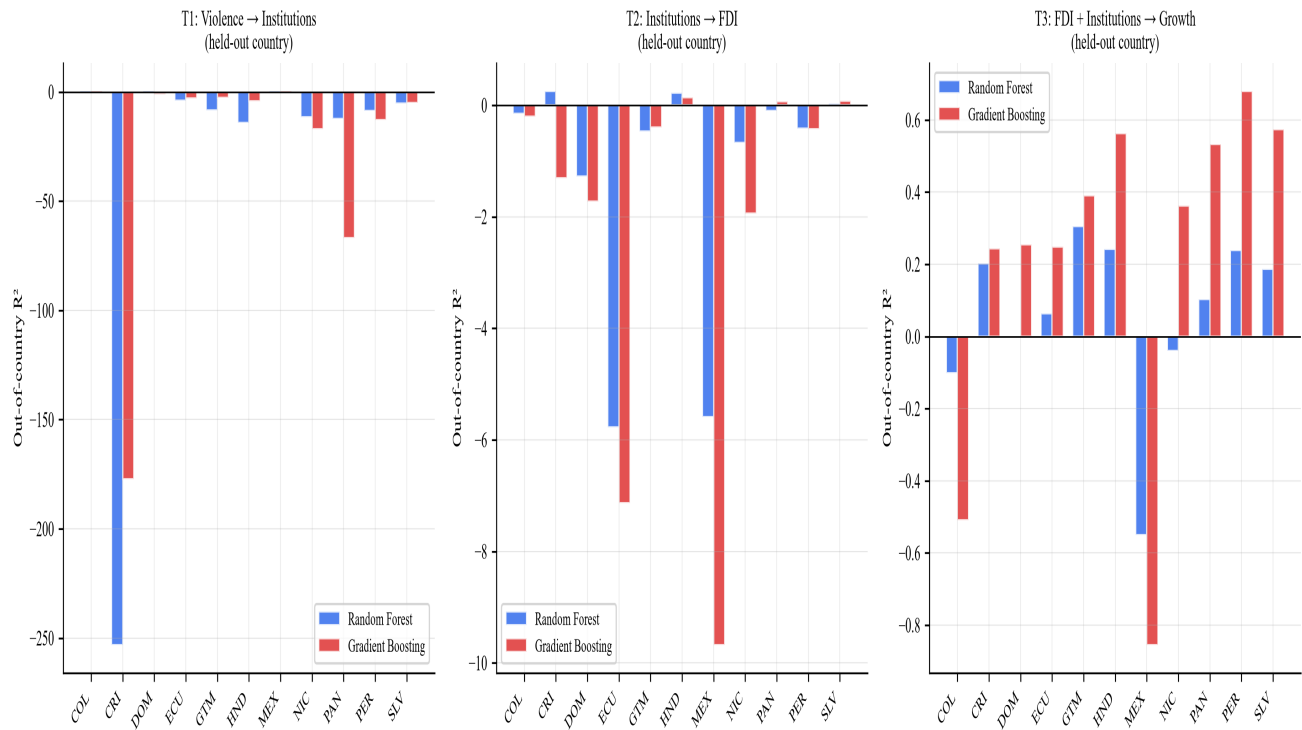
■ Interpretación ML — EQ3

Variable clave rango SHAP: #1. Rango permutación: #2. Una posición consistente en el top-5 del ranking de importancia refuerza la relevancia de la variable, incluso en ausencia de significancia estadística formal bajo los estrictos criterios de inferencia panel.

Figura 8 — LOCO-CV R^2 por país

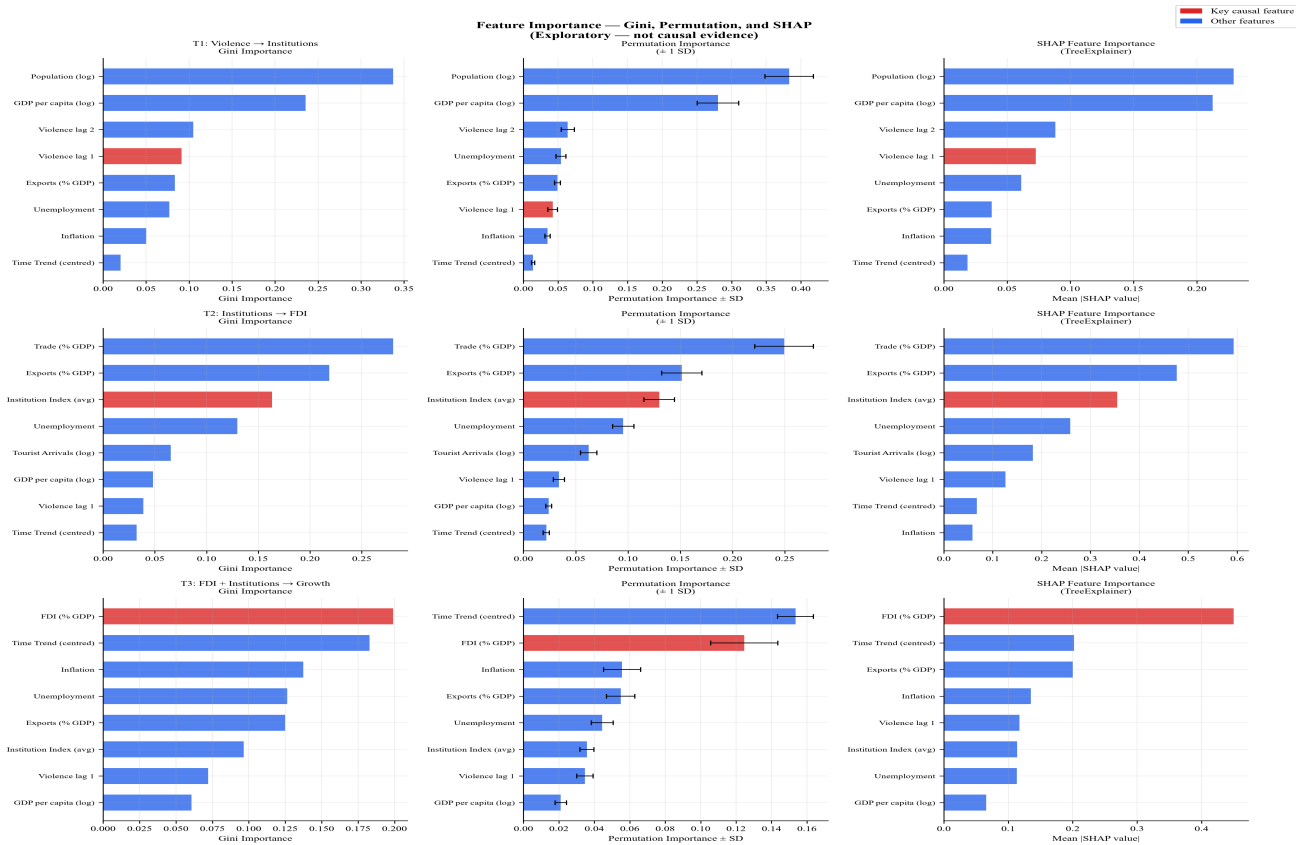
Leave-One-Country-Out Cross-Validation (LOCO-CV)

Random Forest & Gradient Boosting | Note: G=8 folds → wide uncertainty



R^2 de validación cruzada leave-one-country-out para RF y GB en las tres tareas.

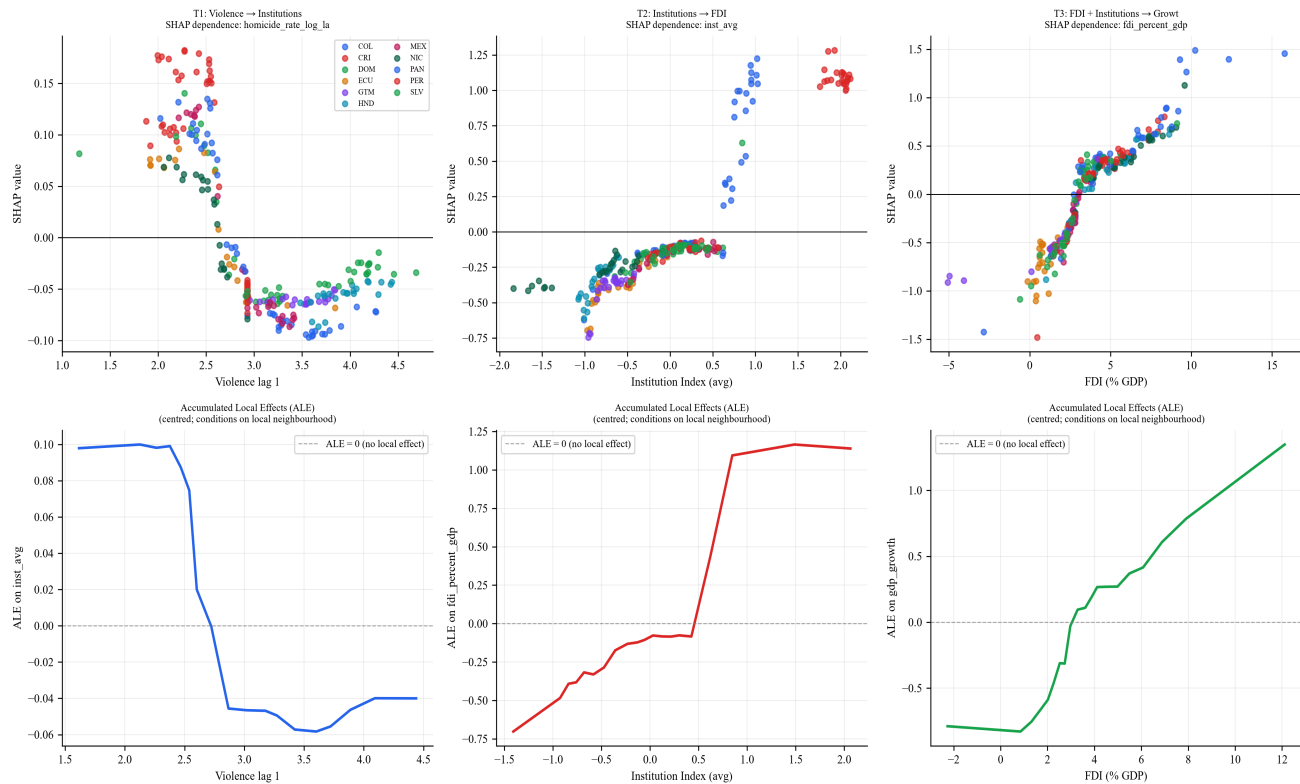
Figura 9 — Importancia de variables



Importancia Gini, permutación y SHAP de las variables en las tres tareas ML.

Figura 10 — SHAP / ALE plots

SHAP Dependence Plots & ALE Plots — Key Causal Variables
(Exploratory — shows non-linearity, not causal effect)
ALE used instead of PDP due to correlated predictors (Apley & Zhu 2020)



Efectos marginales acumulados (ALE) y valores SHAP para los predictores clave.

16. Conclusiones

A continuación se sintetizan las principales conclusiones derivadas estrictamente de los resultados calculados. No se realizan inferencias extrapoladas.

1	Violencia → Instituciones (EQ1): coeficiente negativo, no significativo	$\beta = -0.0836$, p-CL = 0.4624, p-boot = 0.5616
2	Instituciones → FDI (EQ2): coeficiente positivo, significativo (robusto a 3 de 4 SE)	$\beta = 0.6111$, p-CL = 0.0298, p-boot = 0.0380
3	FDI → Crecimiento (EQ3): coeficiente positivo, no significativo	$\beta = 0.2074$, p-CL = 0.0098, p-boot = 0.1081
4	Inflación → Crecimiento: el único efecto estadísticamente robusto en EQ3	$\beta = -0.0478$, p-CL = 0.1089
5	Bajo número de clústeres (G=8) limita el poder estadístico	Bootstrap confirma amplios intervalos de confianza para todos los coeficientes clave

■ Limitaciones metodológicas

Con G=8 clústeres, el poder de los tests convencionales es reducido. La no significancia estadística no implica ausencia de efecto económico — los intervalos de confianza bootstrap son amplios y compatibles tanto con efectos nulos como con efectos moderados. Se recomienda ampliar la muestra geográfica para obtener inferencia más precisa.

Apéndice — Log de Consola

El siguiente es el registro textual completo generado por el pipeline. Contiene todos los mensajes de consola, advertencias y métricas intermedias.

```
=====
RUNNING MODULE 01: 01_data_preparation.py
=====
```

```
01 — DATA LOAD & VALIDATION
```

```
01 — DATA LOAD & VALIDATION
```

```
✓ Panel loaded: 274 observations
```

```
✓ Panel loaded: 274 observations
```

```
✓ Countries (N=11): COL, CRI, DOM, ECU, GTM, HND, MEX, NIC, PAN, PER, SLV
```

```
✓ Countries (N=11): COL, CRI, DOM, ECU, GTM, HND, MEX, NIC, PAN, PER, SLV
```

```
✓ Years: 2000–2024 (25 distinct years)
```

```
✓ Years: 2000–2024 (25 distinct years)
```

```
■■ Balance check ■■
```

```
■■ Balance check ■■
```

```
T_obs year_min year_max
```

```
T_obs year_min year_max
```

```
country_code
```

```
country_code
```

```
COL 25 2000 2024
```

```
COL 25 2000 2024
```

```
CRI 25 2000 2024
```

```
CRI 25 2000 2024
```

```
DOM 25 2000 2024
```

```
DOM 25 2000 2024
```

```
ECU 25 2000 2024
```

```
ECU 25 2000 2024
```

```
GTM 25 2000 2024
```

```
GTM 25 2000 2024
```

```
HND 25 2000 2024
```

```
HND 25 2000 2024
```

```
MEX 25 2000 2024
```

```
MEX 25 2000 2024
```

```
NIC 25 2000 2024
```

```
NIC 25 2000 2024
```

```
PAN 25 2000 2024
```

```
PAN 25 2000 2024
```

```
PER 24 2000 2024
```

```
PER 24 2000 2024
```

```
SLV 25 2000 2024
```

```
SLV 25 2000 2024
```

```
■ Panel is UNBALANCED — accounting for this in all estimators.
```

```
■ Panel is UNBALANCED — accounting for this in all estimators.
```

```
Missing country-year cells:
```

```
Missing country-year cells:
```

```
PER: missing years [2001]
```

PER: missing years [2001]

■■ Column validation ■■

■■ Column validation ■■

✓ All 24 required columns present.

✓ All 24 required columns present.

■■ Missingness ■■

■■ Missingness ■■

■ Missing values:

■ Missing values:

homicide_rate_log_lag3 51 (18.6%)

homicide_rate_log_lag3 51 (18.6%)

homicide_rate_log_lag3 51 (18.6%)

homicide_rate_log_lag3 51 (18.6%)

tourist_arrivals_log 48 (17.5%)

tourist_arrivals_log 48 (17.5%)

homicide_rate_log_lag2 42 (15.3%)

homicide_rate_log_lag2 42 (15.3%)

homicide_rate_log_lag2 42 (15.3%)

homicide_rate_log_lag2 42 (15.3%)

homicide_rate_lag1 33 (12.0%)

homicide_rate_lag1 33 (12.0%)

homicide_rate_log_lag1 33 (12.0%)

homicide_rate_log_lag1 33 (12.0%)

homicide_rate 32 (11.7%)

homicide_rate 32 (11.7%)

homicide_rate_log 32 (11.7%)

homicide_rate_log 32 (11.7%)

regulatory_quality 10 (3.6%)

regulatory_quality 10 (3.6%)

government_effectiveness 10 (3.6%)

government_effectiveness 10 (3.6%)

rule_of_law 10 (3.6%)

rule_of_law 10 (3.6%)

control_corruption 10 (3.6%)

control_corruption 10 (3.6%)

voice_accountability 10 (3.6%)

voice_accountability 10 (3.6%)

political_stability 10 (3.6%)

political_stability 10 (3.6%)

02 — CENTRED TIME TREND

02 — CENTRED TIME TREND

✓ year_c = year – 2012.0 (range: -12 to 12)

✓ year_c = year – 2012.0 (range: -12 to 12)

Rationale: centring eliminates the large-value component of year that

Rationale: centring eliminates the large-value component of year that

inflates the condition number of the design matrix (collinearity).

inflates the condition number of the design matrix (collinearity).

03 — INSTITUTION INDEX: PRIMARY (STANDARDISED AVERAGE)

03 — INSTITUTION INDEX: PRIMARY (STANDARDISED AVERAGE)



Reference: Kaufmann et al. (2010) — WGI percentile ranks 0–100.

Reference: Kaufmann et al. (2010) — WGI percentile ranks 0–100.

All six dimensions are first z-scored (global mean=0, SD=1),

All six dimensions are first z-scored (global mean=0, SD=1),

then averaged. This ensures equal weighting regardless of raw scale.

then averaged. This ensures equal weighting regardless of raw scale.

✓ inst_avg: mean=-0.0000, SD=0.8500, range=[-1.839, 2.115]

✓ inst_avg: mean=-0.0000, SD=0.8500, range=[-1.839, 2.115]

Higher inst_avg → better governance (all six WGI dimensions).

Higher inst_avg → better governance (all six WGI dimensions).

Cronbach's α (internal consistency of index): ✓ $\alpha = 0.922$ (good internal consistency)

Cronbach's α (internal consistency of index): ✓ $\alpha = 0.922$ (good internal consistency)



04 — INSTITUTION INDEX: ROBUSTNESS (PCA)

04 — INSTITUTION INDEX: ROBUSTNESS (PCA)



PCA operates on the same six z-scored variables.

PCA operates on the same six z-scored variables.

PC1 is retained. Sign is aligned so higher = better governance.

PC1 is retained. Sign is aligned so higher = better governance.

✓ KMO = 0.863 $\geq$ 0.6 — adequate for PCA

✓ KMO = 0.863 $\geq$ 0.6 — adequate for PCA

rule_of_law KMO = 0.795

rule_of_law KMO = 0.795

control_corruption KMO = 0.886

control_corruption KMO = 0.886

political_stability KMO = 0.813

political_stability KMO = 0.813

voice_accountability KMO = 0.886

voice_accountability KMO = 0.886

government_effectiveness KMO = 0.924

government_effectiveness KMO = 0.924

regulatory_quality KMO = 0.891

regulatory_quality KMO = 0.891

Bartlett sphericity: ✓ $\chi^2(15)=1450.9$, $p=0.0000$ — reject H_0 , PCA appropriate

Bartlett sphericity: ✓ $\chi^2(15)=1450.9$, $p=0.0000$ — reject H_0 , PCA appropriate

✓ PC1 explains 72.9% of variance in the six WGI dimensions.

✓ PC1 explains 72.9% of variance in the six WGI dimensions.

PC2: 11.7%, PC3: 7.2%, PC4: 4.5%, PC5: 2.5%, PC6: 1.2%

PC2: 11.7%, PC3: 7.2%, PC4: 4.5%, PC5: 2.5%, PC6: 1.2%

PCA loadings (PC1):

PCA loadings (PC1):

rule_of_law 0.4575 ■■■■■■■■■■

rule_of_law 0.4575 ■■■■■■■■■■

control_corruption 0.4261 ■■■■■■■■■■

control_corruption 0.4261 ■■■■■■■■■■

political_stability 0.3128 ■■■■■■■■

political_stability 0.3128 ■■■■■■■■

voice_accountability 0.4499 ■■■■■■■■■■
 voice_accountability 0.4499 ■■■■■■■■■■
 government_effectiveness 0.3805 ■■■■■■■■■■
 government_effectiveness 0.3805 ■■■■■■■■■■
 regulatory_quality 0.4051 ■■■■■■■■■■
 regulatory_quality 0.4051 ■■■■■■■■■■

✓ Correlation(inst_avg, inst_pca) = 0.9992 — high convergence
 ✓ Correlation(inst_avg, inst_pca) = 0.9992 — high convergence

04b — LAGGED INSTITUTIONS / FDI
 04b — LAGGED INSTITUTIONS / FDI

The primary EQ2/EQ3 specifications use contemporaneous inst_avg and
 The primary EQ2/EQ3 specifications use contemporaneous inst_avg and
 fdi_percent_gdp. The theoretical model, however, argues institutions
 fdi_percent_gdp. The theoretical model, however, argues institutions
 move gradually and growth responds with a further lag. To test this
 move gradually and growth responds with a further lag. To test this
 explicitly (see Module 05, Robustness — Lag structure EQ2/EQ3),
 explicitly (see Module 05, Robustness — Lag structure EQ2/EQ3),
 one-year-lagged versions are generated here.
 one-year-lagged versions are generated here.

✓ inst_avg_lag1 created (253 non-null obs)
 ✓ inst_avg_lag1 created (253 non-null obs)
 ✓ inst_pca_lag1 created (253 non-null obs)
 ✓ inst_pca_lag1 created (253 non-null obs)
 ✓ fdi_percent_gdp_lag1 created (263 non-null obs)
 ✓ fdi_percent_gdp_lag1 created (263 non-null obs)

05 — WITHIN / BETWEEN VARIANCE DECOMPOSITION
 05 — WITHIN / BETWEEN VARIANCE DECOMPOSITION

Reference: Wooldridge (2010), *Econometric Analysis of Cross Section
 Reference: Wooldridge (2010), Econometric Analysis of Cross Section
 and Panel Data*, 2nd ed., ch. 10.
 and Panel Data, 2nd ed., ch. 10.

'Within' variance is the variation exploited by Fixed Effects.
 'Within' variance is the variation exploited by Fixed Effects.

Variable Total SD Between SD Within SD Within %
 Variable Total SD Between SD Within SD Within %

homicide_rate_log 0.7151 0.6224 0.4182 58.5% ✓
 homicide_rate_log 0.7151 0.6224 0.4182 58.5% ✓
 inst_avg 0.8500 0.8386 0.2840 33.4% ✓
 inst_avg 0.8500 0.8386 0.2840 33.4% ✓
 inst_pca 2.0954 2.0699 0.6936 33.1% ✓
 inst_pca 2.0954 2.0699 0.6936 33.1% ✓
 fdi_percent_gdp 2.6057 1.9293 1.8387 70.6% ✓

```

fdi_percent_gdp 2.6057 1.9293 1.8387 70.6% ✓
gdp_growth 3.8464 1.1356 3.6899 95.9% ✓
gdp_growth 3.8464 1.1356 3.6899 95.9% ✓
gdp_per_capita_log 0.7036 0.5986 0.4086 58.1% ✓
gdp_per_capita_log 0.7036 0.5986 0.4086 58.1% ✓

```

■ Variables with low within % are primarily identified from cross-section.

■ Variables with low within % are primarily identified from cross-section.

Fixed Effects may not efficiently estimate their coefficients.

Fixed Effects may not efficiently estimate their coefficients.

06 — SUMMARY STATISTICS

06 — SUMMARY STATISTICS

count mean std min 10% 50% 90% max

count mean std min 10% 50% 90% max

Variable

Variable

```

homicide_rate 242.0 25.344 19.735 1.900 7.864 17.475 55.925 107.638
homicide_rate 242.0 25.344 19.735 1.900 7.864 17.475 55.925 107.638
homicide_rate_log 242.0 3.015 0.715 1.065 2.182 2.916 4.042 4.688
homicide_rate_log 242.0 3.015 0.715 1.065 2.182 2.916 4.042 4.688
rule_of_law 264.0 49.225 7.914 31.509 41.001 48.572 58.502 69.627
rule_of_law 264.0 49.225 7.914 31.509 41.001 48.572 58.502 69.627
control_corruption 264.0 37.325 9.161 19.296 27.801 36.502 46.406 68.768
control_corruption 264.0 37.325 9.161 19.296 27.801 36.502 46.406 68.768
political_stability 264.0 59.419 10.652 24.200 49.018 59.054 74.027 87.744
political_stability 264.0 59.419 10.652 24.200 49.018 59.054 74.027 87.744
voice_accountability 264.0 58.668 9.161 25.889 48.513 58.261 69.233 79.459
voice_accountability 264.0 58.668 9.161 25.889 48.513 58.261 69.233 79.459
government_effectiveness 264.0 46.541 8.954 24.107 33.797 47.687 58.403 64.003
government_effectiveness 264.0 46.541 8.954 24.107 33.797 47.687 58.403 64.003
regulatory_quality 264.0 54.645 7.804 32.980 44.787 55.125 64.077 68.904
regulatory_quality 264.0 54.645 7.804 32.980 44.787 55.125 64.077 68.904
inst_avg 264.0 -0.000 0.850 -1.839 -0.889 -0.087 0.999 2.115
inst_avg 264.0 -0.000 0.850 -1.839 -0.889 -0.087 0.999 2.115
inst_pca 264.0 0.000 2.095 -4.790 -2.152 -0.222 2.402 5.206
inst_pca 264.0 0.000 2.095 -4.790 -2.152 -0.222 2.402 5.206
fdi_percent_gdp 274.0 3.785 2.606 -5.088 1.131 3.434 7.379 15.779
fdi_percent_gdp 274.0 3.785 2.606 -5.088 1.131 3.434 7.379 15.779
gdp_growth 274.0 3.652 3.846 -17.821 0.560 3.778 7.496 16.467
gdp_growth 274.0 3.652 3.846 -17.821 0.560 3.778 7.496 16.467
gdp_per_capita_log 274.0 8.409 0.704 6.922 7.488 8.444 9.307 9.861
gdp_per_capita_log 274.0 8.409 0.704 6.922 7.488 8.444 9.307 9.861
inflation 274.0 5.345 7.319 -1.550 0.915 4.179 9.131 96.096
inflation 274.0 5.345 7.319 -1.550 0.915 4.179 9.131 96.096
unemployment 274.0 5.770 2.855 2.143 3.053 5.128 9.326 20.520
unemployment 274.0 5.770 2.855 2.143 3.053 5.128 9.326 20.520
exports_percent_gdp 274.0 32.508 13.019 13.522 18.093 29.163 48.001 78.299
exports_percent_gdp 274.0 32.508 13.019 13.522 18.093 29.163 48.001 78.299
population_log 274.0 16.386 1.015 14.918 15.281 16.155 17.758 18.690
population_log 274.0 16.386 1.015 14.918 15.281 16.155 17.758 18.690
tourist_arrivals_log 226.0 14.735 1.288 13.069 13.584 14.460 15.820 18.476

```

```

tourist_arrivals_log 226.0 14.735 1.288 13.069 13.584 14.460 15.820 18.476
remittances_percent_gdp 274.0 7.526 7.226 0.137 0.990 4.136 19.095 27.001
remittances_percent_gdp 274.0 7.526 7.226 0.137 0.990 4.136 19.095 27.001
✓ LaTeX summary table saved → tables/summary_statistics.tex
✓ LaTeX summary table saved → tables/summary_statistics.tex

```

07 — FIGURES

07 — FIGURES

```

✓ Figure saved → figures/01_panel_overview.png
✓ Figure saved → figures/01_panel_overview.png
✓ Figure saved → figures/02_institution_indices.png
✓ Figure saved → figures/02_institution_indices.png
✓ Figure saved → figures/03_correlation_heatmap.png
✓ Figure saved → figures/03_correlation_heatmap.png

```

08 — EXPORT

08 — EXPORT

```

✓ Enriched panel saved → panel_enriched.csv (274 rows)
✓ Enriched panel saved → panel_enriched.csv (274 rows)
✓ Saved → C:\data_analyzer\econometric_pipeline\pipeline\json\01_metadata.json
✓ Saved → C:\data_analyzer\econometric_pipeline\pipeline\pipeline\json\01_metadata.json

```

Module 01 complete. Enriched dataset and metadata exported.

Module 01 complete. Enriched dataset and metadata exported.

✓ Module 01 completed in 15.8s

=====

RUNNING MODULE 02: 02_panel_estimation.py

=====

02-A — LOAD DATA

02-A — LOAD DATA

```

✓ Data loaded: 274 obs, 11 countries
✓ Data loaded: 274 obs, 11 countries

```

02-B — BENCHMARK: POOLED OLS

02-B — BENCHMARK: POOLED OLS

■ Pooled OLS ignores country fixed effects. Reported for comparison only.

■ Pooled OLS ignores country fixed effects. Reported for comparison only.

Reference: Angrist & Pischke (2009) — benchmark before FE.

Reference: Angrist & Pischke (2009) — benchmark before FE.

EQ1: Violence→Institutions (N=231, R²=0.7582)

EQ1: Violence→Institutions (N=231, $R^2=0.7582$)

homicide_rate_log_lag1 -0.1067 $p=0.337$

homicide_rate_log_lag1 -0.1067 $p=0.337$

gdp_per_capita_log 1.0864 $p=0.000^{***}$

gdp_per_capita_log 1.0864 $p=0.000^{***}$

inflation 0.0053 $p=0.714$

inflation 0.0053 $p=0.714$

exports_percent_gdp -0.0054 $p=0.341$

exports_percent_gdp -0.0054 $p=0.341$

population_log -0.3640 $p=0.003^{***}$

population_log -0.3640 $p=0.003^{***}$

unemployment 0.0564 $p=0.004^{***}$

unemployment 0.0564 $p=0.004^{***}$

year_c -0.0472 $p=0.000^{***}$

year_c -0.0472 $p=0.000^{***}$

EQ2: Institutions→FDI (N=264, $R^2=0.4154$)

EQ2: Institutions→FDI (N=264, $R^2=0.4154$)

inst_avg 1.2247 $p=0.004^{***}$

inst_avg 1.2247 $p=0.004^{***}$

gdp_per_capita_log -1.2594 $p=0.096^*$

gdp_per_capita_log -1.2594 $p=0.096^*$

inflation -0.0133 $p=0.502$

inflation -0.0133 $p=0.502$

exports_percent_gdp 0.1153 $p=0.000^{***}$

exports_percent_gdp 0.1153 $p=0.000^{***}$

population_log 0.1727 $p=0.556$

population_log 0.1727 $p=0.556$

unemployment 0.1259 $p=0.217$

unemployment 0.1259 $p=0.217$

year_c 0.0426 $p=0.357$

year_c 0.0426 $p=0.357$

EQ3: FDI+Inst→GDP Growth (N=264, $R^2=0.1833$)

EQ3: FDI+Inst→GDP Growth (N=264, $R^2=0.1833$)

fdi_percent_gdp 0.5869 $p=0.000^{***}$

fdi_percent_gdp 0.5869 $p=0.000^{***}$

inst_avg -1.4073 $p=0.000^{***}$

inst_avg -1.4073 $p=0.000^{***}$

gdp_per_capita_log 2.3918 $p=0.000^{***}$

gdp_per_capita_log 2.3918 $p=0.000^{***}$

inflation 0.0074 $p=0.743$

inflation 0.0074 $p=0.743$

exports_percent_gdp -0.0201 $p=0.354$

exports_percent_gdp -0.0201 $p=0.354$

population_log -0.7208 $p=0.159$

population_log -0.7208 $p=0.159$

unemployment -0.1700 $p=0.099^*$

unemployment -0.1700 $p=0.099^*$

year_c -0.1569 $p=0.000^{***}$

year_c -0.1569 $p=0.000^{***}$

02-C — PRIMARY: TWO-WAY FIXED EFFECTS

02-C — PRIMARY: TWO-WAY FIXED EFFECTS

```

Estimator: PanelOLS(entity_effects=True, time_effects=True)
Estimator: PanelOLS(entity_effects=True, time_effects=True)
SE reported: (1) Clustered by entity [conventional, G=8 caution]
SE reported: (1) Clustered by entity [conventional, G=8 caution]
(2) Driscoll-Kraay HAC [robust to cross-sect. dep.]
(2) Driscoll-Kraay HAC [robust to cross-sect. dep.]

```

Preferred inference: Wild Cluster Bootstrap — see Module 04.
Preferred inference: Wild Cluster Bootstrap — see Module 04.

■■ EQ1: Violence → Institutional Quality ■■
■■ EQ1: Violence → Institutional Quality ■■

EQ1: homicide_log_lag1 → inst_avg
EQ1: homicide_log_lag1 → inst_avg
N=231 | Within R²=0.2031 | DK bandwidth=2
N=231 | Within R²=0.2031 | DK bandwidth=2

Variable Coef SE (Cl.) p(Cl.) SE (DK) p(DK) Stars
Variable Coef SE (Cl.) p(Cl.) SE (DK) p(DK) Stars

```

homicide_rate_log_lag1 -0.0836 0.1136 0.462 0.0822 0.310
homicide_rate_log_lag1 -0.0836 0.1136 0.462 0.0822 0.310
gdp_per_capita_log 0.3086 0.3176 0.332 0.2038 0.132
gdp_per_capita_log 0.3086 0.3176 0.332 0.2038 0.132
inflation -0.0058 0.0071 0.413 0.0049 0.237
inflation -0.0058 0.0071 0.413 0.0049 0.237
exports_percent_gdp -0.0128 0.0084 0.130 0.0025 0.000 ***
exports_percent_gdp -0.0128 0.0084 0.130 0.0025 0.000 ***
population_log -1.0100 1.2847 0.433 0.4042 0.013 **
population_log -1.0100 1.2847 0.433 0.4042 0.013 **
unemployment -0.0116 0.0167 0.487 0.0169 0.492
unemployment -0.0116 0.0167 0.487 0.0169 0.492
Note: Wild cluster bootstrap p-values in Module 04 (preferred for G=8).
Note: Wild cluster bootstrap p-values in Module 04 (preferred for G=8).

```

■■ EQ2: Institutional Quality → FDI ■■
■■ EQ2: Institutional Quality → FDI ■■

EQ2: inst_avg → fdi_percent_gdp
EQ2: inst_avg → fdi_percent_gdp
N=264 | Within R²=0.0192 | DK bandwidth=2
N=264 | Within R²=0.0192 | DK bandwidth=2

Variable Coef SE (Cl.) p(Cl.) SE (DK) p(DK) Stars
Variable Coef SE (Cl.) p(Cl.) SE (DK) p(DK) Stars

```

inst_avg 0.6111 0.2795 0.030 0.2192 0.006 ***
inst_avg 0.6111 0.2795 0.030 0.2192 0.006 ***
gdp_per_capita_log 0.2138 1.0342 0.836 0.6046 0.724
gdp_per_capita_log 0.2138 1.0342 0.836 0.6046 0.724
inflation -0.0085 0.0113 0.454 0.0110 0.444
inflation -0.0085 0.0113 0.454 0.0110 0.444

```

```

exports_percent_gdp 0.1070 0.0228 0.000 0.0282 0.000 ***
exports_percent_gdp 0.1070 0.0228 0.000 0.0282 0.000 ***
population_log 8.2444 5.2378 0.117 2.9199 0.005 ***
population_log 8.2444 5.2378 0.117 2.9199 0.005 ***

```

unemployment -0.2094 0.1413 0.140 0.1076 0.053 *

unemployment -0.2094 0.1413 0.140 0.1076 0.053 *

Note: Wild cluster bootstrap p-values in Module 04 (preferred for G=8).

Note: Wild cluster bootstrap p-values in Module 04 (preferred for G=8).

■ ■ EQ3: FDI + Institutions → GDP Growth ■ ■

EQ3: $fdi + inst_avg \rightarrow gdp_growth$
EQ3: $fdi + inst_avg \rightarrow gdp_growth$
N=264 | Within $R^2=0.0772$ | DK bandwidth=2
N=264 | Within $R^2=0.0772$ | DK bandwidth=2

Variable Coef SE (Cl.) p(Cl.) SE (DK) p(DK) Stars
Variable Coef SE (Cl.) p(Cl.) SE (DK) p(DK) Stars

[illegible]

```
fdi_percent_gdp 0.2074 0.0796 0.010 0.1443 0.152
fdi_percent_gdp 0.2074 0.0796 0.010 0.1443 0.152
inst_avg -0.5442 0.5652 0.337 0.5012 0.279
inst_avg -0.5442 0.5652 0.337 0.5012 0.279
gdp_per_capita_log 2.4482 1.0844 0.025 1.2929 0.060 *
gdp_per_capita_log 2.4482 1.0844 0.025 1.2929 0.060 *
inflation -0.0478 0.0297 0.109 0.0329 0.148
inflation -0.0478 0.0297 0.109 0.0329 0.148
exports_percent_gdp 0.0512 0.0250 0.042 0.0572 0.371
exports_percent_gdp 0.0512 0.0250 0.042 0.0572 0.371
population_log -3.2666 5.6121 0.561 4.9767 0.512
population_log -3.2666 5.6121 0.561 4.9767 0.512
unemployment -0.1688 0.0809 0.038 0.0769 0.029 **
unemployment -0.1688 0.0809 0.038 0.0769 0.029 **
```

Note: Wild cluster bootstrap p-values in Module 04 (preferred for G=8).

02-D — METHODOLOGICAL NOTE: GENERATED REGRESSORS

- Sequential OLS and the generated regressors problem
- Sequential OLS and the generated regressors problem

The sequential estimation above uses observed values of `inst_avg` in EQ2 and observed `fdi` in EQ3. This is consistent (each equation is estimated independently) but does NOT recover the causal chain from a single identified system.

An alternative using fitted values (*inst_avg_hat* in EQ2, *fdi_hat* in EQ3)

An alternative using fitted values (*inst_avg_hat* in EQ2, *fdi_hat* in EQ3)

is a 'chain of OLS' approach, which suffers from the 'generated regressors' problem (Pagan, 1984, Review of Economic Studies 51(2)):

problem (Pagan, 1984, Review of Economic Studies 51(2)):

— Standard errors in downstream equations are understated because they

— Standard errors in downstream equations are understated because they

ignore sampling uncertainty in the upstream prediction.

ignore sampling uncertainty in the upstream prediction.

Correct approaches:

Correct approaches:

(a) 3SLS: estimate all three equations simultaneously.

(a) 3SLS: estimate all three equations simultaneously.

(b) Causal mediation analysis (Imai et al. 2010) for formal decomposition.

(b) Causal mediation analysis (Imai et al. 2010) for formal decomposition.

(c) IV/2SLS for each upstream equation with valid external instruments.

(c) IV/2SLS for each upstream equation with valid external instruments.

In this pipeline:

In this pipeline:

— Module 02 reports observed-value FE (each equation self-contained).

— Module 02 reports observed-value FE (each equation self-contained).

— Module 04 bootstrap re-estimates the full system jointly, providing

— Module 04 bootstrap re-estimates the full system jointly, providing

honest inference that integrates uncertainty across the chain.

honest inference that integrates uncertainty across the chain.

— IV/2SLS is architecturally prepared but requires external instruments

— IV/2SLS is architecturally prepared but requires external instruments

(cocaine price index, rainfall anomaly) not present in current dataset.

(cocaine price index, rainfall anomaly) not present in current dataset.

02-E — HAUSMAN-TYPE FE vs BETWEEN COMPARISON

02-E — HAUSMAN-TYPE FE vs BETWEEN COMPARISON

Motivation: if FE and Between (cross-sectional average) estimates

Motivation: if FE and Between (cross-sectional average) estimates

diverge, this suggests omitted time-invariant heterogeneity —

diverge, this suggests omitted time-invariant heterogeneity —

supporting the use of Fixed Effects.

supporting the use of Fixed Effects.

Reference: Hausman (1978), *Econometrica* 46(6), 1251–1271.

Reference: Hausman (1978), *Econometrica* 46(6), 1251–1271.

EQ1: *inst_avg*

EQ1: *inst_avg*

Variable FE coef BE coef Δ

Variable FE coef BE coef Δ

homicide_rate_log_lag1 -0.0836 -0.2253 0.1416 ■ ← divergence

homicide_rate_log_lag1 -0.0836 -0.2253 0.1416 ■ ← divergence

gdp_per_capita_log 0.3086 0.9312 -0.6225 ■ ← divergence

gdp_per_capita_log 0.3086 0.9312 -0.6225 ■ ← divergence

inflation -0.0058 -0.0306 0.0248 ■ ← divergence
inflation -0.0058 -0.0306 0.0248 ■ ← divergence
exports_percent_gdp -0.0128 -0.0129 0.0001
exports_percent_gdp -0.0128 -0.0129 0.0001
population_log -1.0100 -0.4349 -0.5751 ■ ← divergence
population_log -1.0100 -0.4349 -0.5751 ■ ← divergence
unemployment -0.0116 0.0889 -0.1006 ■ ← divergence
unemployment -0.0116 0.0889 -0.1006 ■ ← divergence

EQ2: fdi_percent_gdp

EQ2: fdi_percent_gdp

Variable FE coef BE coef Δ

Variable FE coef BE coef Δ

inst_avg 0.6111 0.8854 -0.2743

inst_avg 0.6111 0.8854 -0.2743

gdp_per_capita_log 0.2138 -1.1980 1.4118 ■ ← divergence

gdp_per_capita_log 0.2138 -1.1980 1.4118 ■ ← divergence

inflation -0.0085 0.0604 -0.0688 ■ ← divergence

inflation -0.0085 0.0604 -0.0688 ■ ← divergence

exports_percent_gdp 0.1070 0.1374 -0.0304

exports_percent_gdp 0.1070 0.1374 -0.0304

population_log 8.2444 0.4207 7.8237 ■ ← divergence

population_log 8.2444 0.4207 7.8237 ■ ← divergence

unemployment -0.2094 0.3912 -0.6006 ■ ← divergence

unemployment -0.2094 0.3912 -0.6006 ■ ← divergence

EQ3: gdp_growth

EQ3: gdp_growth

Variable FE coef BE coef Δ

Variable FE coef BE coef Δ

fdi_percent_gdp 0.2074 0.4824 -0.2750 ■ ← divergence

fdi_percent_gdp 0.2074 0.4824 -0.2750 ■ ← divergence

inst_avg -0.5442 -1.3748 0.8306 ■ ← divergence

inst_avg -0.5442 -1.3748 0.8306 ■ ← divergence

gdp_per_capita_log 2.4482 2.2189 0.2293

gdp_per_capita_log 2.4482 2.2189 0.2293

inflation -0.0478 -0.0123 -0.0355 ■ ← divergence

inflation -0.0478 -0.0123 -0.0355 ■ ← divergence

exports_percent_gdp 0.0512 -0.0445 0.0957 ■ ← divergence

exports_percent_gdp 0.0512 -0.0445 0.0957 ■ ← divergence

population_log -3.2666 -0.9260 -2.3406 ■ ← divergence

population_log -3.2666 -0.9260 -2.3406 ■ ← divergence

unemployment -0.1688 -0.0274 -0.1413 ■ ← divergence

unemployment -0.1688 -0.0274 -0.1413 ■ ← divergence

02-F — RESIDUAL FIGURES

02-F — RESIDUAL FIGURES

✓ Figure saved → figures/04_fe_residuals.png

✓ Figure saved → figures/04_fe_residuals.png

02-G — EXPORT

02-G — EXPORT

✓ Dataset with predictions saved → panel_with_predictions.csv

✓ Dataset with predictions saved → panel_with_predictions.csv

✓ Saved → C:\data_analyzer\econometric_pipeline\pipeline\json\02_fe_results.json

✓ Saved → C:\data_analyzer\econometric_pipeline\pipeline\json\02_fe_results.json

✓ FE specifications saved → fe_specs.pkl (for bootstrap module)

✓ FE specifications saved → fe_specs.pkl (for bootstrap module)

Module 02 complete.

Module 02 complete.

✓ Module 02 completed in 11.0s

RUNNING MODULE 03: 03_diagnostics.py

03-A — LOAD

03-A — LOAD

✓ Data loaded: 274 obs, 11 countries

✓ Data loaded: 274 obs, 11 countries

TEST 1 — Wooldridge (2002) Serial Correlation in Panel Residuals

TEST 1 — Wooldridge (2002) Serial Correlation in Panel Residuals

Reference: Wooldridge (2002), *Econometric Analysis of Cross Section*

Reference: Wooldridge (2002), *Econometric Analysis of Cross Section*

and Panel Data, MIT Press, p.282-283.

and Panel Data, MIT Press, p.282-283.

Procedure:

Procedure:

(a) Estimate the model in FIRST DIFFERENCES: $\Delta y_{it} = \Delta x_{it} \cdot \beta + \Delta \varepsilon_{it}$

(a) Estimate the model in FIRST DIFFERENCES: $\Delta y_{it} = \Delta x_{it} \cdot \beta + \Delta \varepsilon_{it}$

(b) Regress $\Delta \varepsilon_{it}$ on $\Delta \varepsilon_{it-1}$ (OLS, pooled, no intercept)

(b) Regress $\Delta \varepsilon_{it}$ on $\Delta \varepsilon_{it-1}$ (OLS, pooled, no intercept)

(c) Test $H_0: \rho = -0.5$

(c) Test $H_0: \rho = -0.5$

Under AR(1) errors in levels, the FD residuals have

Under AR(1) errors in levels, the FD residuals have

autocorrelation exactly -0.5.

autocorrelation exactly -0.5.

A t-test against -0.5 is the Wooldridge test.

A t-test against -0.5 is the Wooldridge test.

LIMITATION: Stata's `xtserial` implements the exact Wooldridge F-test

LIMITATION: Stata's `xtserial` implements the exact Wooldridge F -test using robust standard errors and a slightly different form. The Python using robust standard errors and a slightly different form. The Python implementation below follows the same theoretical derivation but may differ numerically from Stata output.

```
eq1: inst_avg
eq1: inst_avg
ρ = 0.0390 (H0: ρ = -0.5 under AR(1) errors in levels)
ρ = 0.0390 (H0: ρ = -0.5 under AR(1) errors in levels)
t = 7.764, p = 0.0000 ***
t = 7.764, p = 0.0000 ***
N_pairs = 205
N_pairs = 205
→ ■ SERIAL CORRELATION detected (reject H0: ρ = -0.5) → use HAC SE
→ ■ SERIAL CORRELATION detected (reject H0: ρ = -0.5) → use HAC SE
LIMITATION: Differs from Stata `xtserial` F-statistic by construction.
LIMITATION: Differs from Stata `xtserial` F-statistic by construction.
```

```
eq2: fdi_percent_gdp
eq2: fdi_percent_gdp
ρ = -0.3082 (H0: ρ = -0.5 under AR(1) errors in levels)
ρ = -0.3082 (H0: ρ = -0.5 under AR(1) errors in levels)
t = 3.279, p = 0.0012 ***
t = 3.279, p = 0.0012 ***
N_pairs = 232
N_pairs = 232
→ ■ SERIAL CORRELATION detected (reject H0: ρ = -0.5) → use HAC SE
→ ■ SERIAL CORRELATION detected (reject H0: ρ = -0.5) → use HAC SE
LIMITATION: Differs from Stata `xtserial' F-statistic by construction.
LIMITATION: Differs from Stata `xtserial' F-statistic by construction.
```

```
eq3: gdp_growth
eq3: gdp_growth
ρ = -0.3383 (H0: ρ = -0.5 under AR(1) errors in levels)
ρ = -0.3383 (H0: ρ = -0.5 under AR(1) errors in levels)
t = 2.623, p = 0.0093 ***
t = 2.623, p = 0.0093 ***
N_pairs = 232
N_pairs = 232
→ ■ SERIAL CORRELATION detected (reject H0: ρ = -0.5) → use HAC SE
→ ■ SERIAL CORRELATION detected (reject H0: ρ = -0.5) → use HAC SE
```

LIMITATION: Differs from Stata `xtserial' F-statistic by construction.
LIMITATION: Differs from Stata `xtserial' F-statistic by construction.

TEST 2 — Pesaran (2004) Cross-Sectional Dependence (CD) Test

TEST 2 — Pesaran (2004) Cross-Sectional Dependence (CD) Test

Reference: Pesaran (2004), General Diagnostic Tests for Cross Section
Reference: Pesaran (2004), General Diagnostic Tests for Cross Section
Dependence in Panels, Cambridge Working Papers in Economics 0435.
Dependence in Panels, Cambridge Working Papers in Economics 0435.

$$CD = \sqrt{2T \sum_{i < j} (N-1)} \cdot \sum_{i < j} T_{ij} \cdot \rho_{ij}$$

$$CD = \sqrt{2T \sum_{i < j} (N-1)} \cdot \sum_{i < j} T_{ij} \cdot \rho_{ij}$$

where ρ_{ij} is the sample correlation of residuals between units i and j ,

where ρ_{ij} is the sample correlation of residuals between units i and j ,

T_{ij} is the number of common time periods.

T_{ij} is the number of common time periods.

Under H_0 (no CD): $CD \rightarrow N(0,1)$ as $N, T \rightarrow \infty$.

Under H_0 (no CD): $CD \rightarrow N(0,1)$ as $N, T \rightarrow \infty$.

Note: with $N=8$, the asymptotic distribution is approximate.

Note: with $N=8$, the asymptotic distribution is approximate.

eq1: inst_avg

eq1: inst_avg

CD = -59.6764, $p = 0.0000$ ***

CD = -59.6764, $p = 0.0000$ ***

N pairs = 55, $T = 23$

N pairs = 55, $T = 23$

→ Cross-sectional dependence detected → Driscoll-Kraay SE appropriate

→ Cross-sectional dependence detected → Driscoll-Kraay SE appropriate

eq2: fdi_percent_gdp

eq2: fdi_percent_gdp

CD = -69.1112, $p = 0.0000$ ***

CD = -69.1112, $p = 0.0000$ ***

N pairs = 55, $T = 24$

N pairs = 55, $T = 24$

→ Cross-sectional dependence detected → Driscoll-Kraay SE appropriate

→ Cross-sectional dependence detected → Driscoll-Kraay SE appropriate

eq3: gdp_growth

eq3: gdp_growth

CD = -76.5595, $p = 0.0000$ ***

CD = -76.5595, $p = 0.0000$ ***

N pairs = 55, $T = 24$

N pairs = 55, $T = 24$

→ Cross-sectional dependence detected → Driscoll-Kraay SE appropriate

→ Cross-sectional dependence detected → Driscoll-Kraay SE appropriate

TEST 3 — Modified Wald Test for Group-Wise Heteroscedasticity

TEST 3 — Modified Wald Test for Group-Wise Heteroscedasticity

Reference: Greene (2000), *Econometric Analysis*, 4th ed., ch. 11.

Reference: Greene (2000), *Econometric Analysis*, 4th ed., ch. 11.

Tests $H_0: \sigma^2_i = \sigma^2$ for all i (homoscedasticity across groups).

Tests $H_0: \sigma^2_i = \sigma^2$ for all i (homoscedasticity across groups).

Statistic: $W = \sum_i [(\hat{\epsilon}_i' \hat{\epsilon}_i / T_i - \sigma^2)^2 / (2\sigma^4 / T_i)] \sim \chi^2(N-1)$

Statistic: $W = \sum_i [(\hat{\epsilon}_i' \hat{\epsilon}_i / T_i - \sigma^2)^2 / (2\sigma^4 / T_i)] \sim \chi^2(N-1)$

eq1: inst_avg

eq1: inst_avg

$W = 44.488$, $\chi^2(10)$, $p = 0.0000$ ***

$W = 44.488$, $\chi^2(10)$, $p = 0.0000$ ***

σ^2_i by country:

σ^2_i by country:

COL $\sigma^2=0.0772$ $T=23.0$
 COL $\sigma^2=0.0772$ $T=23.0$
 CRI $\sigma^2=0.0109$ $T=23.0$
 CRI $\sigma^2=0.0109$ $T=23.0$
 DOM $\sigma^2=0.0762$ $T=22.0$
 DOM $\sigma^2=0.0762$ $T=22.0$
 ECU $\sigma^2=0.0847$ $T=19.0$
 ECU $\sigma^2=0.0847$ $T=19.0$
 GTM $\sigma^2=0.0125$ $T=21.0$
 GTM $\sigma^2=0.0125$ $T=21.0$
 HND $\sigma^2=0.0144$ $T=23.0$
 HND $\sigma^2=0.0144$ $T=23.0$
 MEX $\sigma^2=0.0356$ $T=23.0$
 MEX $\sigma^2=0.0356$ $T=23.0$
 NIC $\sigma^2=0.0766$ $T=20.0$
 NIC $\sigma^2=0.0766$ $T=20.0$
 PAN $\sigma^2=0.0267$ $T=23.0$
 PAN $\sigma^2=0.0267$ $T=23.0$
 PER $\sigma^2=0.0255$ $T=11.0$
 PER $\sigma^2=0.0255$ $T=11.0$
 SLV $\sigma^2=0.0578$ $T=23.0$
 SLV $\sigma^2=0.0578$ $T=23.0$
 Pooled $\sigma^2 = 0.0454$
 Pooled $\sigma^2 = 0.0454$
 → ■ Heteroscedastic → clustered / DK SE already applied
 → ■ Heteroscedastic → clustered / DK SE already applied
 eq2: fdi_percent_gdp
 eq2: fdi_percent_gdp
 $W = 97.126$, $\chi^2(10)$, $p = 0.0000$ ***
 $W = 97.126$, $\chi^2(10)$, $p = 0.0000$ ***

σ^2_j by country:
 σ^2_j by country:
 COL $\sigma^2=0.9658$ $T=24.0$
 COL $\sigma^2=0.9658$ $T=24.0$
 CRI $\sigma^2=0.8655$ $T=24.0$
 CRI $\sigma^2=0.8655$ $T=24.0$
 DOM $\sigma^2=0.6069$ $T=24.0$
 DOM $\sigma^2=0.6069$ $T=24.0$
 ECU $\sigma^2=2.1311$ $T=24.0$
 ECU $\sigma^2=2.1311$ $T=24.0$
 GTM $\sigma^2=2.5205$ $T=24.0$
 GTM $\sigma^2=2.5205$ $T=24.0$
 HND $\sigma^2=1.3826$ $T=24.0$
 HND $\sigma^2=1.3826$ $T=24.0$
 MEX $\sigma^2=0.5226$ $T=24.0$
 MEX $\sigma^2=0.5226$ $T=24.0$
 NIC $\sigma^2=1.4346$ $T=24.0$
 NIC $\sigma^2=1.4346$ $T=24.0$
 PAN $\sigma^2=6.8248$ $T=24.0$
 PAN $\sigma^2=6.8248$ $T=24.0$
 PER $\sigma^2=1.5679$ $T=24.0$
 PER $\sigma^2=1.5679$ $T=24.0$
 SLV $\sigma^2=3.0953$ $T=24.0$
 SLV $\sigma^2=3.0953$ $T=24.0$

```

Pooled  $\sigma^2 = 1.9925$ 
Pooled  $\sigma^2 = 1.9925$ 
→ ■ Heteroscedastic → clustered / DK SE already applied
→ ■ Heteroscedastic → clustered / DK SE already applied

eq3: gdp_growth
eq3: gdp_growth
W = 47.801,  $\chi^2(10)$ , p = 0.0000 ***
W = 47.801,  $\chi^2(10)$ , p = 0.0000 ***
 $\sigma^2_i$  by country:
 $\sigma^2_i$  by country:
COL  $\sigma^2=1.5063$  T=24.0
COL  $\sigma^2=1.5063$  T=24.0
CRI  $\sigma^2=1.4438$  T=24.0
CRI  $\sigma^2=1.4438$  T=24.0
DOM  $\sigma^2=5.0612$  T=24.0
DOM  $\sigma^2=5.0612$  T=24.0
ECU  $\sigma^2=4.7060$  T=24.0
ECU  $\sigma^2=4.7060$  T=24.0
GTM  $\sigma^2=2.9620$  T=24.0
GTM  $\sigma^2=2.9620$  T=24.0
HND  $\sigma^2=2.0341$  T=24.0
HND  $\sigma^2=2.0341$  T=24.0
MEX  $\sigma^2=2.2581$  T=24.0
MEX  $\sigma^2=2.2581$  T=24.0
NIC  $\sigma^2=4.5205$  T=24.0
NIC  $\sigma^2=4.5205$  T=24.0
PAN  $\sigma^2=8.2602$  T=24.0
PAN  $\sigma^2=8.2602$  T=24.0
PER  $\sigma^2=3.2729$  T=24.0
PER  $\sigma^2=3.2729$  T=24.0
SLV  $\sigma^2=1.1059$  T=24.0
SLV  $\sigma^2=1.1059$  T=24.0
Pooled  $\sigma^2 = 3.3756$ 
Pooled  $\sigma^2 = 3.3756$ 

```

```

→ ■ Heteroscedastic → clustered / DK SE already applied
→ ■ Heteroscedastic → clustered / DK SE already applied

```

```

TEST 4 — Multicollinearity: VIF and Condition Number
TEST 4 — Multicollinearity: VIF and Condition Number

```

```

VIF computed on within-transformed (two-way demeaned) regressors.
VIF computed on within-transformed (two-way demeaned) regressors.
Reference: Belsley, Kuh & Welsch (1980), Regression Diagnostics.
Reference: Belsley, Kuh & Welsch (1980), Regression Diagnostics.

```

```

eq1: inst_avg
eq1: inst_avg
Variable VIF Flag
Variable VIF Flag

```

```

exports_percent_gdp 1.91 ✓ OK
exports_percent_gdp 1.91 ✓ OK

```


TEST 5 — Influence Diagnostics: DFBETA by Country

TEST 5 — Influence Diagnostics: DFBETA by Country

$DFBETA_c = (\beta_{full} - \beta_{-c}) / SE(\beta_{full})$

$DFBETA_c = (\beta_{full} - \beta_{-c}) / SE(\beta_{full})$

where β_{-c} is the coefficient estimated excluding country c.

where β_{-c} is the coefficient estimated excluding country c.

$|DFBETA| > 1$ indicates influential observation.

$|DFBETA| > 1$ indicates influential observation.

Reference: Belsley, Kuh & Welsch (1980).

Reference: Belsley, Kuh & Welsch (1980).

eq1: inst_avg | Key var: homicide_rate_log_lag1

eq1: inst_avg | Key var: homicide_rate_log_lag1

$\beta_{full} = -0.0836$ SE = 0.1136

$\beta_{full} = -0.0836$ SE = 0.1136

Country β_{-c} DFBETA Flag

Country β_{-c} DFBETA Flag

COL -0.0252 -0.5149 ✓ OK

COL -0.0252 -0.5149 ✓ OK

CRI -0.1076 0.2108 ✓ OK

CRI -0.1076 0.2108 ✓ OK

DOM -0.0547 -0.2550 ✓ OK

DOM -0.0547 -0.2550 ✓ OK

ECU -0.1018 0.1600 ✓ OK

ECU -0.1018 0.1600 ✓ OK

GTM -0.0906 0.0608 ✓ OK

GTM -0.0906 0.0608 ✓ OK

HND -0.1018 0.1600 ✓ OK

HND -0.1018 0.1600 ✓ OK

MEX -0.0054 -0.6885 ✓ OK

MEX -0.0054 -0.6885 ✓ OK

NIC -0.1100 0.2322 ✓ OK

NIC -0.1100 0.2322 ✓ OK

PAN -0.0353 -0.4253 ✓ OK

PAN -0.0353 -0.4253 ✓ OK

PER -0.0916 0.0700 ✓ OK

PER -0.0916 0.0700 ✓ OK

SLV -0.3024 1.9257 ✗ Influential ($|DFBETA| > 1$)

SLV -0.3024 1.9257 ✗ Influential ($|DFBETA| > 1$)

eq2: fdi_percent_gdp | Key var: inst_avg

eq2: fdi_percent_gdp | Key var: inst_avg

$\beta_{full} = 0.6111$ SE = 0.2795

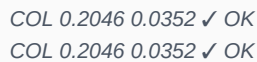
$\beta_{full} = 0.6111$ SE = 0.2795

Country β_{-c} DFBETA Flag

Country β_{-c} DFBETA Flag

COL 0.5987 0.0446 ✓ OK

eq3: gdp_growth | Key var: fdi_percent_gdp
eq3: gdp_growth | Key var: fdi_percent_gdp
 $\beta_{full} = 0.2074$ SE = 0.0796
 $\beta_{full} = 0.2074$ SE = 0.0796
Country β_{-c} DFBETA Flag
Country β_{-c} DFBETA Flag



Página 53

03-F — DIAGNOSTIC FIGURE

✓ Figure saved → figures/05_diagnostics.png

✓ Figure saved → figures/05_diagnostics.png

03-G — REMEDIES SUMMARY

03-G — REMEDIES SUMMARY

Diagnostic Result Remedy applied

Diagnostic Result Remedy applied

Serial correlation See results above Driscoll-Kraay HAC (Mod.02) ✓

Serial correlation See results above Driscoll-Kraay HAC (Mod.02) ✓

Wild bootstrap (Mod.04) ✓

Wild bootstrap (Mod.04) ✓

Cross-sect. dep. See results above Driscoll-Kraay HAC ✓

Cross-sect. dep. See results above Driscoll-Kraay HAC ✓

Group heterosced. See results above Clustered SE ✓

Group heterosced. See results above Clustered SE ✓

Multicollinearity See VIF above year_c centring (Mod.01) ✓

Multicollinearity See VIF above year_c centring (Mod.01) ✓

Influential obs. See DFBETA above Leave-one-out (Mod.05) ✓

Influential obs. See DFBETA above Leave-one-out (Mod.05) ✓

AR(1) in errors Wooldridge test Bootstrap inference (Mod.04) ✓

AR(1) in errors Wooldridge test Bootstrap inference (Mod.04) ✓

✓ Saved → C:\data_analyzer\econometric_pipeline\pipeline\json\03_diagnostics.json

✓ Saved → C:\data_analyzer\econometric_pipeline\pipeline\json\03_diagnostics.json

Module 03 complete.

Module 03 complete.

✓ Module 03 completed in 16.8s

=====

RUNNING MODULE 04: 04_bootstrap_inference.py

=====

04-A — LOAD

04-A — LOAD

✓ Data loaded. G=11 clusters, B=999 bootstrap draws.

✓ Data loaded. G=11 clusters, B=999 bootstrap draws.

Webb weights: [-1.2247 -1. -0.7071 0.7071 1. 1.2247]

Webb weights: [-1.2247 -1. -0.7071 0.7071 1. 1.2247]

■ With G=11, conventional clustered SE over-reject. Wild bootstrap preferred.

■ With G=11, conventional clustered SE over-reject. Wild bootstrap preferred.

04-B — WILD CLUSTER BOOTSTRAP (Webb weights, B=999)

B=999 bootstrap draws | Webb weights | Two-Way FE each iteration
 B=999 bootstrap draws | Webb weights | Two-Way FE each iteration
 Restricted model: PanelOLS without 'inst_avg' (H0: $\beta=0$)
 Restricted model: PanelOLS without 'inst_avg' (H0: $\beta=0$)
 b=200/999 elapsed=22s valid=200 fails=0
 b=200/999 elapsed=22s valid=200 fails=0
 b=400/999 elapsed=44s valid=400 fails=0
 b=400/999 elapsed=44s valid=400 fails=0
 b=600/999 elapsed=66s valid=600 fails=0
 b=600/999 elapsed=66s valid=600 fails=0
 b=800/999 elapsed=87s valid=800 fails=0
 b=800/999 elapsed=87s valid=800 fails=0

Full-sample estimate:

Full-sample estimate:

$\beta = 0.6111$

$\beta = 0.6111$

SE (clustered, conventional) = 0.2795

SE (clustered, conventional) = 0.2795

t = 2.1861

t = 2.1861

Wild cluster bootstrap (Webb, B=999):

Wild cluster bootstrap (Webb, B=999):

p-value (two-sided) = 0.0380 **

p-value (two-sided) = 0.0380 **

CI 95% (percentile) = [-0.5616, 0.6063]

CI 95% (percentile) = [-0.5616, 0.6063]

Valid draws: 999/999 | Fails: 0

Valid draws: 999/999 | Fails: 0

Runtime: 107.4s

Runtime: 107.4s

✓ $\beta(\text{inst_avg})$ significant at 10% (bootstrap p=0.0380)

✓ $\beta(\text{inst_avg})$ significant at 10% (bootstrap p=0.0380)

CR2 (Bell-McCaffrey) SE — Satterthwaite df (Pustejovsky & Tipton 2018):

CR2 (Bell-McCaffrey) SE — Satterthwaite df (Pustejovsky & Tipton 2018):

$\beta = 0.6111$

$\beta = 0.6111$

SE(CR2) = 0.2946

SE(CR2) = 0.2946

t(CR2) = 2.0744

t(CR2) = 2.0744

df(Satterthwaite) = 3.95 [cf. G-1=10 used in v1 — different]

df(Satterthwaite) = 3.95 [cf. G-1=10 used in v1 — different]

p(CR2) = 0.1075

p(CR2) = 0.1075

EQ3: dep=gdp_growth, key=fdi_percent_gdp

EQ3: dep=gdp_growth, key=fdi_percent_gdp

B=999 bootstrap draws | Webb weights | Two-Way FE each iteration

B=999 bootstrap draws | Webb weights | Two-Way FE each iteration

Restricted model: PanelOLS without 'fdi_percent_gdp' (H0: $\beta=0$)

Restricted model: PanelOLS without 'fdi_percent_gdp' (H0: $\beta=0$)

b=200/999 elapsed=20s valid=200 fails=0

b=200/999 elapsed=20s valid=200 fails=0

b=400/999 elapsed=42s valid=400 fails=0

b=400/999 elapsed=42s valid=400 fails=0
 b=600/999 elapsed=62s valid=600 fails=0
 b=600/999 elapsed=62s valid=600 fails=0
 b=800/999 elapsed=82s valid=800 fails=0
 b=800/999 elapsed=82s valid=800 fails=0

Full-sample estimate:

Full-sample estimate:

$\beta_{\text{■}} = 0.2074$

$\beta_{\text{■}} = 0.2074$

SE (clustered, conventional) = 0.0796

SE (clustered, conventional) = 0.0796

t = 2.6052

t = 2.6052

Wild cluster bootstrap (Webb, B=999):

Wild cluster bootstrap (Webb, B=999):

p-value (two-sided) = 0.1081

p-value (two-sided) = 0.1081

CI 95% (percentile) = [-0.2287, 0.2300]

CI 95% (percentile) = [-0.2287, 0.2300]

Valid draws: 999/999 | Fails: 0

Valid draws: 999/999 | Fails: 0

Runtime: 101.7s

Runtime: 101.7s

■ $\beta(\text{fdi_percent_gdp})$ not significant (bootstrap p=0.1081)

■ $\beta(\text{fdi_percent_gdp})$ not significant (bootstrap p=0.1081)

Note: inspect sign and economic magnitude regardless of p-value.

Note: inspect sign and economic magnitude regardless of p-value.

CR2 (Bell-McCaffrey) SE — Satterthwaite df (Pustejovsky & Tipton 2018):

CR2 (Bell-McCaffrey) SE — Satterthwaite df (Pustejovsky & Tipton 2018):

$\beta_{\text{■}} = 0.2074$

$\beta_{\text{■}} = 0.2074$

SE(CR2) = 0.0813

SE(CR2) = 0.0813

t(CR2) = 2.5514

t(CR2) = 2.5514

df(Satterthwaite) = 2.71 [cf. G-1=10 used in v1 — different]

df(Satterthwaite) = 2.71 [cf. G-1=10 used in v1 — different]

p(CR2) = 0.0929 *

p(CR2) = 0.0929 *

04-C — MEDIATION: BOOTSTRAP OF THE FULL-CHAIN INDIRECT EFFECT

04-C — MEDIATION: BOOTSTRAP OF THE FULL-CHAIN INDIRECT EFFECT

Motivation

Motivation

EQ1, EQ2, EQ3 are estimated as three INDEPENDENT single-equation FE

EQ1, EQ2, EQ3 are estimated as three INDEPENDENT single-equation FE

models (see Module 02, "Generated regressors" note) specifically to

models (see Module 02, "Generated regressors" note) specifically to

avoid the Pagan (1984) generated-regressors problem. A side effect is
 avoid the Pagan (1984) generated-regressors problem. A side effect is
 that nothing in the pipeline so far formally tests the hypothesized
 that nothing in the pipeline so far formally tests the hypothesized
 mediation chain itself:
 mediation chain itself:

Homicide $\rightarrow$ Institutions $\rightarrow$ FDI $\rightarrow$ Growth
 Homicide $\rightarrow$ Institutions $\rightarrow$ FDI $\rightarrow$ Growth

Three individually-insignificant coefficients strung together in prose
 Three individually-insignificant coefficients strung together in prose
 ("the results support a transmission mechanism") is NOT a statistical
 ("the results support a transmission mechanism") is NOT a statistical
 test of that mechanism. The standard remedy (Sobel 1982; MacKinnon,
 test of that mechanism. The standard remedy (Sobel 1982; MacKinnon,
 Lockhart & Hoffman 2004; Preacher & Hayes 2008) is to test the PRODUCT
 Lockhart & Hoffman 2004; Preacher & Hayes 2008) is to test the PRODUCT
 of the path coefficients, indirect = $\beta_1 * \gamma_1 * \delta_1$, against
 of the path coefficients, indirect = $\beta_1 * \gamma_1 * \delta_1$, against

H0: indirect = 0. The product's sampling distribution is not normal
 H0: indirect = 0. The product's sampling distribution is not normal
 even when each component is, so it must be obtained by resampling
 even when each component is, so it must be obtained by resampling
 rather than the delta method.
 rather than the delta method.

Procedure (cluster bootstrap over countries, not observations)
 Procedure (cluster bootstrap over countries, not observations)

Because observations within a country are serially correlated, the
 Because observations within a country are serially correlated, the
 resampling unit is the COUNTRY (all of its years move together), as
 resampling unit is the COUNTRY (all of its years move together), as
 in the SHAP cluster bootstrap of Module 06 and the Wild Cluster
 in the SHAP cluster bootstrap of Module 06 and the Wild Cluster
 Bootstrap above (Cameron & Miller 2015).
 Bootstrap above (Cameron & Miller 2015).

1. Resample G=8 countries WITH replacement.
1. Resample G=8 countries WITH replacement.
2. Relabel resampled countries uniquely per draw (e.g. "SLV__0",
 2. Relabel resampled countries uniquely per draw (e.g. "SLV__0",
 "SLV__1") so that a country drawn more than once contributes as
 "SLV__1") so that a country drawn more than once contributes as
 distinct panel entities with a valid (entity, time) index, and
 distinct panel entities with a valid (entity, time) index, and
 so entity/time Two-Way FE are genuinely re-estimated on the
 so entity/time Two-Way FE are genuinely re-estimated on the
 resampled panel rather than reused from the original fit.
 resampled panel rather than reused from the original fit.
3. Refit EQ1, EQ2, EQ3 (identical Two-Way FE spec as Module 02) on
 3. Refit EQ1, EQ2, EQ3 (identical Two-Way FE spec as Module 02) on
 the resampled panel; record β_1 , γ_1 , δ_1 and their
 the resampled panel; record β_1 , γ_1 , δ_1 and their
 product.
 product.

B=500 draws (lower than the B=999 wild bootstrap above because each draw now costs three PanelOLS fits instead of one).

Literature

Literature

■■■■■■■■■■

■■■■■■■■■■

Sobel (1982) — Asymptotic confidence intervals for indirect effects.

Sobel (1982) — Asymptotic confidence intervals for indirect effects.

MacKinnon, Lockhart & Hoffman (2004) — Confidence limits for the indirect effect: distribution of the product methods.

indirect effect: distribution of the product methods.

Preacher & Hayes (2008) — Asymptotic and resampling strategies for assessing and comparing indirect effects, Behavior Research Methods.

assessing and comparing indirect effects, Behavior Research Methods.

Cameron & Miller (2015) — cluster-robust / cluster-bootstrap inference.

Cameron & Miller (2015) — cluster-robust / cluster-bootstrap inference.

b=100/500 valid=100 fails=0

b=100/500 valid=100 fails=0

b=200/500 valid=200 fails=0

b=200/500 valid=200 fails=0

b=300/500 valid=300 fails=0

b=300/500 valid=300 fails=0

b=400/500 valid=400 fails=0

b=400/500 valid=400 fails=0

b=500/500 valid=500 fails=0

b=500/500 valid=500 fails=0

Point estimates (original panel, no resampling):

Point estimates (original panel, no resampling):

beta1 (homicide_rate_log_lag1 → inst_avg) = -0.0836

beta1 (homicide_rate_log_lag1 → inst_avg) = -0.0836

gamma1 (inst_avg → fdi_percent_gdp) = 0.6111

gamma1 (inst_avg → fdi_percent_gdp) = 0.6111

delta1 (fdi_percent_gdp → gdp_growth) = 0.2074

delta1 (fdi_percent_gdp → gdp_growth) = 0.2074

indirect = beta1 × gamma1 × delta1 = -0.010604

indirect = beta1 × gamma1 × delta1 = -0.010604

Cluster bootstrap over countries (B=500):

Cluster bootstrap over countries (B=500):

Mean(indirect*) = -0.015213 SD(indirect*) = 0.029542

Mean(indirect*) = -0.015213 SD(indirect*) = 0.029542

95% percentile CI = [-0.112620, 0.020498]

95% percentile CI = [-0.112620, 0.020498]

Bootstrap p-value (H0: indirect=0) = 0.5480

Bootstrap p-value (H0: indirect=0) = 0.5480

Valid draws: 500/500 | Fails: 0

Valid draws: 500/500 | Fails: 0

■ 95% CI includes zero — the FULL three-stage chain is NOT statistically distinguishable from no effect at conventional levels.

■ 95% CI includes zero — the FULL three-stage chain is NOT statistically distinguishable from no effect at conventional levels.

This is the honest headline result for the 'mecanismo de transmisión' claim: each link is individually weak, and the chain as a whole does not clear a formal significance test either. Report the point estimate and CI, not a causal claim.

This is the honest headline result for the 'mecanismo de transmisión' claim: each link is individually weak, and the chain as a whole does not clear a formal significance test either. Report the point estimate and CI, not a causal claim.

✓ Saved → C:\data_analyzer\econometric_pipeline\pipeline\json\04_mediation.json

✓ Saved → C:\data_analyzer\econometric_pipeline\pipeline\json\04_mediation.json

✓ Figure saved → figures\06b_mediation_bootstrap.png

✓ Figure saved → figures\06b_mediation_bootstrap.png

04-D — SE COMPARISON TABLE

04-D — SE COMPARISON TABLE

Equation Dep. var. Key var. β SE(CI.) p(CI.) SE(DK) SE(CR2) df_satt p(WCB)

Equation Dep. var. Key var. β SE(CI.) p(CI.) SE(DK) SE(CR2) df_satt p(WCB)

eq1 inst_avg homicide_rate_log_lag1 -0.0836 0.1136 0.462 0.0822 0.1374 2.12 0.562

eq1 inst_avg homicide_rate_log_lag1 -0.0836 0.1136 0.462 0.0822 0.1374 2.12 0.562

eq2 fdi_percent_gdp inst_avg 0.6111 0.2795 0.030 0.2192 0.2946 3.95 0.038**

eq2 fdi_percent_gdp inst_avg 0.6111 0.2795 0.030 0.2192 0.2946 3.95 0.038**

eq3 gdp_growth fdi_percent_gdp 0.2074 0.0796 0.010 0.1443 0.0813 2.71 0.108

eq3 gdp_growth fdi_percent_gdp 0.2074 0.0796 0.010 0.1443 0.0813 2.71 0.108

Columns:

Columns:

SE(CI.) — conventional clustered SE [UNDERSIZED with G=8, use for reference only]

SE(CI.) — conventional clustered SE [UNDERSIZED with G=8, use for reference only]

SE(DK) — Driscoll-Kraay HAC SE (robust to cross-sectional dependence)

SE(DK) — Driscoll-Kraay HAC SE (robust to cross-sectional dependence)

SE(CR2) — Bell-McCaffrey bias-corrected SE (Bell & McCaffrey 2002)

SE(CR2) — Bell-McCaffrey bias-corrected SE (Bell & McCaffrey 2002)

df(Satt.) — Satterthwaite effective df for t-test (Pustejovsky & Tipton 2018)

df(Satt.) — Satterthwaite effective df for t-test (Pustejovsky & Tipton 2018)

p(WCB) — wild cluster bootstrap p-value [PREFERRED for G=8]

p(WCB) — wild cluster bootstrap p-value [PREFERRED for G=8]

Preferred inference: p(WCB) from wild cluster bootstrap (Webb weights).

Preferred inference: p(WCB) from wild cluster bootstrap (Webb weights).

SE(CR2) with df_Satterthwaite as secondary check.

SE(CR2) with df_Satterthwaite as secondary check.

WRITE-UP WARNING — do not cherry-pick p(CI.):

WRITE-UP WARNING — do not cherry-pick p(CI.):

p(CI.) is the LEAST conservative column here by construction (this is

p(CI.) is the LEAST conservative column here by construction (this is

exactly why Modules 04's bootstrap and CR2 exist in the first place —

exactly why Modules 04's bootstrap and CR2 exist in the first place —

see docstring, "Why not conventional clustered SE?"). Any claim in the

see docstring, "Why not conventional clustered SE?"). Any claim in the

paper that a coefficient is "robust across specifications" must be

paper that a coefficient is "robust across specifications" must be

checked against p(DK), p(CR2) AND p(WCB), not just p(CI.). A table with

checked against p(DK), p(CR2) AND p(WCB), not just p(CI.). A table with

all four is exported to tables/se_comparison.tex for direct inclusion

all four is exported to tables/se_comparison.tex for direct inclusion

in the manuscript.

in the manuscript.

✓ LaTeX SE-comparison table saved → tables/se_comparison.tex (use this instead of quoting a single p-value in the manuscript)

✓ LaTeX SE-comparison table saved → tables/se_comparison.tex (use this instead of quoting a single p-value in the manuscript)

04-E — BOOTSTRAP FIGURES

04-E — BOOTSTRAP FIGURES

✓ Figure saved → figures/06_bootstrap_distributions.png

✓ Figure saved → figures/06_bootstrap_distributions.png

✓ Saved → C:\data_analyzer\econometric_pipeline\pipeline\json\04_bootstrap.json

✓ Saved → C:\data_analyzer\econometric_pipeline\pipeline\json\04_bootstrap.json

Module 04 complete.

Module 04 complete.

✓ Module 04 completed in 500.6s

RUNNING MODULE 05: 05_robustness.py

05-A — LOAD

05-A — LOAD

✓ Data: 274 obs | 11 countries

✓ Data: 274 obs | 11 countries

ROBUSTNESS 1 — Leave-One-Country-Out (LOCO)

ROBUSTNESS 1 — Leave-One-Country-Out (LOCO)

Bias addressed: results driven by a single influential country.

Bias addressed: results driven by a single influential country.

EQ1: inst_avg

EQ1: inst_avg

Specification N β SE p Stars $\Delta\beta$

Specification N β SE p Stars $\Delta\beta$

Full sample (baseline) 231 -0.0836 0.1136 0.462 +0.0000

Full sample (baseline) 231 -0.0836 0.1136 0.462 +0.0000

Excl. COL 208 -0.0252 0.0790 0.751 +0.0585

Excl. COL 208 -0.0252 0.0790 0.751 +0.0585

Excl. CRI 208 -0.1076 0.1173 0.360 -0.0239

Excl. CRI 208 -0.1076 0.1173 0.360 -0.0239

Excl. DOM 209 -0.0547 0.0998 0.584 +0.0290

Excl. DOM 209 -0.0547 0.0998 0.584 +0.0290

Excl. ECU 212 -0.1018 0.1331 0.445 -0.0182
 Excl. ECU 212 -0.1018 0.1331 0.445 -0.0182
 Excl. GTM 210 -0.0906 0.1194 0.449 -0.0069
 Excl. GTM 210 -0.0906 0.1194 0.449 -0.0069
 Excl. HND 208 -0.1018 0.1194 0.395 -0.0182
 Excl. HND 208 -0.1018 0.1194 0.395 -0.0182
 Excl. MEX 208 -0.0054 0.0793 0.945 +0.0782
 Excl. MEX 208 -0.0054 0.0793 0.945 +0.0782
 Excl. NIC 211 -0.1100 0.1211 0.365 -0.0264
 Excl. NIC 211 -0.1100 0.1211 0.365 -0.0264
 Excl. PAN 208 -0.0353 0.0954 0.712 +0.0483
 Excl. PAN 208 -0.0353 0.0954 0.712 +0.0483
 Excl. PER 220 -0.0916 0.1176 0.437 -0.0080
 Excl. PER 220 -0.0916 0.1176 0.437 -0.0080
 Excl. SLV 208 -0.3024 0.1134 0.008 *** -0.2187
 Excl. SLV 208 -0.3024 0.1134 0.008 *** -0.2187

■ 0 sign reversal(s), 4 large shift(s) — check influential countries.

■ 0 sign reversal(s), 4 large shift(s) — check influential countries.

EQ2: fdi_percent_gdp

EQ2: fdi_percent_gdp

Specification N β SE p Stars Δβ

Specification N β SE p Stars Δβ



Full sample (baseline) 264 0.6111 0.2795 0.030 ** +0.0000

Full sample (baseline) 264 0.6111 0.2795 0.030 ** +0.0000

Excl. COL 240 0.5987 0.4213 0.157 -0.0125

Excl. COL 240 0.5987 0.4213 0.157 -0.0125

Excl. CRI 240 0.6247 0.3009 0.039 ** +0.0135

Excl. CRI 240 0.6247 0.3009 0.039 ** +0.0135

Excl. DOM 240 0.5162 0.3186 0.107 -0.0949

Excl. DOM 240 0.5162 0.3186 0.107 -0.0949

Excl. ECU 240 0.5120 0.3367 0.130 -0.0991

Excl. ECU 240 0.5120 0.3367 0.130 -0.0991

Excl. GTM 240 0.5933 0.3346 0.078 * -0.0178

Excl. GTM 240 0.5933 0.3346 0.078 * -0.0178

Excl. HND 240 0.5012 0.2651 0.060 * -0.1099

Excl. HND 240 0.5012 0.2651 0.060 * -0.1099

Excl. MEX 240 0.4651 0.2637 0.079 * -0.1460

Excl. MEX 240 0.4651 0.2637 0.079 * -0.1460

Excl. NIC 240 0.8832 0.3732 0.019 ** +0.2721

Excl. NIC 240 0.8832 0.3732 0.019 ** +0.2721

Excl. PAN 240 0.5384 0.3704 0.148 -0.0727

Excl. PAN 240 0.5384 0.3704 0.148 -0.0727

Excl. PER 240 0.7081 0.2495 0.005 *** +0.0970

Excl. PER 240 0.7081 0.2495 0.005 *** +0.0970

Excl. SLV 240 0.7361 0.3410 0.032 ** +0.1250

Excl. SLV 240 0.7361 0.3410 0.032 ** +0.1250

✓ Robust: no sign reversals, no large shifts across LOCO.

✓ Robust: no sign reversals, no large shifts across LOCO.

EQ3: gdp_growth

EQ3: gdp_growth

Specification N β SE p Stars $\Delta\beta$

Specification N β SE p Stars $\Delta\beta$

Full sample (baseline) 264 0.2074 0.0796 0.010 *** +0.0000

Full sample (baseline) 264 0.2074 0.0796 0.010 *** +0.0000

Excl. COL 240 0.2046 0.0806 0.012 ** -0.0028

Excl. COL 240 0.2046 0.0806 0.012 ** -0.0028

Excl. CRI 240 0.2114 0.0743 0.005 *** +0.0039

Excl. CRI 240 0.2114 0.0743 0.005 *** +0.0039

Excl. DOM 240 0.2331 0.0691 0.001 *** +0.0256

Excl. DOM 240 0.2331 0.0691 0.001 *** +0.0256

Excl. ECU 240 0.1671 0.0885 0.061 * -0.0404

Excl. ECU 240 0.1671 0.0885 0.061 * -0.0404

Excl. GTM 240 0.2319 0.0832 0.006 *** +0.0245

Excl. GTM 240 0.2319 0.0832 0.006 *** +0.0245

Excl. HND 240 0.2336 0.0736 0.002 *** +0.0262

Excl. HND 240 0.2336 0.0736 0.002 *** +0.0262

Excl. MEX 240 0.2125 0.0875 0.016 ** +0.0050

Excl. MEX 240 0.2125 0.0875 0.016 ** +0.0050

Excl. NIC 240 0.1944 0.0918 0.035 ** -0.0131

Excl. NIC 240 0.1944 0.0918 0.035 ** -0.0131

Excl. PAN 240 0.1382 0.1074 0.200 -0.0692

Excl. PAN 240 0.1382 0.1074 0.200 -0.0692

Excl. PER 240 0.1681 0.0937 0.074 * -0.0394

Excl. PER 240 0.1681 0.0937 0.074 * -0.0394

Excl. SLV 240 0.2468 0.0896 0.006 *** +0.0394

Excl. SLV 240 0.2468 0.0896 0.006 *** +0.0394

✓ Robust: no sign reversals, no large shifts across LOCO.

✓ Robust: no sign reversals, no large shifts across LOCO.

ROBUSTNESS 2 — Time Windows

ROBUSTNESS 2 — Time Windows

Bias addressed: period-specific shocks, structural breaks.

Bias addressed: period-specific shocks, structural breaks.

EQ1: inst_avg

EQ1: inst_avg

Specification N β SE p Stars $\Delta\beta$

Specification N β SE p Stars $\Delta\beta$

Full sample 231 -0.0836 0.1136 0.462 +0.0000

Full sample 231 -0.0836 0.1136 0.462 +0.0000

Pre-2010 (pre-GFC) 76 -0.1240 0.1344 0.360 -0.0404

Pre-2010 (pre-GFC) 76 -0.1240 0.1344 0.360 -0.0404

Post-2010 155 -0.0130 0.0579 0.822 +0.0706

Post-2010 155 -0.0130 0.0579 0.822 +0.0706

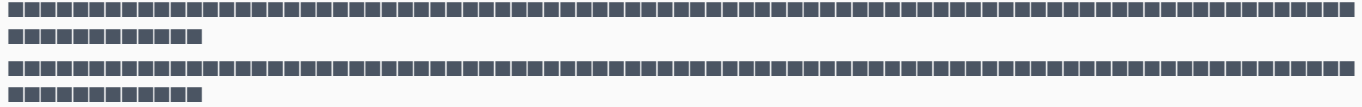
Excl. COVID (2020–21) 210 -0.0554 0.1036 0.593 +0.0282
 Excl. COVID (2020–21) 210 -0.0554 0.1036 0.593 +0.0282
 Post-peace (2014+) 113 0.0267 0.0546 0.626 +0.1104
 Post-peace (2014+) 113 0.0267 0.0546 0.626 +0.1104

EQ2: fdi_percent_gdp

EQ2: fdi_percent_gdp

Specification N β SE p Stars $\Delta\beta$

Specification N β SE p Stars $\Delta\beta$



Full sample 264 0.6111 0.2795 0.030 ** +0.0000

Full sample 264 0.6111 0.2795 0.030 ** +0.0000

Pre-2010 (pre-GFC) 99 2.2119 1.8548 0.237 +1.6008

Pre-2010 (pre-GFC) 99 2.2119 1.8548 0.237 +1.6008

Post-2010 165 1.2632 0.4252 0.004 *** +0.6521

Post-2010 165 1.2632 0.4252 0.004 *** +0.6521

Excl. COVID (2020–21) 242 0.6968 0.3489 0.047 ** +0.0857

Excl. COVID (2020–21) 242 0.6968 0.3489 0.047 ** +0.0857

Post-peace (2014+) 121 1.5415 0.6329 0.017 ** +0.9304

Post-peace (2014+) 121 1.5415 0.6329 0.017 ** +0.9304

EQ3: gdp_growth

EQ3: gdp_growth

Specification N β SE p Stars $\Delta\beta$

Specification N β SE p Stars $\Delta\beta$



Full sample 264 0.2074 0.0796 0.010 *** +0.0000

Full sample 264 0.2074 0.0796 0.010 *** +0.0000

Pre-2010 (pre-GFC) 99 0.1976 0.1268 0.124 -0.0098

Pre-2010 (pre-GFC) 99 0.1976 0.1268 0.124 -0.0098

Post-2010 165 0.2540 0.1540 0.101 +0.0466

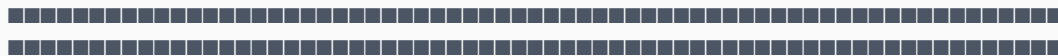
Post-2010 165 0.2540 0.1540 0.101 +0.0466

Excl. COVID (2020–21) 242 0.1118 0.0753 0.139 -0.0956

Excl. COVID (2020–21) 242 0.1118 0.0753 0.139 -0.0956

Post-peace (2014+) 121 0.2592 0.2084 0.217 +0.0517

Post-peace (2014+) 121 0.2592 0.2084 0.217 +0.0517



ROBUSTNESS 3 — Alternative Lag Structures (EQ1: Violence → Institutions)

ROBUSTNESS 3 — Alternative Lag Structures (EQ1: Violence → Institutions)



Uses pre-computed lag columns from dataset (not re-computed).

Uses pre-computed lag columns from dataset (not re-computed).

Bias addressed: uncertainty about transmission delay of violence.

Bias addressed: uncertainty about transmission delay of violence.

EQ1: Alternative Lags

EQ1: Alternative Lags

Specification N β SE p Stars $\Delta\beta$

Specification N β SE p Stars $\Delta\beta$

```

Contemporaneous (lag 0) 232 -0.0215 0.1223 0.861 +0.0622
Contemporaneous (lag 0) 232 -0.0215 0.1223 0.861 +0.0622
Lag 1 (primary) 231 -0.0836 0.1136 0.462 +0.0000
Lag 1 (primary) 231 -0.0836 0.1136 0.462 +0.0000
Lag 2 232 -0.2114 0.1324 0.112 -0.1277
Lag 2 232 -0.2114 0.1324 0.112 -0.1277
Lag 3 223 -0.3059 0.1369 0.027 ** -0.2222
Lag 3 223 -0.3059 0.1369 0.027 ** -0.2222

```

ROBUSTNESS 3b — Lag Structure for EQ2 and EQ3
 ROBUSTNESS 3b — Lag Structure for EQ2 and EQ3

Motivation: the theoretical model (paper Introduction) argues that
 Motivation: the theoretical model (paper Introduction) argues that
 security, institutions and growth adjust at DIFFERENT speeds --
 security, institutions and growth adjust at DIFFERENT speeds --

institutions evolve gradually and growth responds with a further lag.
 institutions evolve gradually and growth responds with a further lag.
 Yet the PRIMARY specifications of EQ2 (institutions -> FDI) and EQ3
 Yet the PRIMARY specifications of EQ2 (institutions -> FDI) and EQ3
 (FDI + institutions -> growth) use CONTEMPORANEOUS regressors, which
 (FDI + institutions -> growth) use CONTEMPORANEOUS regressors, which
 is inconsistent with that narrative and leaves the door open to
 is inconsistent with that narrative and leaves the door open to
 reverse causality / simultaneity (e.g. growth expectations attracting
 reverse causality / simultaneity (e.g. growth expectations attracting
 same-year FDI). This block re-estimates EQ2 and EQ3 replacing the
 same-year FDI). This block re-estimates EQ2 and EQ3 replacing the
 contemporaneous institution/FDI regressors with their one-year lag
 contemporaneous institution/FDI regressors with their one-year lag
 (inst_avg_lag1, fdi_percent_gdp_lag1 -- built in Module 01) to test
 (inst_avg_lag1, fdi_percent_gdp_lag1 -- built in Module 01) to test
 whether the results survive a specification that matches the stated
 whether the results survive a specification that matches the stated
 theory.
 theory.

EQ2: Contemporaneous vs Lagged Institutions

EQ2: Contemporaneous vs Lagged Institutions

Specification N β SE p Stars $\Delta\beta$

Specification N β SE p Stars $\Delta\beta$

```

Contemporaneous inst_avg (primary) 264 0.6111 0.2795 0.030 ** +0.0000
Contemporaneous inst_avg (primary) 264 0.6111 0.2795 0.030 ** +0.0000
Lag 1: inst_avg_lag1 253 0.3802 0.3796 0.318 -0.2309
Lag 1: inst_avg_lag1 253 0.3802 0.3796 0.318 -0.2309

```

EQ3: Contemporaneous vs Lagged FDI+Institutions

EQ3: Contemporaneous vs Lagged FDI+Institutions

Specification N β SE p Stars $\Delta\beta$

Specification N β SE p Stars $\Delta\beta$

Contemporaneous fdi+inst (primary) 264 0.2074 0.0796 0.010 *** +0.0000

Contemporaneous fdi+inst (primary) 264 0.2074 0.0796 0.010 *** +0.0000

Lag 1: fdi_lag1+inst_lag1 253 -0.0703 0.0582 0.228 -0.2777

Lag 1: fdi_lag1+inst_lag1 253 -0.0703 0.0582 0.228 -0.2777

■ Note: 'key' here is `fdi_percent_gap` (or its lag) -- the coefficient printed. `inst_avg`'s own coefficient in this lagged spec is in the JSON export, not this table.

■ Note: 'key' here is `fdi_percent_gdp` (or its lag) -- the coefficient printed. `inst_avg`'s own coefficient in this lagged spec is in the JSON export, not this table.

ROBUSTNESS 4 — Alternative Institution Indices

ROBUSTNESS 4 — Alternative Institution Indices

Bias addressed: sensitivity to index construction methodology.

Bias addressed: sensitivity to index construction methodology.

EQ1: Alternative Institution Indices

EQ1: Alternative Institution Indices

Specification N β SE p Stars $\Delta\beta$

Specification N β SE p Stars $\Delta\beta$

inst_avg (primary) 231 -0.0836 0.1136 0.462 +0.0000

inst_avg (primary) 231 -0.0836 0.1136 0.462 +0.0000

inst_pca (PCA-based) 231 -0.1829 0.2796 0.514 -0.0992

inst_pca (PCA-based) 231 -0.1829 0.2796 0.514 -0.0992

rule_of_law 231 -1.9609 0.9337 0.037 ** -1.8772

rule_of_law 231 -1.9609 0.9337 0.037 ** -1.8772

control_corruption 231 -0.3954 0.8575 0.645 -0.3118

control_corruption 231 -0.3954 0.8575 0.645 -0.3118

political_stability 231 -2.6715 1.7807 0.135 -2.5879

political_stability 231 -2.6715 1.7807 0.135 -2.5879

voice accountability 231 1.7207 1.7964 0.339 +1.8043

voice accountability 231 1.7207 1.7964 0.339 +1.8043

government effectiveness 231 -0.9994 0.8707 0.253 -0.9157

government effectiveness 231 -0.9994 0.8707 0.253 -0.9157

regulatory_quality 231 -0.2765 1.1070 0.803 -0.1929

regulatory_quality 231 -0.2765 1.1070 0.803 -0.1929

■ *Mixed signs across institution specifications — investigate.*

- Mixed signs across institution specifications — investigate.

ROBUSTNESS 5 — Alternative Control Sets

ROBUSTNESS 5 — Alternative Control Sets

Bias addressed: over-controlling or omitted variable bias.

Bias addressed: over-controlling or omitted variable bias.

■ 'Excl. GDP p.c. (bad-control check)': *gdp_per_capita_log* is plausibly

■ 'Excl. GDP p.c. (bad-control check)': *gdp_per_capita_log* is plausibly

DOWNSTREAM of institutions/FDI (part of the very causal channel this
DOWNSTREAM of institutions/FDI (part of the very causal channel this

pipeline studies), not an exogenous confounder. Including it as a

pipeline studies), not an exogenous confounder. Including it as a

control in EQ1/EQ2 risks a post-treatment ('bad control') bias that

control in EQ1/EQ2 risks a post-treatment ('bad control') bias that

partials out some of the effect of interest (Angrist & Pischke 2009,

partials out some of the effect of interest (Angrist & Pischke 2009,

ch. 3). This variant drops it while keeping the rest of the baseline

ch. 3). This variant drops it while keeping the rest of the baseline

controls, to see whether the key coefficient is sensitive to its

controls, to see whether the key coefficient is sensitive to its

inclusion.

inclusion.

■ 'Incl. remittances (bad-control check)': *remittances_percent_gdp* (added

■ 'Incl. remittances (bad-control check)': *remittances_percent_gdp* (added

September 2026) is plausibly a BAD CONTROL of a different kind -- it is

September 2026) is plausibly a BAD CONTROL of a different kind -- it is

a candidate mediator/collider between violence (emigration push) and

a candidate mediator/collider between violence (emigration push) and

growth (remittance-funded consumption), not a clean exogenous control.

growth (remittance-funded consumption), not a clean exogenous control.

This variant ADDS it to the baseline controls (rather than treating it

This variant ADDS it to the baseline controls (rather than treating it

as part of the primary specification) to see whether the key

as part of the primary specification) to see whether the key

coefficient is sensitive to its inclusion, kept separate from

coefficient is sensitive to its inclusion, kept separate from

Module 01's core institution-index change for the same reason

Module 01's core institution-index change for the same reason

gdp_per_capita_log is tested as a variant rather than assumed safe.

gdp_per_capita_log is tested as a variant rather than assumed safe.

EQ1: *inst_avg*

EQ1: *inst_avg*

Specification N β ■ SE p Stars $\Delta\beta$

Specification N β ■ SE p Stars $\Delta\beta$

Baseline controls 231 -0.0836 0.1136 0.462 +0.0000

Baseline controls 231 -0.0836 0.1136 0.462 +0.0000

Extended controls 191 -0.2131 0.0682 0.002 *** -0.1295

Extended controls 191 -0.2131 0.0682 0.002 *** -0.1295

Minimal controls 231 -0.0987 0.1182 0.405 -0.0150

Minimal controls 231 -0.0987 0.1182 0.405 -0.0150

Excl. GDP p.c. (bad-control check) 231 -0.0999 0.1250 0.425 -0.0163

Excl. GDP p.c. (bad-control check) 231 -0.0999 0.1250 0.425 -0.0163

Incl. remittances (bad-control check) 231 -0.1554 0.1152 0.179 -0.0718
 Incl. remittances (bad-control check) 231 -0.1554 0.1152 0.179 -0.0718
 No controls (FE only) 231 -0.1466 0.1628 0.369 -0.0630
 No controls (FE only) 231 -0.1466 0.1628 0.369 -0.0630

EQ2: fdi_percent_gdp

EQ2: fdi_percent_gdp

Specification N β SE p Stars $\Delta\beta$

Specification N β SE p Stars $\Delta\beta$

Baseline controls 264 0.6111 0.2795 0.030 ** +0.0000
 Baseline controls 264 0.6111 0.2795 0.030 ** +0.0000
 Extended controls 216 -1.0665 0.9786 0.277 -1.6776
 Extended controls 216 -1.0665 0.9786 0.277 -1.6776
 Minimal controls 264 0.1358 0.7602 0.858 -0.4753
 Minimal controls 264 0.1358 0.7602 0.858 -0.4753
 Excl. GDP p.c. (bad-control check) 264 0.6331 0.2545 0.014 ** +0.0220
 Excl. GDP p.c. (bad-control check) 264 0.6331 0.2545 0.014 ** +0.0220
 Incl. remittances (bad-control check) 264 0.9671 0.2976 0.001 *** +0.3560
 Incl. remittances (bad-control check) 264 0.9671 0.2976 0.001 *** +0.3560
 No controls (FE only) 264 -0.2344 0.6450 0.717 -0.8456
 No controls (FE only) 264 -0.2344 0.6450 0.717 -0.8456

EQ3: gdp_growth

EQ3: gdp_growth

Specification N β SE p Stars $\Delta\beta$

Specification N β SE p Stars $\Delta\beta$

Baseline controls 274 0.2052 0.0862 0.018 ** +0.0000
 Baseline controls 274 0.2052 0.0862 0.018 ** +0.0000
 Extended controls 226 0.3650 0.1130 0.001 *** +0.1598
 Extended controls 226 0.3650 0.1130 0.001 *** +0.1598
 Minimal controls 274 0.2830 0.1130 0.013 ** +0.0778
 Minimal controls 274 0.2830 0.1130 0.013 ** +0.0778
 Excl. GDP p.c. (bad-control check) 274 0.2135 0.0798 0.008 *** +0.0083
 Excl. GDP p.c. (bad-control check) 274 0.2135 0.0798 0.008 *** +0.0083
 Incl. remittances (bad-control check) 274 0.2028 0.0896 0.025 ** -0.0024
 Incl. remittances (bad-control check) 274 0.2028 0.0896 0.025 ** -0.0024
 No controls (FE only) 274 0.2699 0.1083 0.013 ** +0.0647
 No controls (FE only) 274 0.2699 0.1083 0.013 ** +0.0647

ROBUSTNESS 6 — Placebo Test (permuted homicide)

ROBUSTNESS 6 — Placebo Test (permuted homicide)

Procedure: randomly permute homicide_rate_log_lag1 within year cells.

Procedure: randomly permute homicide_rate_log_lag1 within year cells.

This breaks the country-specific causal link while preserving the

This breaks the country-specific causal link while preserving the

cross-sectional and temporal distribution of the variable.

cross-sectional and temporal distribution of the variable.

H0 (placebo): permuted violence has no effect on institutions.

H0 (placebo): permuted violence has no effect on institutions.

If the TRUE estimate is more extreme than 95% of placebo estimates,

If the TRUE estimate is more extreme than 95% of placebo estimates,

this confirms the result is not driven by spurious correlation.

this confirms the result is not driven by spurious correlation.

N_permutations = 500.

N_permutations = 500.

True estimate: $\beta = -0.0836$

True estimate: $\beta = -0.0836$

Placebo distribution: mean=0.0016, SD=0.0236, 5th pctl=-0.0374

Placebo distribution: mean=0.0016, SD=0.0236, 5th pctl=-0.0374

Placebo p-value: 0.0000

Placebo p-value: 0.0000

✓ Placebo test PASSED — true estimate more extreme than 95% of random permutations.

✓ Placebo test PASSED — true estimate more extreme than 95% of random permutations.

ROBUSTNESS 7 — Country-Specific Linear Trends

ROBUSTNESS 7 — Country-Specific Linear Trends

Motivation: Two-Way FE (entity + time) removes country LEVELS and
 Motivation: Two-Way FE (entity + time) removes country LEVELS and
 common year shocks, but NOT country-specific trends. Several series
 common year shocks, but NOT country-specific trends. Several series
 in this panel are strongly trending over 25 years (GDP per capita,
 in this panel are strongly trending over 25 years (GDP per capita,
 population, and especially El Salvador's homicide collapse from >100
 population, and especially El Salvador's homicide collapse from >100
 to <10 per 100,000). If two trending series happen to move together
 to <10 per 100,000). If two trending series happen to move together
 for reasons unrelated to the hypothesized mechanism, Two-Way FE alone
 for reasons unrelated to the hypothesized mechanism, Two-Way FE alone
 will not catch it -- this is the classic 'spurious panel regression'
 will not catch it -- this is the classic 'spurious panel regression'
 risk (Granger & Newbold 1974; Kao 1999; Phillips & Moon 1999).
 risk (Granger & Newbold 1974; Kao 1999; Phillips & Moon 1999).

This check re-estimates the PRIMARY specification of each equation
 This check re-estimates the PRIMARY specification of each equation
 adding one linear trend per country (country_dummy x year_c). If the
 adding one linear trend per country (country_dummy x year_c). If the
 key coefficient survives with the same sign and comparable magnitude,
 key coefficient survives with the same sign and comparable magnitude,
 the baseline result is not merely two coincidentally-trending series.
 the baseline result is not merely two coincidentally-trending series.

Reference: Wooldridge (2010), ch. 10.5 (unit-specific trends).

Reference: Wooldridge (2010), ch. 10.5 (unit-specific trends).

EQ1: inst_avg

EQ1: inst_avg

Specification N β SE p Stars $\Delta\beta$

Specification N β SE p Stars $\Delta\beta$



Baseline (no country trends) 231 -0.0836 0.1136 0.462 +0.0000

Baseline (no country trends) 231 -0.0836 0.1136 0.462 +0.0000

+ 11 country-specific trends 231 -0.0690 0.0485 0.157 +0.0147

+ 11 country-specific trends 231 -0.0690 0.0485 0.157 +0.0147

✓ Survives country-specific trends ($\Delta=18\%$, same sign).

✓ Survives country-specific trends ($\Delta=18\%$, same sign).

EQ2: fdi_percent_gdp

EQ2: fdi_percent_gdp

Specification N β SE p Stars $\Delta\beta$

Specification N β SE p Stars $\Delta\beta$



Baseline (no country trends) 264 0.6111 0.2795 0.030 ** +0.0000

Baseline (no country trends) 264 0.6111 0.2795 0.030 ** +0.0000

+ 11 country-specific trends 264 0.9104 0.8053 0.260 +0.2993

+ 11 country-specific trends 264 0.9104 0.8053 0.260 +0.2993

✓ Survives country-specific trends ($\Delta=49\%$, same sign).

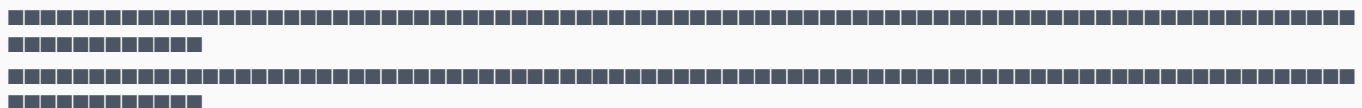
✓ Survives country-specific trends ($\Delta=49\%$, same sign).

EQ3: gdp_growth

EQ3: gdp_growth

Specification N β SE p Stars $\Delta\beta$

Specification N β SE p Stars $\Delta\beta$



Baseline (no country trends) 264 0.2074 0.0796 0.010 *** +0.0000

Baseline (no country trends) 264 0.2074 0.0796 0.010 *** +0.0000

+ 11 country-specific trends 264 0.1901 0.1061 0.075 * -0.0173

+ 11 country-specific trends 264 0.1901 0.1061 0.075 * -0.0173

✓ Survives country-specific trends ($\Delta=8\%$, same sign).

✓ Survives country-specific trends ($\Delta=8\%$, same sign).



ROBUSTNESS 8 — Panel Unit Root Test (Fisher-ADF, Maddala & Wu 1999)

ROBUSTNESS 8 — Panel Unit Root Test (Fisher-ADF, Maddala & Wu 1999)



No unit-root test was previously implemented anywhere in this pipeline

No unit-root test was previously implemented anywhere in this pipeline

despite 25-year trending macro series. This runs an Augmented

despite 25-year trending macro series. This runs an Augmented

Dickey-Fuller test SEPARATELY for each country's time series (T=25 per

Dickey-Fuller test SEPARATELY for each country's time series (T=25 per

country, so maxlag=1, no automatic lag selection -- higher lags are

country, so maxlag=1, no automatic lag selection -- higher lags are

not identifiable with so few observations) and combines the p-values

not identifiable with so few observations) and combines the p -values with the Fisher (1932) / Maddala-Wu (1999) combination test:
with the Fisher (1932) / Maddala-Wu (1999) combination test:

$P = -2 * \sum(\log(p_i)) \sim \chi^2(2N)$ under H_0 : ALL panels have a unit root.

$P = -2 * \sum(\log(p_i)) \sim \chi^2(2N)$ under H_0 : ALL panels have a unit root.

CAVEAT (consistent with this pipeline's other small-sample warnings):

CAVEAT (consistent with this pipeline's other small-sample warnings):

with $T \sim 25$ observations per country, per-country ADF tests have very low power. Failing to reject H_0 here is only weak evidence of a unit

low power. Failing to reject H_0 here is only weak evidence of a unit root, not proof of one. This test should be read as a flag for

root, not proof of one. This test should be read as a flag for

further caution, not a definitive diagnosis -- see Robustness 7 above

further caution, not a definitive diagnosis -- see Robustness 7 above

(country-specific trends) as a partial, practical remedy regardless of

(country-specific trends) as a partial, practical remedy regardless of

the formal test outcome.

the formal test outcome.

Reference: Maddala & Wu (1999), A comparative study of unit root tests

Reference: Maddala & Wu (1999), A comparative study of unit root tests

with panel data and a new simple test, Oxford Bulletin of Economics

with panel data and a new simple test, Oxford Bulletin of Economics

and Statistics 61(S1), 631-652.

and Statistics 61(S1), 631-652.

Variable N countries Fisher χ^2 df p (Fisher) Flag

Variable N countries Fisher χ^2 df p (Fisher) Flag

Variable N countries Fisher χ^2 df p (Fisher) Flag

Variable N countries Fisher χ^2 df p (Fisher) Flag

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Variable N countries Fisher χ^2 df p (Fisher) Flag

✓ Saved → C:\data_analyzer\econometric_pipeline\pipeline\json\05_robustness.json
 ✓ Saved → C:\data_analyzer\econometric_pipeline\pipeline\json\05_robustness.json

Module 05 complete.

Module 05 complete.

✓ Module 05 completed in 79.4s

=====

RUNNING MODULE 06: 06_ml_triangular.py

=====

06-A — LOAD

06-A — LOAD

✓ Data: 274 obs | 11 countries

✓ Data: 274 obs | 11 countries

CV strategy: Leave-One-Country-Out (11 folds)

CV strategy: Leave-One-Country-Out (11 folds)

■ All ML results are EXPLORATORY. They do not imply causation.

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■ G=11 LOCO folds → CV performance estimates carry wide uncertainty. Interpret means alongside SDs, not in isolation.

■ G=11 LOCO folds → CV performance estimates carry wide uncertainty. Interpret means alongside SDs, not in isolation.

06-B — ML ESTIMATION (LOCO-CV, SHAP, ALE)

06-B — ML ESTIMATION (LOCO-CV, SHAP, ALE)

■■ T1: Violence → Institutions ■■

■■ T1: Violence → Institutions ■■

Missing summary:

Missing summary:

homicide_rate_log_lag2 0.153285

homicide_rate_log_lag2 0.153285

homicide_rate_log_lag1 0.120438

homicide_rate_log_lag1 0.120438

inst_avg 0.036496

inst_avg 0.036496

gdp_per_capita_log 0.000000

gdp_per_capita_log 0.000000

inflation 0.000000

inflation 0.000000

unemployment 0.000000

unemployment 0.000000

exports_percent_gdp 0.000000

exports_percent_gdp 0.000000

population_log 0.000000

population_log 0.000000

year_c 0.000000

year_c 0.000000

dtype: float64

dtype: float64

Rows before dropna: 274

Rows before dropna: 274

Rows after dropna: 226

Rows after dropna: 226

Model R^2 mean R^2 SD RMSE mean Valid folds

Model R^2 mean R^2 SD RMSE mean Valid folds

Random Forest -28.8166 71.1718 0.6801 11

Random Forest -28.8166 71.1718 0.6801 11

Gradient Boosting -26.2960 51.1953 0.7083 11

Gradient Boosting -26.2960 51.1953 0.7083 11

■ Note: G=11 LOCO folds → R^2 estimates have wide uncertainty. SD across folds is as informative as the mean.

■ Note: G=11 LOCO folds → R^2 estimates have wide uncertainty. SD across folds is as informative as the mean.

RF vs GB Wilcoxon signed-rank (paired LOCO R^2):

RF vs GB Wilcoxon signed-rank (paired LOCO R^2):

N pairs: 11 | W=28.0 p=0.7001953125

N pairs: 11 | W=28.0 p=0.7001953125

■ G=11 pairs → very low power. Treat p-value as descriptive only.

■ G=11 pairs → very low power. Treat p-value as descriptive only.

RF OOB R^2 (internal diagnostic — in-sample, optimistic): 0.7919

RF OOB R^2 (internal diagnostic — in-sample, optimistic): 0.7919

■ OOB R^2 is a training-data statistic, NOT a LOCO out-of-country R^2 . Do not cite OOB R^2 as evidence of predictive performance.

■ OOB R^2 is a training-data statistic, NOT a LOCO out-of-country R^2 . Do not cite OOB R^2 as evidence of predictive performance.

Computing SHAP cluster bootstrap stability (B=200, cluster=country)...

Computing SHAP cluster bootstrap stability (B=200, cluster=country)...

Feature Gini Perm±SD |SHAP| SHAP rank Rank SD P(top3) P(top5)

Feature Gini Perm±SD |SHAP| SHAP rank Rank SD P(top3) P(top5)

Population (log) 0.3373 0.3803±0.0331 0.2292 1.44 0.73 0.99 0.99

Population (log) 0.3373 0.3803±0.0331 0.2292 1.44 0.73 0.99 0.99

GDP per capita (log) 0.2355 0.2834±0.0290 0.2125 2.12 1.18 0.88 0.97

GDP per capita (log) 0.2355 0.2834±0.0290 0.2125 2.12 1.18 0.88 0.97

Violence lag 2 0.1049 0.0638±0.0089 0.0882 3.44 1.21 0.64 0.93

Violence lag 2 0.1049 0.0638±0.0089 0.0882 3.44 1.21 0.64 0.93

Violence lag 1 0.0911 0.0424±0.0064 0.0727 4.89 1.12 0.06 0.69

Violence lag 1 0.0911 0.0424±0.0064 0.0727 4.89 1.12 0.06 0.69

Unemployment 0.0770 0.0545±0.0065 0.0612 5.08 1.62 0.20 0.68

Unemployment 0.0770 0.0545±0.0065 0.0612 5.08 1.62 0.20 0.68

Exports (% GDP) 0.0835 0.0492±0.0046 0.0380 5.23 1.50 0.14 0.46

Exports (% GDP) 0.0835 0.0492±0.0046 0.0380 5.23 1.50 0.14 0.46

Inflation 0.0502 0.0348±0.0038 0.0375 6.03 1.37 0.10 0.27

Inflation 0.0502 0.0348±0.0038 0.0375 6.03 1.37 0.10 0.27

Time Trend (centred) 0.0205 0.0137±0.0022 0.0187 7.78 0.61 0.00 0.01

Time Trend (centred) 0.0205 0.0137±0.0022 0.0187 7.78 0.61 0.00 0.01

Key feature 'homicide_rate_log_lag1':

Key feature 'homicide_rate_log_lag1':

Permutation rank: #6/8

Permutation rank: #6/8

SHAP rank: #4/8

SHAP rank: #4/8

year_c 0.000000

[illegible]

Permutation rank: #2/8

Permutation rank: #2/8

SHAP rank: #1/8

SHAP rank: #1/8

✓ Key feature ranks in top 3 by both importance metrics.

✓ Key feature ranks in top 3 by both importance metrics.

06-C — CONVERGENCE TABLE: FE vs ML

06-C — CONVERGENCE TABLE: FE vs ML

Convergence principle (Athey & Imbens 2019):

Convergence principle (Athey & Imbens 2019):

If the variables that matter causally (significant in FE) also rank

If the variables that matter causally (significant in FE) also rank

highly in ML importance, this triangulation strengthens the narrative.

highly in ML importance, this triangulation strengthens the narrative.

Divergence signals potential non-linearities or confounders.

Divergence signals potential non-linearities or confounders.

Task FE coef FE p-val FE sig Direction Perm rank SHAP rank Convergence

Task FE coef FE p-val FE sig Direction Perm rank SHAP rank Convergence

T1: Violence → Institutions -0.0836 0.4624 Neg 6 4 ■ Partial

T1: Violence → Institutions -0.0836 0.4624 Neg 6 4 ■ Partial

T2: Institutions → FDI 0.6111 0.0298 ** Pos 3 3 ✓ Strong

T2: Institutions → FDI 0.6111 0.0298 ** Pos 3 3 ✓ Strong

T3: FDI + Institutions → Growth 0.2074 0.0098 *** Pos 2 1 ✓ Strong

T3: FDI + Institutions → Growth 0.2074 0.0098 *** Pos 2 1 ✓ Strong

Notes:

Notes:

- FE coef and p-value are cluster-robust estimates from Module 02.

- FE coef and p-value are cluster-robust estimates from Module 02.

- FE sig: * p<0.10 ** p<0.05 *** p<0.01.

- FE sig: * p<0.10 ** p<0.05 *** p<0.01.

- Direction: sign of the FE coefficient for the key feature.

- Direction: sign of the FE coefficient for the key feature.

- Perm/SHAP rank: rank among all features in the ML model (1 = most important).

- Perm/SHAP rank: rank among all features in the ML model (1 = most important).

- Convergence: Strong = key feature ranks top 3 by both metrics.

- Convergence: Strong = key feature ranks top 3 by both metrics.

06-D — FIGURES

06-D — FIGURES

✓ Figure saved → figures/08_ml_loco_cv.png

✓ Figure saved → figures/08_ml_loco_cv.png

✓ Figure saved → figures/09_feature_importance.png

✓ Figure saved → figures/09_feature_importance.png

✓ Figure saved → figures/10_shap_ale.png

✓ Figure saved → figures/10_shap_ale.png

06-E — EXPORT

06-E — EXPORT

✓ Saved → C:\data_analyzer\econometric_pipeline\pipeline\json\06_ml_results.json

✓ Saved → C:\data_analyzer\econometric_pipeline\pipeline\json\06_ml_results.json

Module 06 complete. Pipeline finished.

Module 06 complete. Pipeline finished.

■ IMPORTANT REMINDER ■

■ IMPORTANT REMINDER ■

■ ML results (SHAP, permutation importance, LOCO R^2) ■

■ ML results (SHAP, permutation importance, LOCO R^2) ■

■ are EXPLORATORY and NON-CAUSAL. ■

■ are EXPLORATORY and NON-CAUSAL. ■

■ They triangulate — they do not identify causal effects. ■

■ They triangulate — they do not identify causal effects. ■

■ Primary causal evidence: Module 02 (Two-Way FE) and ■

■ Primary causal evidence: Module 02 (Two-Way FE) and ■

■ Module 04 (Wild Cluster Bootstrap inference). ■

■ Module 04 (Wild Cluster Bootstrap inference). ■

■ ■

■ ■

■ SHAP limitation: computed on the full-sample model. ■

■ SHAP limitation: computed on the full-sample model. ■

■ Values reflect in-sample feature associations and may not ■

■ Values reflect in-sample feature associations and may not ■

■ generalise to held-out countries. ■

■ generalise to held-out countries. ■

■ ■

■ ■

■ Bootstrap stability limitation: $G=8$ clusters → discrete ■

■ Bootstrap stability limitation: $G=8$ clusters → discrete ■

■ bootstrap distribution. Treat rank SDs and hit rates as ■

■ bootstrap distribution. Treat rank SDs and hit rates as ■

■ approximate lower bounds on true uncertainty. ■

■ approximate lower bounds on true uncertainty. ■

✓ Module 06 completed in 1268.7s

=====

RUNNING MODULE 07: 07_cointegration.py

=====

07-A — LOAD

07-A — LOAD

✓ Data loaded: 274 obs, 11 countries

✓ Data loaded: 274 obs, 11 countries

Scope reminder: this module only tests pairs of series that Module 05's

Scope reminder: this module only tests pairs of series that Module 05's

Fisher-ADF test found to be $I(1)$ (non-stationary): homicide_rate_log,

Fisher-ADF test found to be $I(1)$ (non-stationary): homicide_rate_log,

inst_avg, gdp_per_capita_log, fdi_percent_gdp and gdp_growth already

inst_avg, gdp_per_capita_log, fdi_percent_gdp and gdp_growth already

reject a unit root there (they are $I(0)$) and are intentionally excluded

reject a unit root there (they are $I(0)$) and are intentionally excluded

from cointegration testing -- see module docstring.

from cointegration testing -- see module docstring.

07-B — STEP 1: LONG-RUN RELATIONSHIP & RESIDUAL COINTEGRATION TEST

07-B — STEP 1: LONG-RUN RELATIONSHIP & RESIDUAL COINTEGRATION TEST

■ T1: Homicide & Institutions ■

■ T1: Homicide & Institutions ■

Long-run regression: $inst_avg_it = a_i + I_t + \beta * homicide_rate_log_it + e_{it}$

Long-run regression: $inst_avg_it = a_i + I_t + \beta * homicide_rate_log_it + e_{it}$

$\beta = -0.1330$ SE(c) = 0.1733 p(c) = 0.4436 N = 232

$\beta = -0.1330$ SE(c) = 0.1733 p(c) = 0.4436 N = 232

Fisher-ADF on residuals: $\chi^2(22)=19.15$, $p=0.6360$

Fisher-ADF on residuals: $\chi^2(22)=19.15$, $p=0.6360$

■ Cannot reject unit root in residuals ($p=0.6360$) -& NO evidence of cointegration; levels relationship may be spurious

■ Cannot reject unit root in residuals ($p=0.6360$) -& NO evidence of cointegration; levels relationship may be spurious

■ T2: Institutions & Income ■

■ T2: Institutions & Income ■

Long-run regression: $inst_avg_it = a_i + I_t + \beta * gdp_per_capita_log_it + e_{it}$

Long-run regression: $inst_avg_it = a_i + I_t + \beta * gdp_per_capita_log_it + e_{it}$

$\beta = 0.7733$ SE(c) = 0.3483 p(c) = 0.0274 ** N = 264

$\beta = 0.7733$ SE(c) = 0.3483 p(c) = 0.0274 ** N = 264

Fisher-ADF on residuals: $\chi^2(22)=23.16$, $p=0.3929$

Fisher-ADF on residuals: $\chi^2(22)=23.16$, $p=0.3929$

■ Cannot reject unit root in residuals ($p=0.3929$) -& NO evidence of cointegration; levels relationship may be spurious

■ Cannot reject unit root in residuals ($p=0.3929$) -& NO evidence of cointegration; levels relationship may be spurious

■ T3: Homicide & Income ■

■ T3: Homicide & Income ■

Long-run regression: $gdp_per_capita_log_it = a_i + I_t + \beta * homicide_rate_log_it + e_{it}$

Long-run regression: $gdp_per_capita_log_it = a_i + I_t + \beta * homicide_rate_log_it + e_{it}$

$\beta = -0.0911$ SE(c) = 0.0962 p(c) = 0.3447 N = 242

$\beta = -0.0911$ SE(c) = 0.0962 p(c) = 0.3447 N = 242

Fisher-ADF on residuals: $\chi^2(22)=39.94$, $p=0.0110$

Fisher-ADF on residuals: $\chi^2(22)=39.94$, $p=0.0110$

✓ REJECT unit root in residuals ($p=0.0110$) -& evidence of cointegration

✓ REJECT unit root in residuals ($p=0.0110$) -& evidence of cointegration

07-C — STEP 2: ERROR-CORRECTION MODEL (cointegrated pairs only)
 07-C — STEP 2: ERROR-CORRECTION MODEL (cointegrated pairs only)

■ ■ T1: Homicide <-> Institutions ■ ■

■ ■ T1: Homicide <-> Institutions ■ ■

■ Not cointegrated in Step 1 -- ECM skipped (would be a regression of one $I(1)$ -differenced series on an unreliable, non-mean-reverting 'equilibrium error' -- not meaningful).

■ Not cointegrated in Step 1 -- ECM skipped (would be a regression of one $I(1)$ -differenced series on an unreliable, non-mean-reverting 'equilibrium error' -- not meaningful).

■ ■ T2: Institutions <-> Income ■ ■

■ ■ T2: Institutions <-> Income ■ ■

■ Not cointegrated in Step 1 -- ECM skipped (would be a regression of one $I(1)$ -differenced series on an unreliable, non-mean-reverting 'equilibrium error' -- not meaningful).

■ Not cointegrated in Step 1 -- ECM skipped (would be a regression of one $I(1)$ -differenced series on an unreliable, non-mean-reverting 'equilibrium error' -- not meaningful).

■ ■ T3: Homicide <-> Income ■ ■

■ ■ T3: Homicide <-> Income ■ ■

$\Delta \text{gdp_per_capita_log_it} = \alpha_i + \lambda_t + \phi_i \hat{e}_{it-1} + \gamma \Delta \text{homicide_rate_log_it} + \epsilon_{it}$

$\Delta \text{gdp_per_capita_log_it} = \alpha_i + \lambda_t + \phi_i \hat{e}_{it-1} + \gamma \Delta \text{homicide_rate_log_it} + \epsilon_{it}$

ϕ (speed of adjustment) = -0.1092 SE(cl)=0.0391 p(cl)=0.0058 p(dk)=0.0000 ***

ϕ (speed of adjustment) = -0.1092 SE(cl)=0.0391 p(cl)=0.0058 p(dk)=0.0000 ***

γ (short-run effect) = -0.0333 SE(cl)=0.0351 p(cl)=0.3440 p(dk)=0.1855

γ (short-run effect) = -0.0333 SE(cl)=0.0351 p(cl)=0.3440 p(dk)=0.1855

N = 227 Within $R^2 = 0.0265$

N = 227 Within $R^2 = 0.0265$

✓ ϕ < 0 and significant: $\text{gdp_per_capita_log}$ corrects 10.9% of a deviation from long-run equilibrium with homicide_rate_log each year.

✓ ϕ < 0 and significant: $\text{gdp_per_capita_log}$ corrects 10.9% of a deviation from long-run equilibrium with homicide_rate_log each year.

07-D — FIGURES

07-D — FIGURES

✓ Figure saved -> figures/11_cointegration_residuals.png

✓ Figure saved -> figures/11_cointegration_residuals.png

07-E — EXPORT

07-E — EXPORT

✓ Saved → C:\data_analyzer\econometric_pipeline\pipeline\json\07_cointegration.json

✓ Saved → C:\data_analyzer\econometric_pipeline\pipeline\json\07_cointegration.json

Module 07 complete.

Module 07 complete.

✓ Module 07 completed in 11.0s

=====
 RUNNING MODULE 08: 08_growth_ceiling_risk.py

=====
 g++ not available, if using conda: `conda install gxx`

g++ not available, if using conda: `conda install gxx`

Initializing NUTS using jitter+adapt_diag...

```

Initializing NUTS using jitter+adapt_diag...
Sequential sampling (4 chains in 1 job)
Sequential sampling (4 chains in 1 job)
NUTS: [alpha_mu, alpha_sigma, z, beta, gamma, sigma]
NUTS: [alpha_mu, alpha_sigma, z, beta, gamma, sigma]
Sampling 4 chains for 1_000 tune and 1_000 draw iterations (4_000 + 4_000 draws total) took 197 seconds.
Sampling 4 chains for 1_000 tune and 1_000 draw iterations (4_000 + 4_000 draws total) took 197 seconds.
Initializing NUTS using jitter+adapt_diag...
Initializing NUTS using jitter+adapt_diag...
Sequential sampling (4 chains in 1 job)
Sequential sampling (4 chains in 1 job)
NUTS: [alpha_mu, alpha_sigma, z, beta, gamma, sigma]
NUTS: [alpha_mu, alpha_sigma, z, beta, gamma, sigma]
Sampling 4 chains for 1_000 tune and 1_000 draw iterations (4_000 + 4_000 draws total) took 78 seconds.
Sampling 4 chains for 1_000 tune and 1_000 draw iterations (4_000 + 4_000 draws total) took 78 seconds.
Initializing NUTS using jitter+adapt_diag...
Initializing NUTS using jitter+adapt_diag...
Sequential sampling (4 chains in 1 job)
Sequential sampling (4 chains in 1 job)
NUTS: [alpha_mu, alpha_sigma, z, beta, gamma, sigma]
NUTS: [alpha_mu, alpha_sigma, z, beta, gamma, sigma]
Sampling 4 chains for 1_000 tune and 1_000 draw iterations (4_000 + 4_000 draws total) took 69 seconds.
Sampling 4 chains for 1_000 tune and 1_000 draw iterations (4_000 + 4_000 draws total) took 69 seconds.
Initializing NUTS using jitter+adapt_diag...
Initializing NUTS using jitter+adapt_diag...
Sequential sampling (4 chains in 1 job)
Sequential sampling (4 chains in 1 job)
NUTS: [alpha_mu, alpha_sigma, z, beta, gamma, sigma]
NUTS: [alpha_mu, alpha_sigma, z, beta, gamma, sigma]
Sampling 4 chains for 1_000 tune and 1_000 draw iterations (4_000 + 4_000 draws total) took 56 seconds.
Sampling 4 chains for 1_000 tune and 1_000 draw iterations (4_000 + 4_000 draws total) took 56 seconds.
Initializing NUTS using jitter+adapt_diag...
Initializing NUTS using jitter+adapt_diag...
Sequential sampling (4 chains in 1 job)
Sequential sampling (4 chains in 1 job)
NUTS: [alpha_mu, alpha_sigma, z, beta, gamma, sigma]
NUTS: [alpha_mu, alpha_sigma, z, beta, gamma, sigma]
Sampling 4 chains for 1_000 tune and 1_000 draw iterations (4_000 + 4_000 draws total) took 83 seconds.
Sampling 4 chains for 1_000 tune and 1_000 draw iterations (4_000 + 4_000 draws total) took 83 seconds.
Initializing NUTS using jitter+adapt_diag...
Initializing NUTS using jitter+adapt_diag...
Sequential sampling (4 chains in 1 job)
Sequential sampling (4 chains in 1 job)
NUTS: [alpha_mu, alpha_sigma, z, beta, gamma, sigma]
NUTS: [alpha_mu, alpha_sigma, z, beta, gamma, sigma]
Sampling 4 chains for 1_000 tune and 1_000 draw iterations (4_000 + 4_000 draws total) took 179 seconds.
Sampling 4 chains for 1_000 tune and 1_000 draw iterations (4_000 + 4_000 draws total) took 179 seconds.
Initializing NUTS using jitter+adapt_diag...
Initializing NUTS using jitter+adapt_diag...
Sequential sampling (4 chains in 1 job)
Sequential sampling (4 chains in 1 job)
NUTS: [alpha_mu, alpha_sigma, z, beta, gamma, sigma]
NUTS: [alpha_mu, alpha_sigma, z, beta, gamma, sigma]
Sampling 4 chains for 1_000 tune and 1_000 draw iterations (4_000 + 4_000 draws total) took 228 seconds.
Sampling 4 chains for 1_000 tune and 1_000 draw iterations (4_000 + 4_000 draws total) took 228 seconds.

```

Initializing NUTS using jitter+adapt_diag...
 Initializing NUTS using jitter+adapt_diag...
 Sequential sampling (4 chains in 1 job)
 Sequential sampling (4 chains in 1 job)
 NUTS: [alpha_mu, alpha_sigma, z, beta, gamma, sigma]
 NUTS: [alpha_mu, alpha_sigma, z, beta, gamma, sigma]
 Sampling 4 chains for 1_000 tune and 1_000 draw iterations (4_000 + 4_000 draws total) took 66 seconds.
 Sampling 4 chains for 1_000 tune and 1_000 draw iterations (4_000 + 4_000 draws total) took 66 seconds.

Initializing NUTS using jitter+adapt_diag...
 Initializing NUTS using jitter+adapt_diag...
 Sequential sampling (4 chains in 1 job)
 Sequential sampling (4 chains in 1 job)
 NUTS: [alpha_mu, alpha_sigma, z, beta, gamma, sigma]
 NUTS: [alpha_mu, alpha_sigma, z, beta, gamma, sigma]
 Sampling 4 chains for 1_000 tune and 1_000 draw iterations (4_000 + 4_000 draws total) took 120 seconds.
 Sampling 4 chains for 1_000 tune and 1_000 draw iterations (4_000 + 4_000 draws total) took 120 seconds.

08-A — LOAD

08-A — LOAD

✓ Primary sample loaded: 241 obs, 11 countries, key regressor = homicide_rate_log_lag1
 ✓ Primary sample loaded: 241 obs, 11 countries, key regressor = homicide_rate_log_lag1
 ✓ Secondary (institutions) sample loaded: 264 obs, 11 countries
 ✓ Secondary (institutions) sample loaded: 264 obs, 11 countries

Motivation: an exploratory frequentist quantile regression (see the
 Motivation: an exploratory frequentist quantile regression (see the
 README's "Exploratory Finding" section) found lagged violence has a
 README's "Exploratory Finding" section) found lagged violence has a
 large, LOCO-robust NEGATIVE effect on the UPPER tail ($q=0.90/0.95$) of
 large, LOCO-robust NEGATIVE effect on the UPPER tail ($q=0.90/0.95$) of
 GDP growth, and essentially no effect near the median -- a
 GDP growth, and essentially no effect near the median -- a
 "growth-ceiling" pattern, the mirror image of the classic lower-tail
 "growth-ceiling" pattern, the mirror image of the classic lower-tail
 Growth-at-Risk framing. This module re-estimates that pattern with a
 Growth-at-Risk framing. This module re-estimates that pattern with a
 hierarchical Bayesian model (partial pooling across 11 countries,
 hierarchical Bayesian model (partial pooling across 11 countries,
 full posterior uncertainty via MCMC) instead of LSDV dummies and
 full posterior uncertainty via MCMC) instead of LSDV dummies and
 asymptotic quantile-regression standard errors.
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08-B — BAYESIAN HIERARCHICAL QUANTILE REGRESSION (MCMC)

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■ q = 0.05 (x = homicide_rate_log_lag1) ■

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Frequentist (LSDV): coef = 2.2099 SE = 1.0706 p = 0.0401 **

Frequentist (LSDV): coef = 2.2099 SE = 1.0706 p = 0.0401 **

Bayesian (hierarchy): mean = 1.3050 94% HDI [-0.0255, 2.5276] P(beta<0|data) = 0.027

Bayesian (hierarchy): mean = 1.3050 94% HDI [-0.0255, 2.5276] P(beta<0|data) = 0.027

✓ MCMC diagnostics clean

✓ MCMC diagnostics clean

■ q = 0.10 (x = homicide_rate_log_lag1) ■

■ q = 0.10 (x = homicide_rate_log_lag1) ■

Frequentist (LSDV): coef = 0.2522 SE = 1.0982 p = 0.8185

Frequentist (LSDV): coef = 0.2522 SE = 1.0982 p = 0.8185

Bayesian (hierarchy): mean = 0.4923 94% HDI [-0.5294, 1.4378] P(beta<0|data) = 0.171

Bayesian (hierarchy): mean = 0.4923 94% HDI [-0.5294, 1.4378] P(beta<0|data) = 0.171

✓ MCMC diagnostics clean

✓ MCMC diagnostics clean

■ q = 0.25 (x = homicide_rate_log_lag1) ■

■ q = 0.25 (x = homicide_rate_log_lag1) ■

Frequentist (LSDV): coef = 0.7559 SE = 0.3746 p = 0.0448 **

Frequentist (LSDV): coef = 0.7559 SE = 0.3746 p = 0.0448 **

Bayesian (hierarchy): mean = 0.1927 94% HDI [-0.4646, 0.8430] P(beta<0|data) = 0.294

Bayesian (hierarchy): mean = 0.1927 94% HDI [-0.4646, 0.8430] P(beta<0|data) = 0.294

✓ MCMC diagnostics clean

✓ MCMC diagnostics clean

■ q = 0.50 (x = homicide_rate_log_lag1) ■

■ q = 0.50 (x = homicide_rate_log_lag1) ■

Frequentist (LSDV): coef = -0.0825 SE = 0.3758 p = 0.8265

Frequentist (LSDV): coef = -0.0825 SE = 0.3758 p = 0.8265

Bayesian (hierarchy): mean = -0.2312 94% HDI [-0.7692, 0.3725] P(beta<0|data) = 0.790

Bayesian (hierarchy): mean = -0.2312 94% HDI [-0.7692, 0.3725] P(beta<0|data) = 0.790

✓ MCMC diagnostics clean

✓ MCMC diagnostics clean

■ q = 0.75 (x = homicide_rate_log_lag1) ■

■ q = 0.75 (x = homicide_rate_log_lag1) ■

Frequentist (LSDV): coef = -0.1950 SE = 0.4294 p = 0.6502

Frequentist (LSDV): coef = -0.1950 SE = 0.4294 p = 0.6502

Bayesian (hierarchy): mean = -0.4463 94% HDI [-1.2126, 0.2622] P(beta<0|data) = 0.880

Bayesian (hierarchy): mean = -0.4463 94% HDI [-1.2126, 0.2622] P(beta<0|data) = 0.880

✓ MCMC diagnostics clean

✓ MCMC diagnostics clean

■ q = 0.90 (x = homicide_rate_log_lag1) ■

■ q = 0.90 (x = homicide_rate_log_lag1) ■

Frequentist (LSDV): coef = -2.2319 SE = 0.6277 p = 0.0005 ***

Frequentist (LSDV): coef = -2.2319 SE = 0.6277 p = 0.0005 ***

Bayesian (hierarchy): mean = -1.7621 94% HDI [-2.9338, -0.5859] P(beta<0|data) = 0.997

Bayesian (hierarchy): mean = -1.7621 94% HDI [-2.9338, -0.5859] P(beta<0|data) = 0.997

✓ MCMC diagnostics clean

✓ MCMC diagnostics clean

■ q = 0.95 (x = homicide_rate_log_lag1) ■

■ q = 0.95 (x = homicide_rate_log_lag1) ■

Frequentist (LSDV): coef = -2.0645 SE = 0.7160 p = 0.0043 ***

Frequentist (LSDV): coef = -2.0645 SE = 0.7160 p = 0.0043 ***

Bayesian (hierarchy): mean = -2.0792 94% HDI [-3.0179, -1.1073] P(beta<0|data) = 1.000

Bayesian (hierarchy): mean = -2.0792 94% HDI [-3.0179, -1.1073] P(beta<0|data) = 1.000

✓ MCMC diagnostics clean

✓ MCMC diagnostics clean

08-C — SECONDARY CHECK: institutions (inst_avg) at the tail quantiles

08-C — SECONDARY CHECK: institutions (inst_avg) at the tail quantiles

The exploratory frequentist check also tested inst_avg as the

The exploratory frequentist check also tested inst_avg as the

conditioning variable and found nothing at any quantile (all

conditioning variable and found nothing at any quantile (all

$p > 0.27$). Repeated here at the two headline tail quantiles under the

$p > 0.27$). Repeated here at the two headline tail quantiles under the

same hierarchical Bayesian model, to confirm the ceiling effect is

same hierarchical Bayesian model, to confirm the ceiling effect is

specific to violence rather than a generic feature of any regressor

specific to violence rather than a generic feature of any regressor

in this panel.

in this panel.

■ q = 0.90 (x = inst_avg) ■

■ q = 0.90 (x = inst_avg) ■

Frequentist (LSDV): coef = -0.3443 p = 0.7594

Frequentist (LSDV): coef = -0.3443 p = 0.7594

Bayesian (hierarchy): mean = 0.2014 94% HDI [-1.0963, 1.3643] $P(\beta \leq 0 | \text{data}) = 0.366$

Bayesian (hierarchy): mean = 0.2014 94% HDI [-1.0963, 1.3643] $P(\beta \leq 0 | \text{data}) = 0.366$

✓ Posterior is roughly symmetric around zero -- no ceiling effect from institutions, consistent with the exploratory frequentist check.

✓ Posterior is roughly symmetric around zero -- no ceiling effect from institutions, consistent with the exploratory frequentist check.

■ q = 0.95 (x = inst_avg) ■

■ q = 0.95 (x = inst_avg) ■

Frequentist (LSDV): coef = -1.2969 p = 0.3034

Frequentist (LSDV): coef = -1.2969 p = 0.3034

Bayesian (hierarchy): mean = -0.1551 94% HDI [-1.4014, 1.0624] $P(\beta \leq 0 | \text{data}) = 0.597$

Bayesian (hierarchy): mean = -0.1551 94% HDI [-1.4014, 1.0624] $P(\beta \leq 0 | \text{data}) = 0.597$

✓ Posterior is roughly symmetric around zero -- no ceiling effect from institutions, consistent with the exploratory frequentist check.

✓ Posterior is roughly symmetric around zero -- no ceiling effect from institutions, consistent with the exploratory frequentist check.

08-D — GROWTH-CEILING-AT-RISK SCENARIO

08-D — GROWTH-CEILING-AT-RISK SCENARIO

Low-violence scenario (p10 of homicide_rate_log_lag1): 2.185

Low-violence scenario (p10 of homicide_rate_log_lag1): 2.185

High-violence scenario (p90 of homicide_rate_log_lag1): 4.042

High-violence scenario (p90 of homicide_rate_log_lag1): 4.042

Both evaluated at the average country (α_{μ}) and average year (year_c=0).

Both evaluated at the average country (α_{μ}) and average year (year_c=0).

q=0.9: ceiling at low violence = 8.79 pts of growth

q=0.9: ceiling at low violence = 8.79 pts of growth

ceiling at high violence = 5.52 pts of growth

ceiling at high violence = 5.52 pts of growth

drop = 3.27 pts 94% HDI [1.09, 5.45] $P(\text{drop} > 0 | \text{data}) = 0.997$

drop = 3.27 pts 94% HDI [1.09, 5.45] $P(\text{drop} > 0 | \text{data}) = 0.997$

$q=0.95$: ceiling at low violence = 10.29 pts of growth
 $q=0.95$: ceiling at low violence = 10.29 pts of growth
 ceiling at high violence = 6.43 pts of growth
 ceiling at high violence = 6.43 pts of growth
 drop = 3.86 pts 94% HDI [2.06, 5.61] $P(\text{drop} > 0 | \text{data}) = 1.000$
 drop = 3.86 pts 94% HDI [2.06, 5.61] $P(\text{drop} > 0 | \text{data}) = 1.000$

08-E — FIGURES

08-E — FIGURES

✓ Figure saved -> figures/12_growth_ceiling_bayesian.png
 ✓ Figure saved -> figures/12_growth_ceiling_bayesian.png
 ✓ Figure saved -> figures/13_growth_ceiling_scenario.png
 ✓ Figure saved -> figures/13_growth_ceiling_scenario.png

08-F — LATEX TABLE

08-F — LATEX TABLE

✓ LaTeX table saved -> tables/growth_ceiling_bayesian.tex
 ✓ LaTeX table saved -> tables/growth_ceiling_bayesian.tex

08-G — EXPORT

08-G — EXPORT

✓ Saved → C:\data_analyzer\econometric_pipeline\pipeline\json\08_growth_ceiling_risk.json
 ✓ Saved → C:\data_analyzer\econometric_pipeline\pipeline\json\08_growth_ceiling_risk.json

Module 08 complete.

Module 08 complete.

✓ Module 08 completed in 1112.8s

PIPELINE SUMMARY

✓ Module 01: Data Preparation & Institution Index
 ✓ Module 02: Panel Estimation (Two-Way FE)
 ✓ Module 03: Panel Diagnostics
 ✓ Module 04: Wild Cluster Bootstrap Inference
 ✓ Module 05: Robustness Checks
 ✓ Module 06: ML Triangulation

✓ Module 07: Panel Cointegration & ECM

✓ Module 08: Growth-Ceiling-at-Risk (Bayesian MCMC)

Total runtime: 3016.1s

Output directories:

Figures → C:\data_analyzer\econometric_pipeline\pipeline\figures/

Tables → C:\data_analyzer\econometric_pipeline\pipeline\tables/

JSON → C:\data_analyzer\econometric_pipeline\pipeline\json/